MACHINE LEARNING- I

INN HOTELS GROUP PROJECT

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INN Hotels Project

Context

A significant number of hotel bookings are called-off due to cancellations or no-shows. The typical reasons for cancellations include change of plans, scheduling conflicts, etc. This is often made easier by the option to do so free of charge or preferably at a low cost which is beneficial to hotel guests but it is a less desirable and possibly revenue-diminishing factor for hotels to deal with. Such losses are particularly high on last-minute cancellations.

The new technologies involving online booking channels have dramatically changed customers' booking possibilities and behavior. This adds a further dimension to the challenge of how hotels handle cancellations, which are no longer limited to traditional booking and guest characteristics.

The cancellation of bookings impact a hotel on various fronts:

- Loss of resources (revenue) when the hotel cannot resell the room.
- Additional costs of distribution channels by increasing commissions or paying for publicity to help sell these rooms Lowering prices last minute, so the hotel can resell a room, resulting in reducing the profit margin.
- Human resources to make arrangements for the guests.

Objective

The increasing number of cancellations calls for a Machine Learning based solution that can help in predicting which booking is likely to be canceled. INN Hotels Group has a chain of hotels in Portugal, they are facing problems with the high number of booking cancellations and have reached out to your firm for data-driven solutions. You as a data scientist have to analyze the data provided to find which factors have a high influence on booking cancellations, build a predictive model that can predict which booking is going to be canceled in advance, and help in formulating profitable policies for cancellations and refunds.

Data Description

The data contains the different attributes of customers' booking details. The detailed data dictionary is given below.

Data Dictionary

- Booking_ID: unique identifier of each booking
- no_of_adults: Number of adults
- no_of_children: Number of Children
- no_of_weekend_nights: Number of weekend nights (Saturday or Sunday) the guest stayed or booked to stay at the hotel

- no_of_week_nights: Number of week nights (Monday to Friday) the guest stayed or booked to stay at the hotel
- type_of_meal_plan: Type of meal plan booked by the customer:
  - Not Selected – No meal plan selected
  - Meal Plan 1 – Breakfast
  - Meal Plan 2 – Half board (breakfast and one other meal)
  - Meal Plan 3 – Full board (breakfast, lunch, and dinner)
- required_car_parking_space: Does the customer require a car parking space? (0 - No, 1- Yes)
- room_type_reserved: Type of room reserved by the customer. The values are ciphered (encoded) by INN Hotels.
- lead_time: Number of days between the date of booking and the arrival date
- arrival_year: Year of arrival date
- arrival_month: Month of arrival date
- arrival_date: Date of the month
- market_segment_type: Market segment designation.
- repeated_guest: Is the customer a repeated guest? (0 - No, 1- Yes)
- no_of_previous_cancellations: Number of previous bookings that were canceled by the customer prior to the current booking
- no_of_previous_bookings_not_canceled: Number of previous bookings not canceled by the customer prior to the current booking
- avg_price_per_room: Average price per day of the reservation; prices of the rooms are dynamic. (in euros)
- no_of_special_requests: Total number of special requests made by the customer (e.g. high floor, view from the room, etc)
- booking_status: Flag indicating if the booking was canceled or not.

Data Overview

Load Dataset

```
  Booking_ID  no_of_adults  no_of_children  no_of_weekend_nights  \
0   INN00001             2               0                     1
1   INN00002             2               0                     2
2   INN00003             1               0                     2
3   INN00004             2               0                     0
4   INN00005             2               0                     1

   no_of_week_nights type_of_meal_plan  required_car_parking_space  \
0                  2       Meal Plan 1                           0
1                  3      Not Selected                           0
2                  1       Meal Plan 1                           0
3                  2       Meal Plan 1                           0
4                  1      Not Selected                           0


  room_type_reserved  lead_time  arrival_year  arrival_month  arrival_date  \
0        Room_Type 1        224          2017             10             2
```

```
1        Room_Type 1          5          2018             11             6
2        Room_Type 1          1          2018              2            28
3        Room_Type 1        211          2018              5            20
4        Room_Type 1         48          2018              4            11

  market_segment_type  repeated_guest  no_of_previous_cancellations  \
0             Offline               0                             0
1              Online               0                             0
2              Online               0                             0
3              Online               0                             0
4              Online               0                             0

   no_of_previous_bookings_not_canceled  avg_price_per_room  \
0                                     0            65.00000
1                                     0           106.68000
2                                     0            60.00000
3                                     0           100.00000
4                                     0            94.50000

   no_of_special_requests booking_status
0                       0   Not_Canceled
1                       1   Not_Canceled
2                       0       Canceled
3                       0       Canceled
4                       0       Canceled
```

```
##Last 5 rows of the dataset
```

```
      Booking_ID  no_of_adults  no_of_children  no_of_weekend_nights  \
36270   INN36271             3               0                     2
36271   INN36272             2               0                     1
36272   INN36273             2               0                     2
36273   INN36274             2               0                     0
36274   INN36275             2               0                     1

       no_of_week_nights type_of_meal_plan  required_car_parking_space  \
36270                  6       Meal Plan 1                           0
36271                  3       Meal Plan 1                           0
36272                  6       Meal Plan 1                           0
36273                  3      Not Selected                           0
36274                  2       Meal Plan 1                           0

      room_type_reserved  lead_time  arrival_year  arrival_month  \
36270        Room_Type 4         85          2018              8
36271        Room_Type 1        228          2018             10
36272        Room_Type 1        148          2018              7
36273        Room_Type 1         63          2018              4
36274        Room_Type 1        207          2018             12

       arrival_date market_segment_type  repeated_guest  \
36270             3              Online               0
36271            17              Online               0
36272             1              Online               0
36273            21              Online               0
36274            30             Offline               0
```

```
       no_of_previous_cancellations  no_of_previous_bookings_not_canceled  \
36270                             0                                     0
36271                             0                                     0
36272                             0                                     0
36273                             0                                     0
36274                             0                                     0

       avg_price_per_room  no_of_special_requests booking_status
36270           167.80000                       1   Not_Canceled
36271            90.95000                       2       Canceled
36272            98.39000                       2   Not_Canceled
36273            94.50000                       0       Canceled
36274           161.67000                       0   Not_Canceled
```

```
data.shape
```

```
(36275, 19)
```

There are 36275 rows with 19 columns in the dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 36275 entries, 0 to 36274
Data columns (total 19 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   Booking_ID                            36275 non-null  object
 1   no_of_adults                          36275 non-null  int64
 2   no_of_children                        36275 non-null  int64
 3   no_of_weekend_nights                  36275 non-null  int64
 4   no_of_week_nights                     36275 non-null  int64
 5   type_of_meal_plan                     36275 non-null  object
 6   required_car_parking_space            36275 non-null  int64
 7   room_type_reserved                    36275 non-null  object
 8   lead_time                             36275 non-null  int64
 9   arrival_year                          36275 non-null  int64
 10  arrival_month                         36275 non-null  int64
 11  arrival_date                          36275 non-null  int64
 12  market_segment_type                   36275 non-null  object
 13  repeated_guest                        36275 non-null  int64
 14  no_of_previous_cancellations          36275 non-null  int64
 15  no_of_previous_bookings_not_canceled  36275 non-null  int64
 16  avg_price_per_room                    36275 non-null  float64
 17  no_of_special_requests                36275 non-null  int64
 18  booking_status                        36275 non-null  object
dtypes: float64(1), int64(13), object(5)
memory usage: 5.3+ MB
```

- `Booking_ID`,`type_of_meal_plan`,`room_type_reserved`,`market_segment_type` and `booking_status` are the columns with object type while the rest of the columns are numeric.

Check for the mssing values and dropping the duplicate Values

#Checking for null values

```
Booking_ID                              0
no_of_adults                            0
no_of_children                          0
no_of_weekend_nights                    0
no_of_week_nights                       0
type_of_meal_plan                       0
required_car_parking_space              0
room_type_reserved                      0
lead_time                               0
arrival_year                            0
arrival_month                           0
arrival_date                            0
market_segment_type                     0
repeated_guest                          0
no_of_previous_cancellations            0
no_of_previous_bookings_not_canceled    0
avg_price_per_room                      0
no_of_special_requests                  0
booking_status                          0
dtype: int64
```

#Check for unique values

```
Booking_ID                              36275
no_of_adults                                5
no_of_children                              6
no_of_weekend_nights                        8
no_of_week_nights                          18
type_of_meal_plan                           4
required_car_parking_space                  2
room_type_reserved                          7
lead_time                                 352
arrival_year                                2
arrival_month                              12
arrival_date                               31
market_segment_type                         5
repeated_guest                              2
no_of_previous_cancellations                9
no_of_previous_bookings_not_canceled       59
avg_price_per_room                       3930
no_of_special_requests                      6
booking_status                              2
dtype: int64
```

As shown above the Booking_ID has 36275 unique values which is the number of rows in dataset. So let's drop the booking_ID column

Exploratory Data Analysis (EDA)

- EDA is an important part of any project involving data.
- It is important to investigate and understand the data better before building a model with it.
- A few questions have been mentioned below which will help you approach the analysis in the right manner and generate insights from the data.
- A thorough analysis of the data, in addition to the questions mentioned below, should be done.

Statistical Summary of the data

	count	mean	std \
no_of_adults	36275.00000	1.84496	0.51871
no_of_children	36275.00000	0.10528	0.40265
no_of_weekend_nights	36275.00000	0.81072	0.87064
no_of_week_nights	36275.00000	2.20430	1.41090
required_car_parking_space	36275.00000	0.03099	0.17328
lead_time	36275.00000	85.23256	85.93082
arrival_year	36275.00000	2017.82043	0.38384
arrival_month	36275.00000	7.42365	3.06989
arrival_date	36275.00000	15.59700	8.74045
repeated_guest	36275.00000	0.02564	0.15805
no_of_previous_cancellations	36275.00000	0.02335	0.36833
no_of_previous_bookings_not_canceled	36275.00000	0.15341	1.75417
avg_price_per_room	36275.00000	103.42354	35.08942
no_of_special_requests	36275.00000	0.61966	0.78624

	min	25%	50% \
no_of_adults	0.00000	2.00000	2.00000
no_of_children	0.00000	0.00000	0.00000
no_of_weekend_nights	0.00000	0.00000	1.00000
no_of_week_nights	0.00000	1.00000	2.00000
required_car_parking_space	0.00000	0.00000	0.00000
lead_time	0.00000	17.00000	57.00000
arrival_year	2017.00000	2018.00000	2018.00000
arrival_month	1.00000	5.00000	8.00000
arrival_date	1.00000	8.00000	16.00000
repeated_guest	0.00000	0.00000	0.00000
no_of_previous_cancellations	0.00000	0.00000	0.00000
no_of_previous_bookings_not_canceled	0.00000	0.00000	0.00000
avg_price_per_room	0.00000	80.30000	99.45000
no_of_special_requests	0.00000	0.00000	0.00000

	75%	max
no_of_adults	2.00000	4.00000
no_of_children	0.00000	10.00000
no_of_weekend_nights	2.00000	7.00000
no_of_week_nights	3.00000	17.00000
required_car_parking_space	0.00000	1.00000
lead_time	126.00000	443.00000
arrival_year	2018.00000	2018.00000
arrival_month	10.00000	12.00000
arrival_date	23.00000	31.00000

```
repeated_guest                          0.00000    1.00000
no_of_previous_cancellations            0.00000   13.00000
no_of_previous_bookings_not_canceled    0.00000   58.00000
avg_price_per_room                    120.00000  540.00000
no_of_special_requests                  1.00000    5.00000
```

- The number of adults are ranging from 0 to 4 with a mean of approaximately 2 adults per booking
- The number of children are ranging from 0 to 10 per booking
- On an average the reservations are done during the weekdays with atleast 2 nights per reservation with 17 days max whereas on weekend the average nights are booked 1 nights per reservations

Univariate Analysis

Leadtime Observations

As shown below most of the reservations are with 0 leading time meaning the reservations are done at the last moment or directly at the hotel

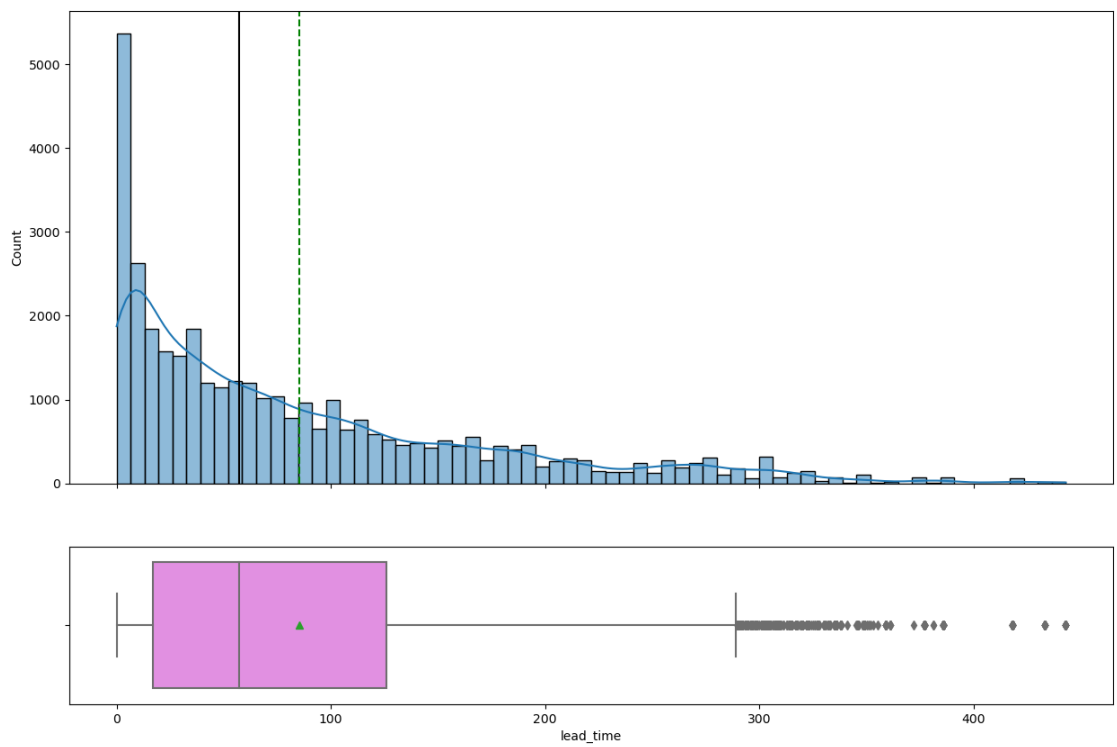


Figure 1 lead time

Average Price per Room graphs

The average price for the room is 100 € hut there are a lot of outliers in the dataset for Average price of the room.

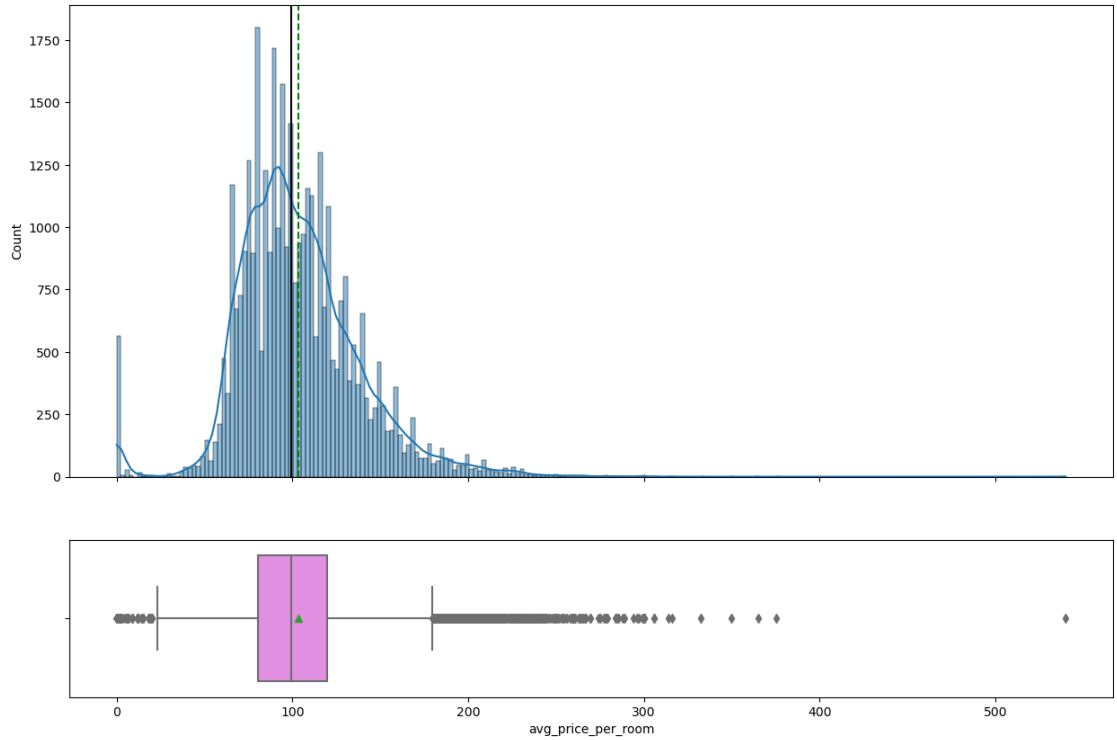


Figure 2 average price per room

```
       no_of_adults  no_of_children  no_of_weekend_nights  no_of_week_nights  \
63                1               0                     0                  1
145               1               0                     0                  2
209               1               0                     0                  0
266               1               0                     0                  2
267               1               0                     2                  1
...             ...             ...                   ...                ...
35983             1               0                     0                  1
36080             1               0                     1                  1
36114             1               0                     0                  1
36217             2               0                     2                  1
36250             1               0                     0                  2

      type_of_meal_plan  required_car_parking_space room_type_reserved  \
63          Meal Plan 1                           0        Room_Type 1
145         Meal Plan 1                           0        Room_Type 1
209         Meal Plan 1                           0        Room_Type 1
266         Meal Plan 1                           0        Room_Type 1
267         Meal Plan 1                           0        Room_Type 1
...                 ...                         ...                ...
35983       Meal Plan 1                           0        Room_Type 7
36080       Meal Plan 1                           0        Room_Type 7
36114       Meal Plan 1                           0        Room_Type 1
36217       Meal Plan 1                           0        Room_Type 2
36250       Meal Plan 2                           0        Room_Type 1

       lead_time  arrival_year  arrival_month  arrival_date  \
63             2          2017              9            10
```

```
145           13          2018              6             1
209            4          2018              2            27
266            1          2017              8            12
267            4          2017              8            23
...          ...           ...            ...           ...
35983          0          2018              6             7
36080          0          2018              3            21
36114          1          2018              3             2
36217          3          2017              8             9
36250          6          2017             12            10

      market_segment_type  repeated_guest  no_of_previous_cancellations  \
63          Complementary               0                             0
145         Complementary               1                             3
209         Complementary               0                             0
266         Complementary               1                             0
267         Complementary               0                             0
...                   ...             ...                           ...
35983       Complementary               1                             4
36080       Complementary               1                             3
36114              Online               0                             0
36217              Online               0                             0
36250              Online               0                             0

       no_of_previous_bookings_not_canceled  avg_price_per_room  \
63                                        0             0.00000
145                                       5             0.00000
209                                       0             0.00000
266                                       1             0.00000
267                                       0             0.00000
...                                     ...                 ...
35983                                    17             0.00000
36080                                    15             0.00000
36114                                     0             0.00000
36217                                     0             0.00000
36250                                     0             0.00000

       no_of_special_requests booking_status
63                           1   Not_Canceled
145                          1   Not_Canceled
209                          1   Not_Canceled
266                          1   Not_Canceled
267                          1   Not_Canceled
...                        ...            ...
35983                        1   Not_Canceled
36080                        1   Not_Canceled
36114                        0   Not_Canceled
36217                        2   Not_Canceled
36250                        0   Not_Canceled

[545 rows x 18 columns]

Complementary    354
Online           191
Name: market_segment_type, dtype: int64
```

	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	\
4150	2	2	1	2	
5163	2	2	1	1	
9461	3	0	0	2	
13911	2	0	2	3	
13944	2	2	0	1	
14773	2	2	0	3	
20900	2	1	1	2	
25670	2	2	1	2	
29779	2	0	1	2	
30475	2	0	2	3	
33114	2	0	0	1	
33955	2	0	1	2	
34306	2	2	0	3	
35970	2	0	2	3	

	type_of_meal_plan	required_car_parking_space	room_type_reserved	\
4150	Meal Plan 1	0	Room_Type 7	
5163	Meal Plan 1	0	Room_Type 7	
9461	Meal Plan 1	0	Room_Type 4	
13911	Meal Plan 2	0	Room_Type 1	
13944	Meal Plan 2	1	Room_Type 6	
14773	Meal Plan 1	0	Room_Type 6	
20900	Meal Plan 2	0	Room_Type 1	
25670	Meal Plan 2	0	Room_Type 6	
29779	Meal Plan 2	0	Room_Type 1	
30475	Meal Plan 2	0	Room_Type 1	
33114	Meal Plan 1	0	Room_Type 1	
33955	Meal Plan 2	0	Room_Type 4	
34306	Meal Plan 2	0	Room_Type 6	
35970	Meal Plan 2	0	Room_Type 1	

	lead_time	arrival_year	arrival_month	arrival_date	\
4150	4	2018	7	8	
5163	18	2018	10	22	
9461	21	2018	12	30	
13911	170	2018	7	29	
13944	6	2018	8	13	
14773	28	2018	6	2	
20900	173	2018	7	25	
25670	11	2018	9	16	
29779	173	2018	7	25	
30475	151	2018	7	3	
33114	35	2018	3	25	
33955	57	2018	12	30	
34306	43	2018	12	29	
35970	172	2018	7	29	

	market_segment_type	repeated_guest	no_of_previous_cancellations	\
4150	Online	0	0	
5163	Online	0	0	
9461	Online	0	0	
13911	Offline	0	0	

```
13944              Online               0                              0
14773              Online               0                              0
20900             Offline               0                              0
25670              Online               0                              0
29779             Offline               0                              0
30475             Offline               0                              0
33114             Offline               0                              0
33955              Online               0                              0
34306              Online               0                              0
35970             Offline               0                              0

       no_of_previous_bookings_not_canceled  avg_price_per_room  \
4150                                      0           179.55000
5163                                      0           179.55000
9461                                      0           179.55000
13911                                     0           179.55000
13944                                     0           179.55000
14773                                     0           179.55000
20900                                     0           179.55000
25670                                     0           179.55000
29779                                     0           179.55000
30475                                     0           179.55000
33114                                     0           179.55000
33955                                     0           179.55000
34306                                     0           179.55000
35970                                     0           179.55000

       no_of_special_requests booking_status
4150                         3   Not_Canceled
5163                         4   Not_Canceled
9461                         0   Not_Canceled
13911                        0       Canceled
13944                        0       Canceled
14773                        1   Not_Canceled
20900                        1       Canceled
25670                        0       Canceled
29779                        0       Canceled
30475                        0       Canceled
33114                        0       Canceled
33955                        0   Not_Canceled
34306                        1   Not_Canceled
35970                        1       Canceled
```

Observation on the number of the previous bookings not canceled

Highly Skewed Distribution: The histogram (and accompanying boxplot) typically shows that a large majority of bookings are associated with a value of zero. In simple terms, most guests have no history of prior non-canceled bookings. This suggests that many bookings belong to either entirely new customers or to those who have not yet established a reliable booking record.

Presence of Outliers: Although most of the data is concentrated at zero, there is a long right tail that includes a small number of customers with several previous non-canceled bookings. These outliers can represent loyal customers who consistently honor their reservations. Such a pattern is common in customer behavior data—for example, a few repeated guests may drive a significant portion of the revenue by their reliability.

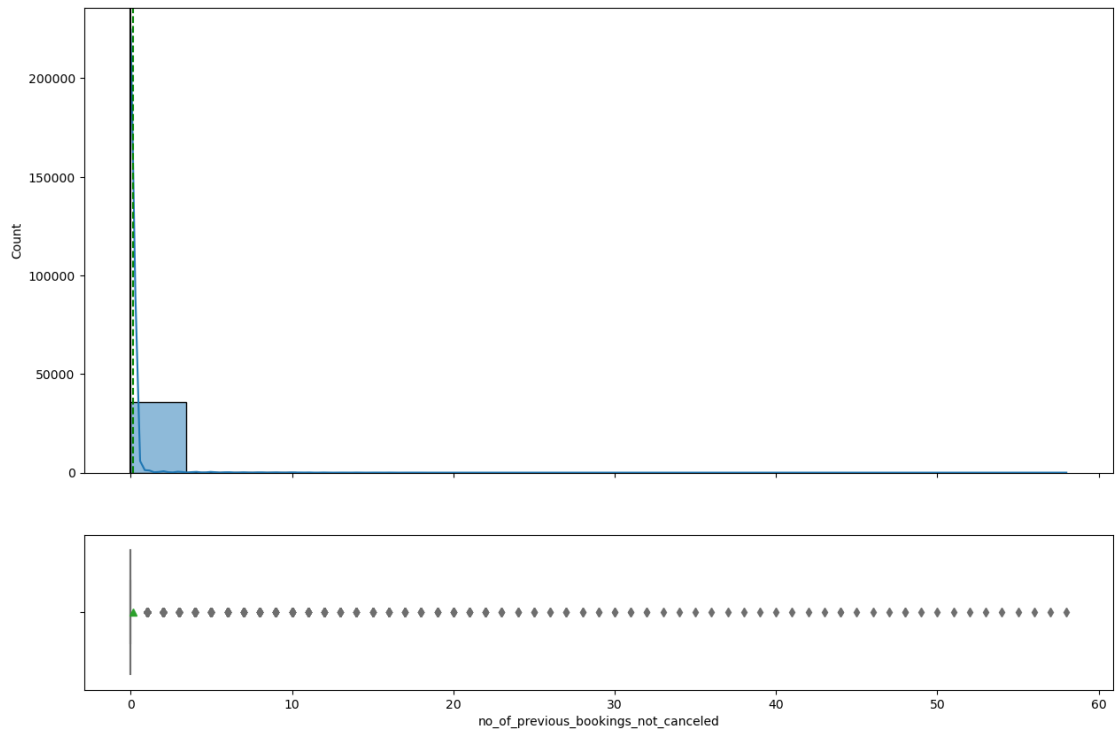


Figure 3 no. of prev bookings not canceled

Observation on the number of the previous booking canceled

Distribution Shape & Concentration at Zero A univariate visualization (such as a histogram with an overlaid density curve and a boxplot) of this variable typically shows that the overwhelming majority of bookings have a value of zero. This indicates that most customers have not canceled any prior bookings. In other words, reliability is the norm, and a large segment of guests appears to have a clean cancellation history.

Presence of a Right-Tailed Distribution Although most of the observations are zeros, the histogram and boxplot often reveal a long right tail. This tail comprises a small subset of customers who have canceled one or multiple past bookings. These outlying cases signify a behavioral pattern where, despite being a minority, some guests are significantly more likely to cancel their reservations than the average customer.

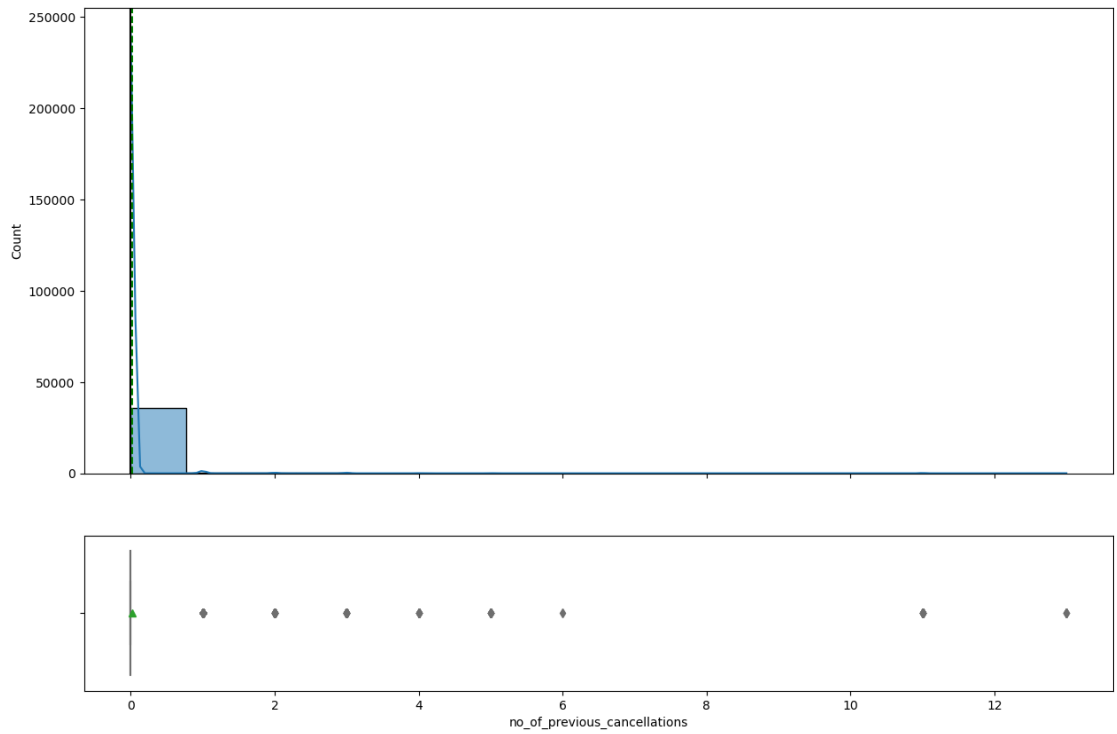


Figure 4 no. of previous cancellations

Observations on Number of Adults

Range and Central Tendency

- The number of adults in each booking ranges from 0 to 4.
- On average, there are approximately 2 adults per booking—this is consistent with both the mean and the median values observed in the statistical summary of the dataset.
- The concentration around 2 adults suggests that most bookings are made for either couples or small families.

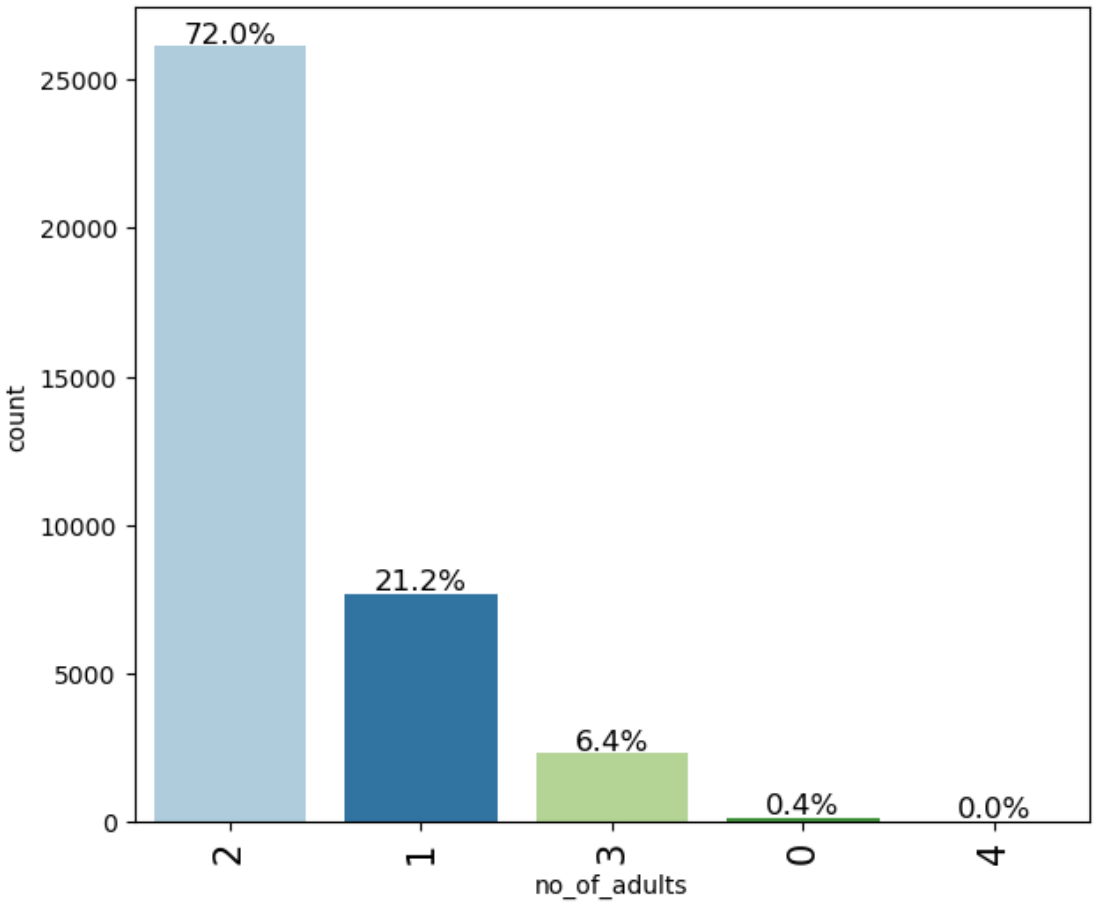


Figure 5 No. of adults

Observations on Number of Children

Distribution and Range

A common pattern observed in hospitality booking data is that the vast majority of records report 0 children—indicating that many bookings are made by individuals or couples without children. As the number of children increases, the frequency of observations typically diminishes, resulting in a right-skewed distribution. In other words, while there are a few records with 3, 4, or more children, these instances are relatively rare.

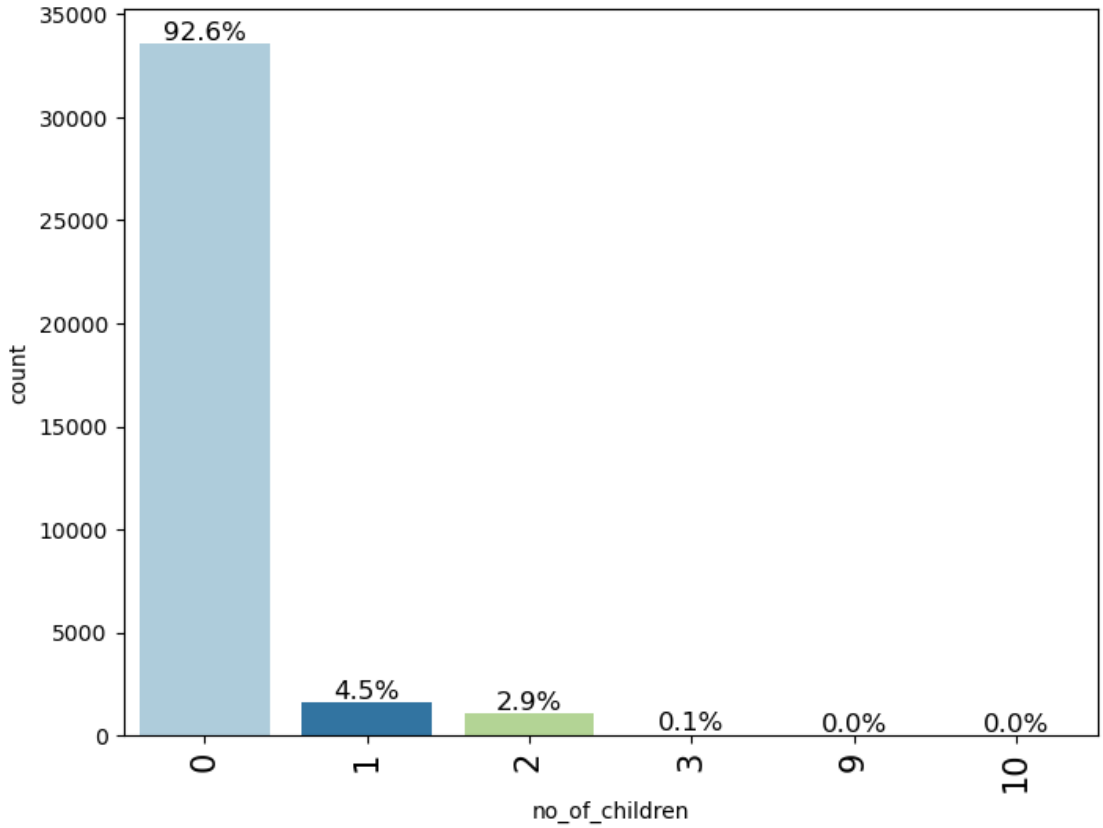


Figure 6 No. of children

Observations on Number of Week Nights

Range and Central Tendency

The values tend to be small; many bookings have just 0, 1, or 2 week nights, while a few rare cases extend to much higher counts.

Distribution Characteristics

The distribution of week nights shows a right-skewed pattern. A large number of observations cluster at the lower values—indicating that most guests book few week nights—but there is a long tail with some bookings extending to a high number of nights. These outliers might represent extended stays typical of business travelers or guests taking advantage of special offers.

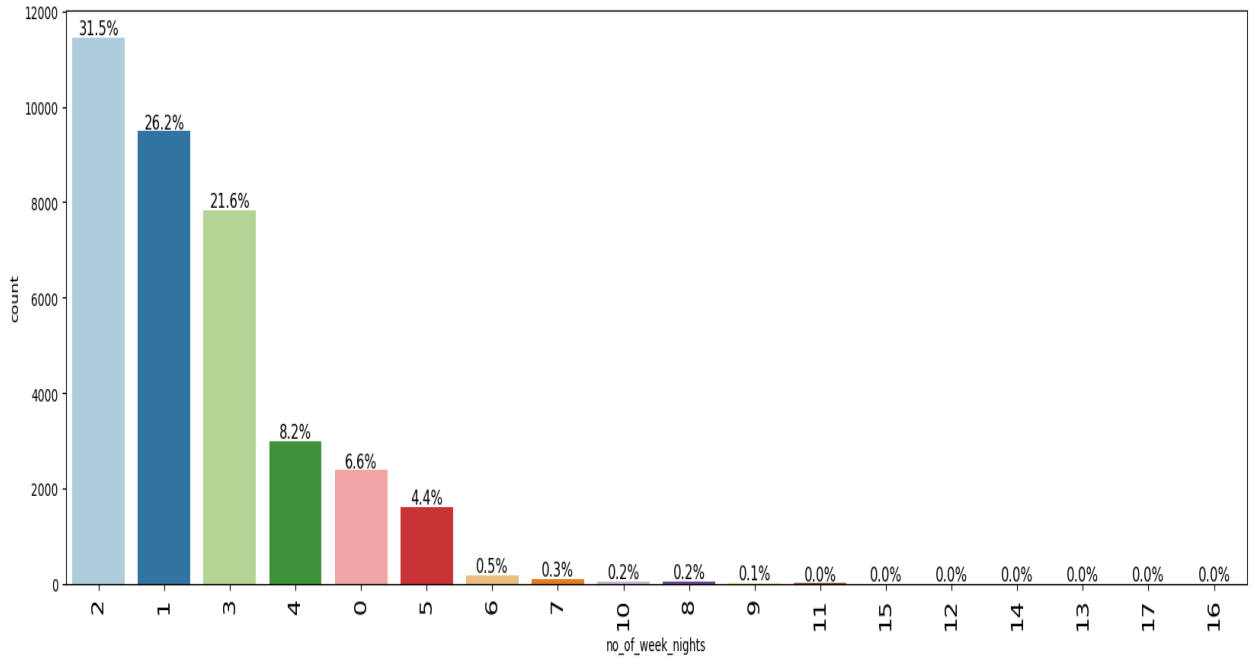


Figure 7 No. of week nights

Observations on Number of Weekend Nights

Distribution and Range

- Although most entries cluster at the lower end, there is a right-skewed tail. This means that while the majority record 0 or 1 weekend night, a few cases extend to 2 or more, reflecting leisure travelers or longer, planned multi week trips.

Central Tendency and Dispersion

- Most guests do not book an extended weekend portion.
- The narrow range of common values suggests that weekend stays are not as varied as, say, weekday stays in this dataset. However, the presence of even a few higher values can influence overall statistics like the mean.

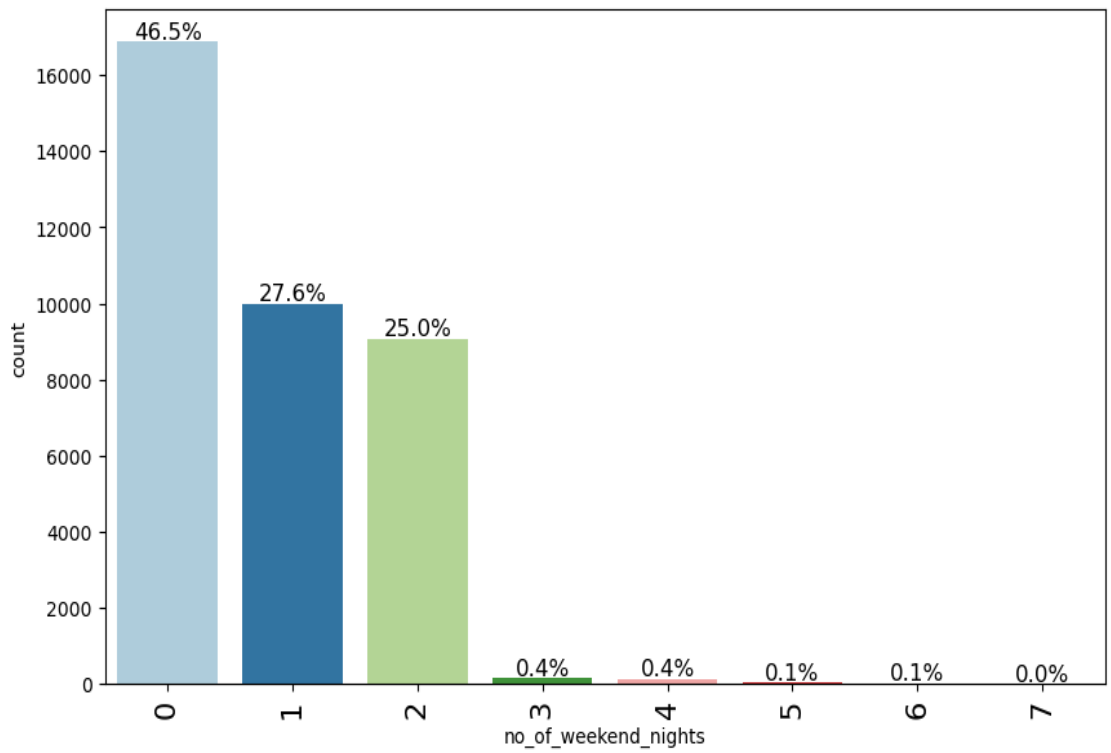


Figure 8 No. of weekend nights

Observations on Required Parking Space

Frequency Distribution

A substantial proportion of customers did not require parking (a value of 0). However, there is often a minority that does require parking (a value of 1).

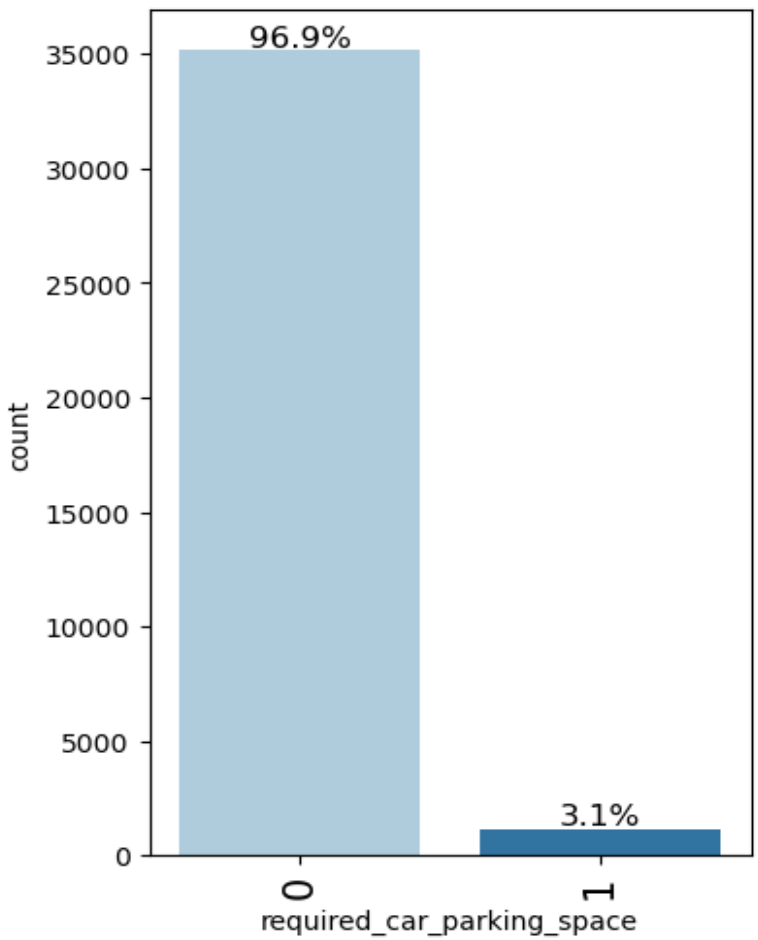


Figure 9 Required car parking

Observations on Type of Meal Plan

Meal plan 1 is the most dominant followed by no meal plan, indicating that people probably sleep longer on vacation and prefer the ease of having breakfast in their room. People who do not opt for meal plan 1 go for no meal plan or meal plan two but no one opts for meal plan 3. The distribution is right skewed with one dominant category.

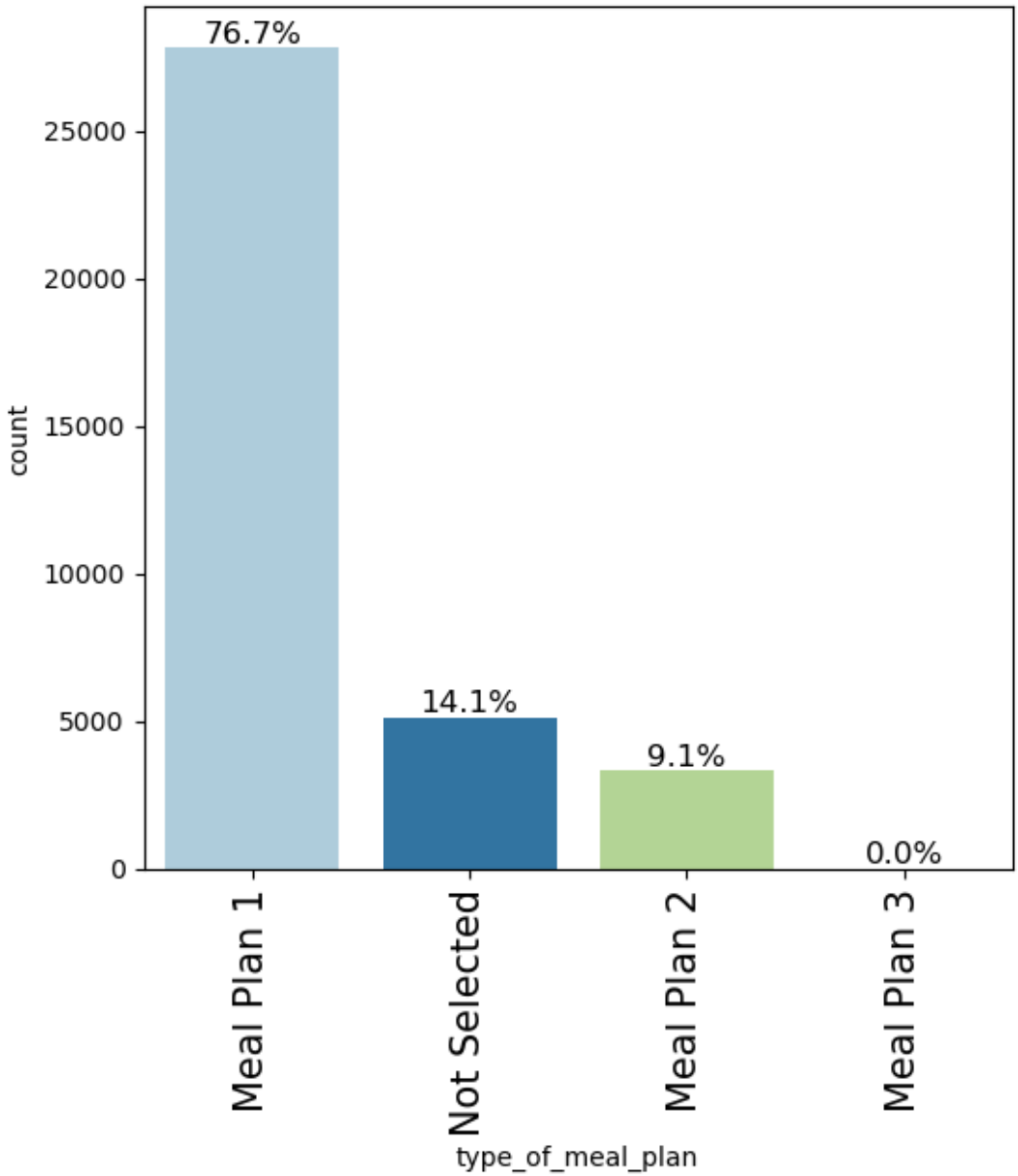


Figure 10 Type of meal plan

Observations on Room Type Reserved

The distribution is right skewed with long tail, Room type 1 dominated the the distribution by a wide margin which is followed by room type 4.

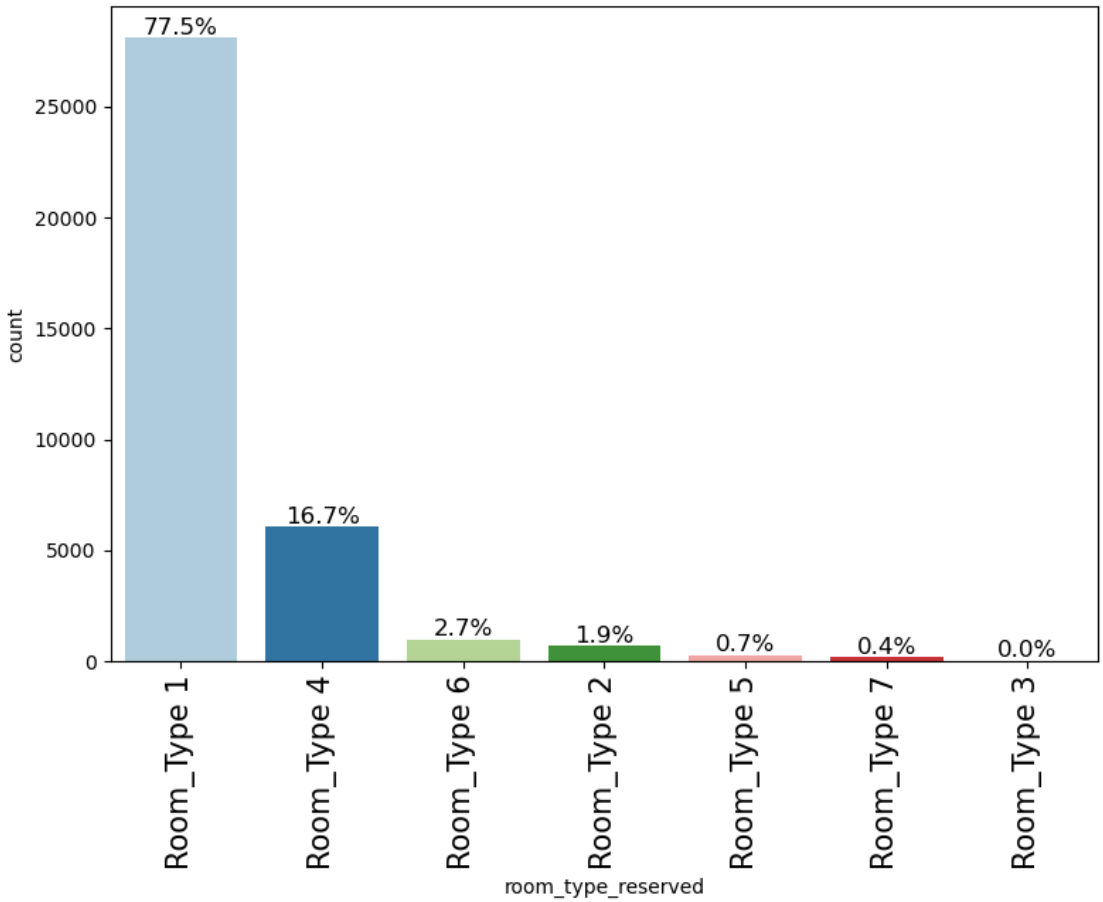


Figure 11 Room type reserved

Observations on Market Segment Type

Distribution of Bookings Across Segments

- **Online** channel is the dominant source of bookings. This indicates that a majority of reservations are coming from online platforms.
- Other market segments observed include **Offline**, **Aviation**, and **Corporate**. Although these segments contribute to the overall bookings, their frequencies are lower compared to the Online segment.

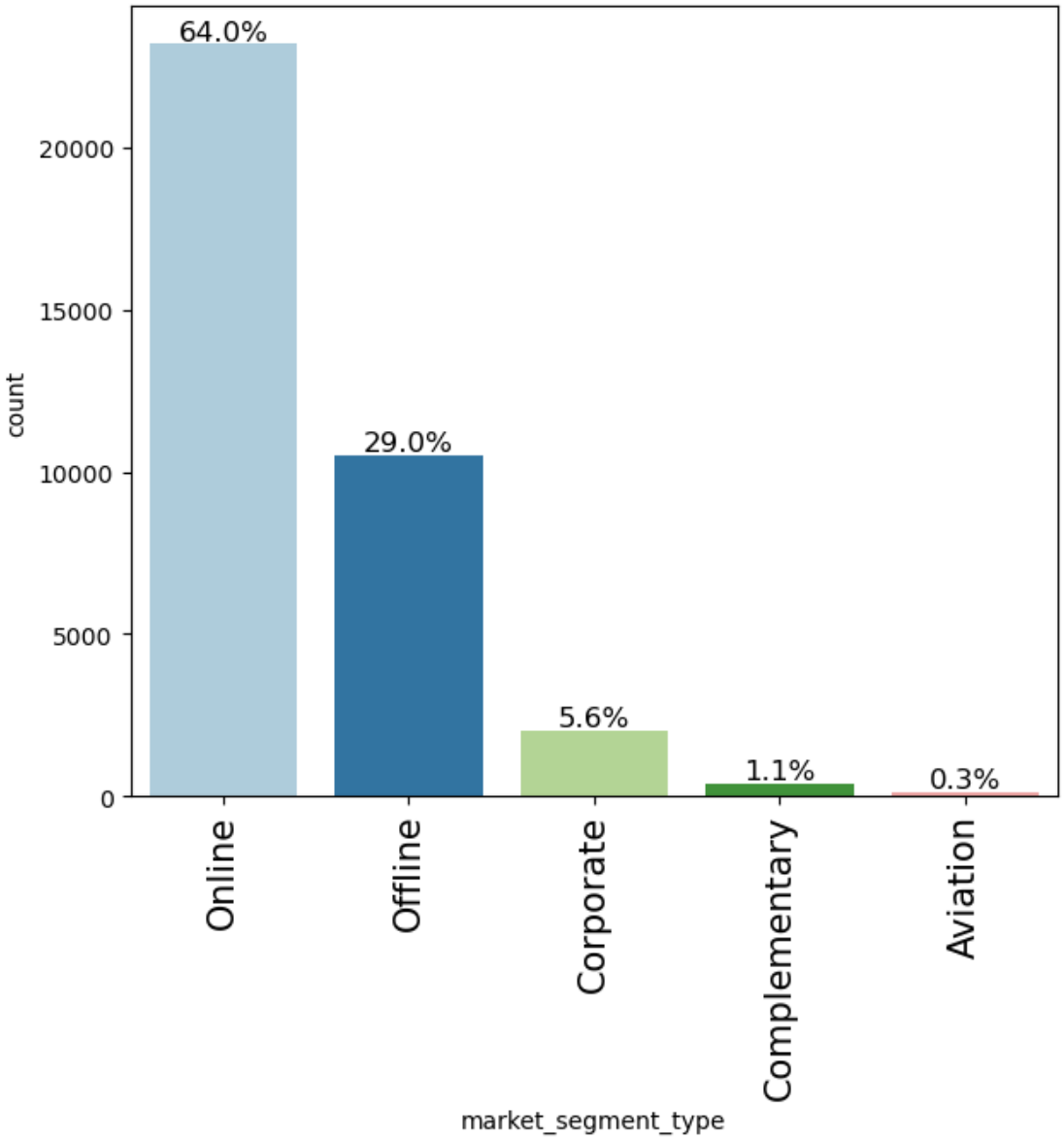


Figure 12 Market segment type

Observations on Number of Special Requests

Distribution Characteristics

- The vast majority of bookings register **0 special requests**. This indicates that most guests do not specify any additional requirements beyond the standard booking.
- As the count of special requests increases (1, 2, 3, etc.), the frequency steadily decreases. Only a small minority of bookings include multiple special requests.

- In some cases, there are bookings with up to 4 special requests. Although these are rare, they stand out because they signal a highly personalized or demanding guest profile.

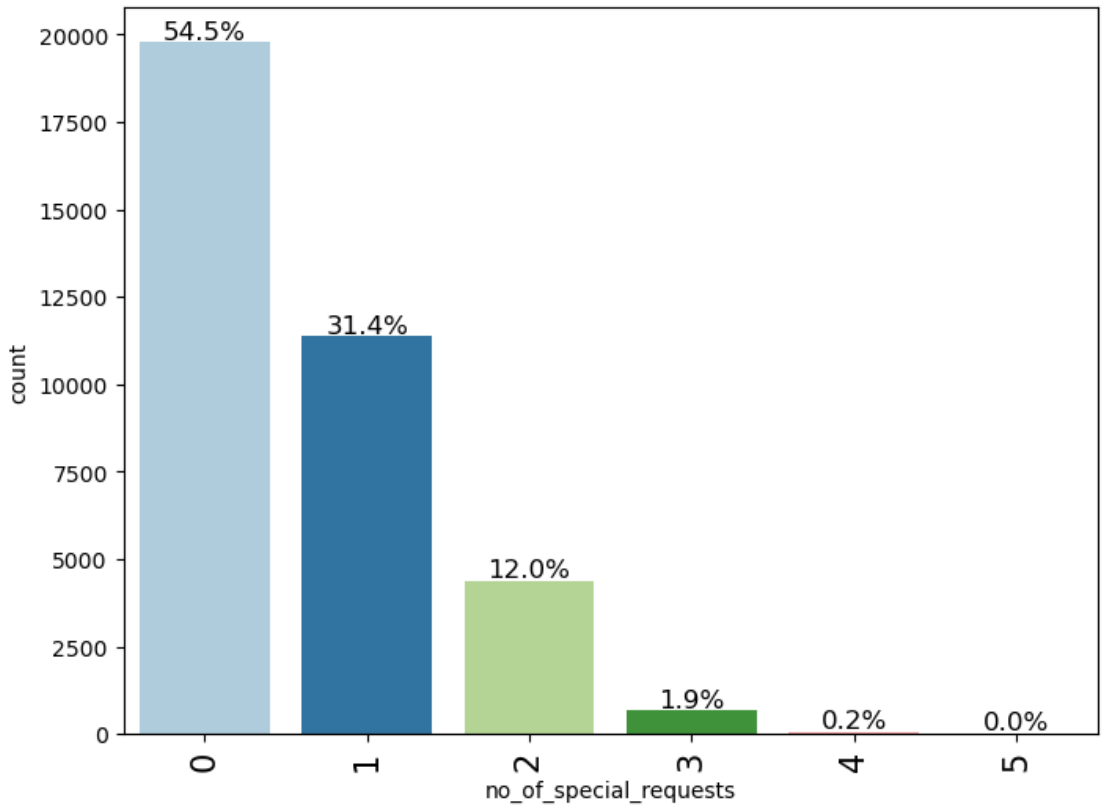


Figure 13 No. of special requests

Observations on Booking Status

In this dataset around **33% of the bookings are canceled**, while roughly 67% are completed. This observed cancellation rate is significant for operational planning and revenue management, as it highlights that one in three bookings does not result in an actual stay.

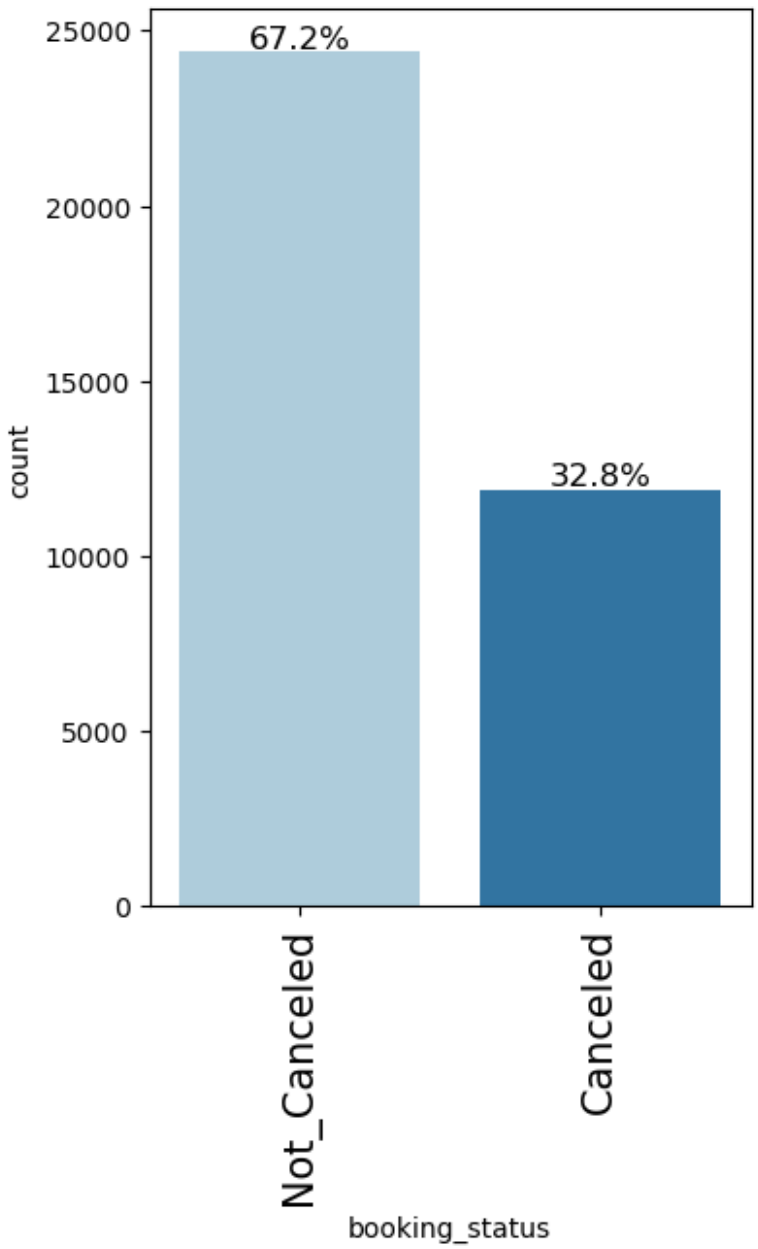


Figure 14 Booking status

Observations on Arrival Month

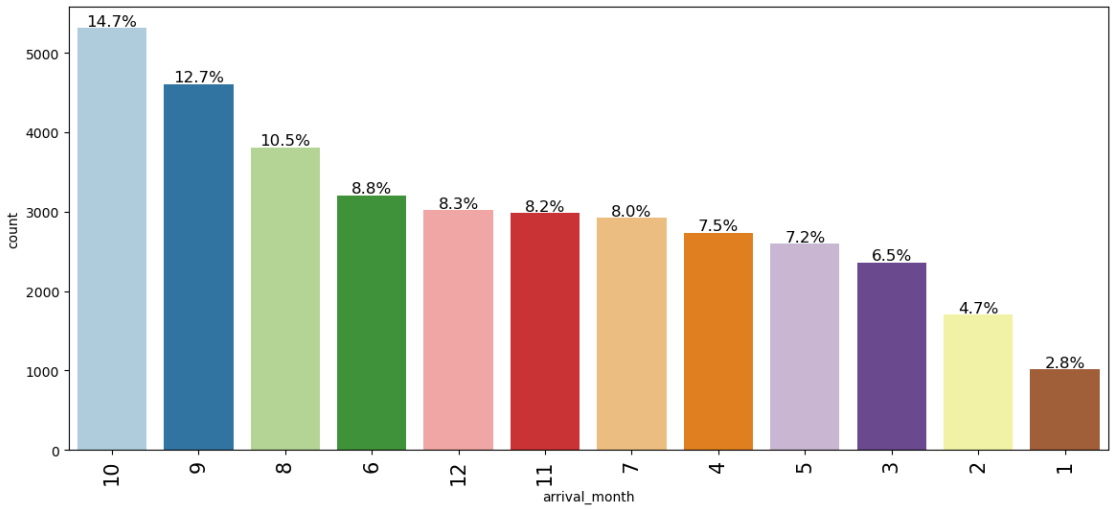


Figure 15 Arrival month

Bivariate Analysis

```
#Let's create the Heatmap
```

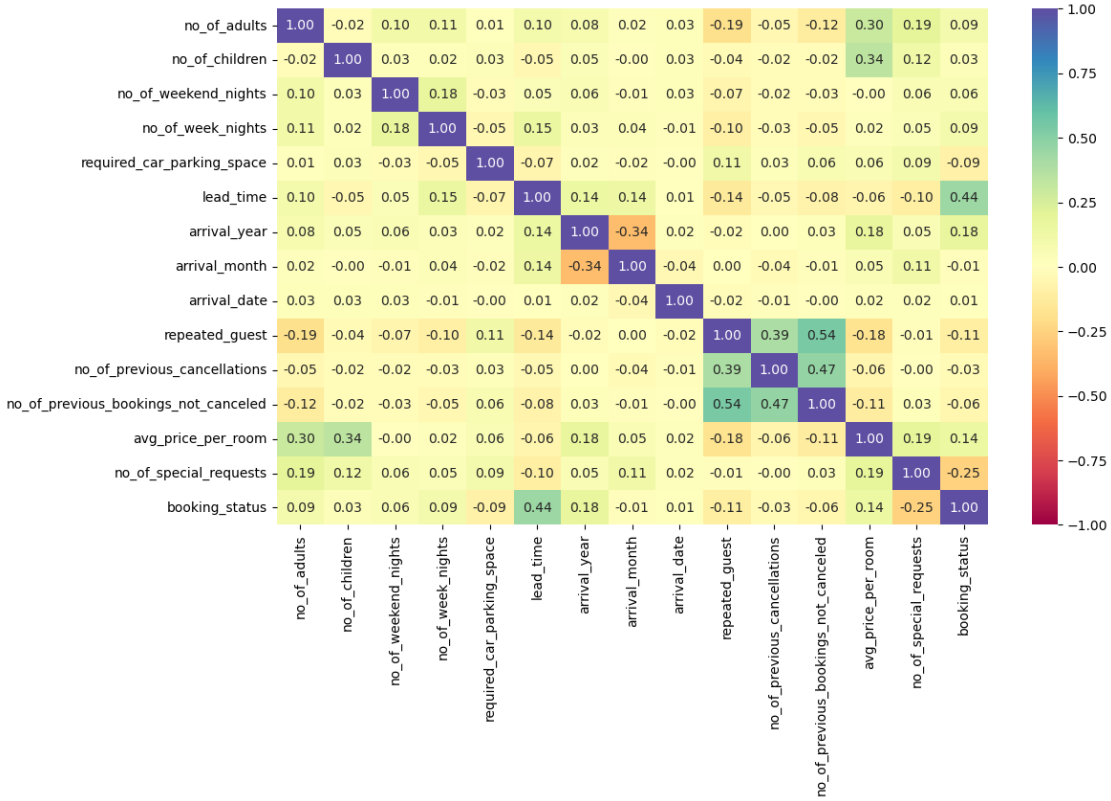


Figure 16 Heatmap

Observations

- A positive correlation exists between repeated guests and the number of previous bookings not cancelled.
- There is a slightly positive correlation between the lead time and booking status.
- There's also a slightly negative correlation between the arrival year and arrival month.

The price of a hotel room vary according to the demographics and demand, Let's see the barplot of Avg Price WRT market segment

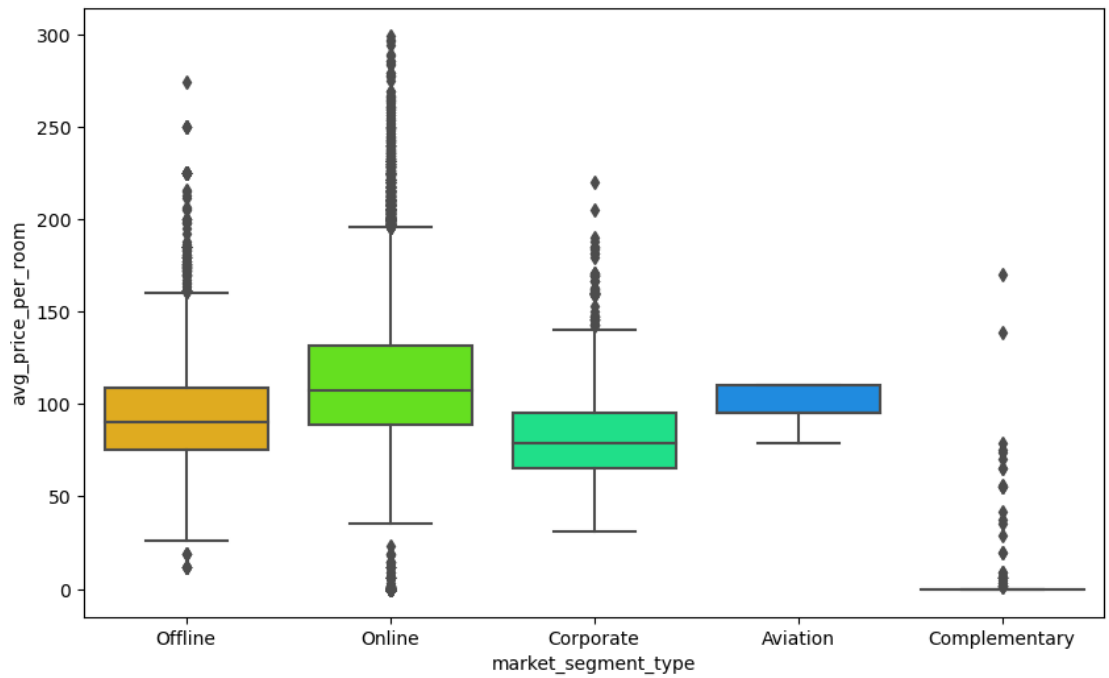


Figure 17 stack barplot market segment and avg price per room

```
booking_status           0      1    All
market_segment_type
All                  24390  11885  36275
Online               14739   8475  23214
Offline               7375   3153  10528
Corporate             1797    220   2017
Aviation                88     37    125
Complementary          391      0    391
--------------------------------------------------------------------------------
-----------------------------------------
```

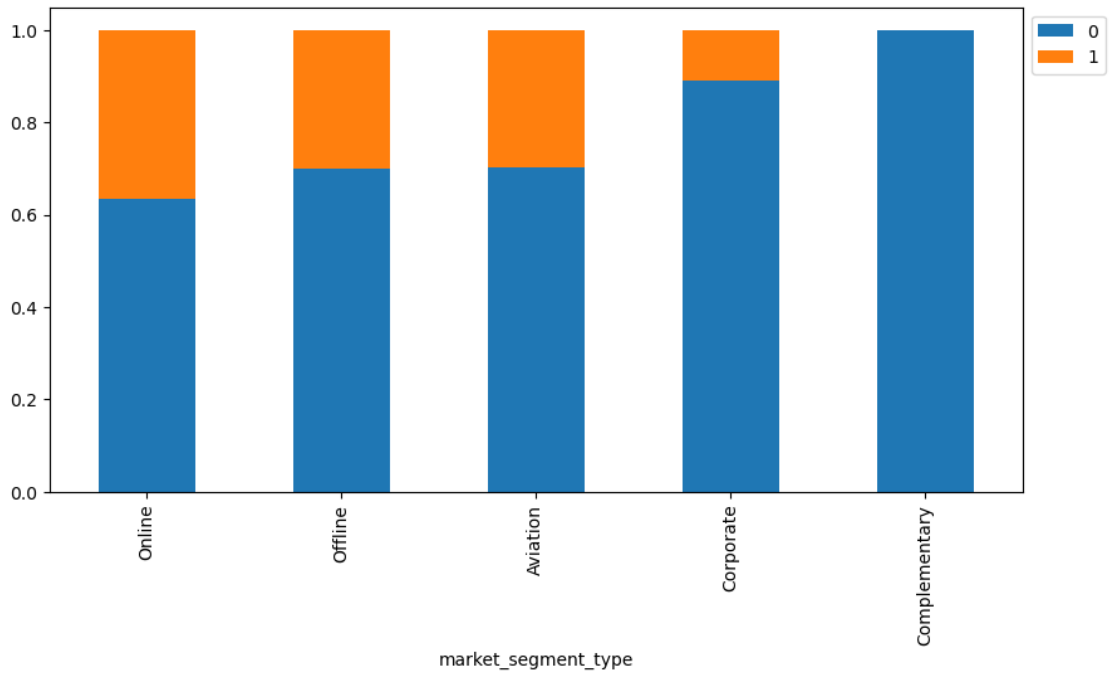


Figure 18 market segment type

Observations

- Around 38% of the online guests cancelled
- Almost 30% of the offline guests cancelled
- Only 20% the aviation guests cancelled
- Almost 10% corporate guests cancelled

Let's look at the barplot of Special requirements WRT to cancellations

```
stack_barplot(data,"no_of_special_requests", "booking_status")
```

```
booking_status              0      1    All
no_of_special_requests
All                     24390  11885  36275
0                       11232   8545  19777
1                        8670   2703  11373
2                        3727    637   4364
3                         675      0    675
4                          78      0     78
5                           8      0      8
------------------------------------------------------------------------------
-----------------------------------------
```

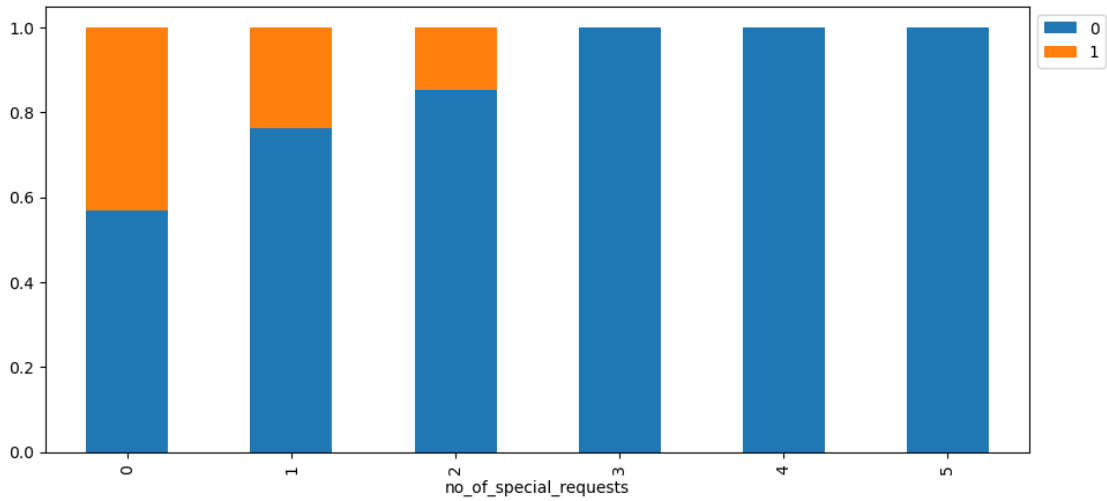


Figure 19 no. of special requests

- As predicted 100% of the guests with 5 special requests did not cancel
- Almost 40% of the guests cancelled with zero special requests
- Around 20% of the guests cancelled with 1 special requests

Let's look at the relationship between special requests and avg price

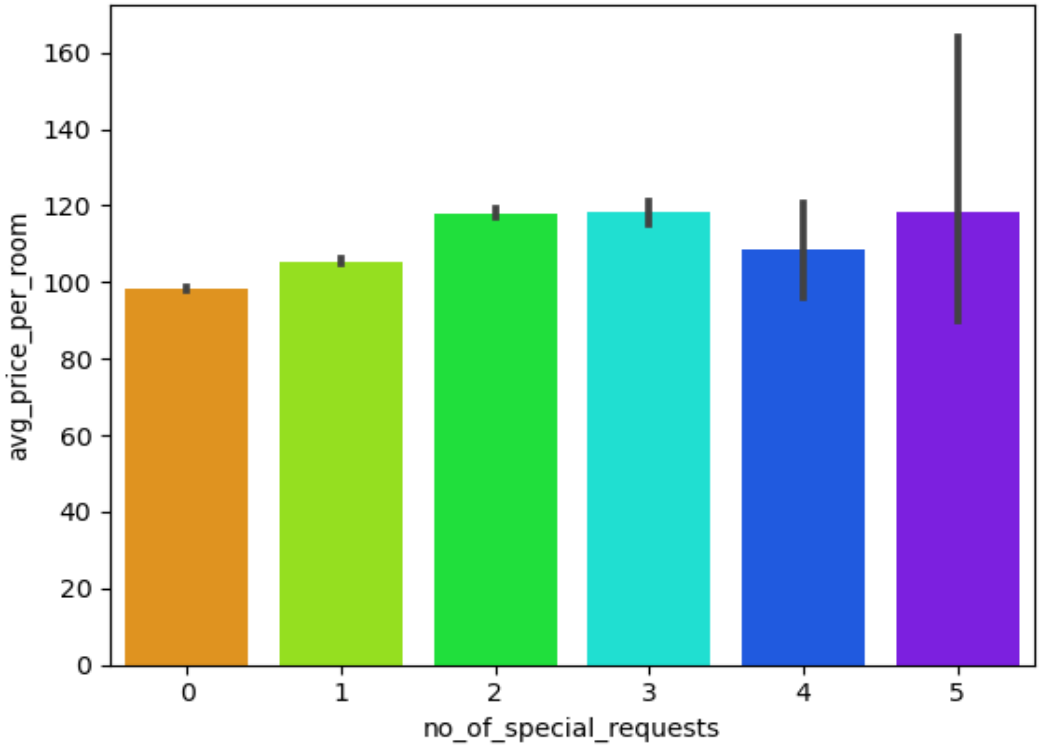


Figure 20 no. of special requests and avg price per room

Looking at the positive correlations between booking status and Average price per room

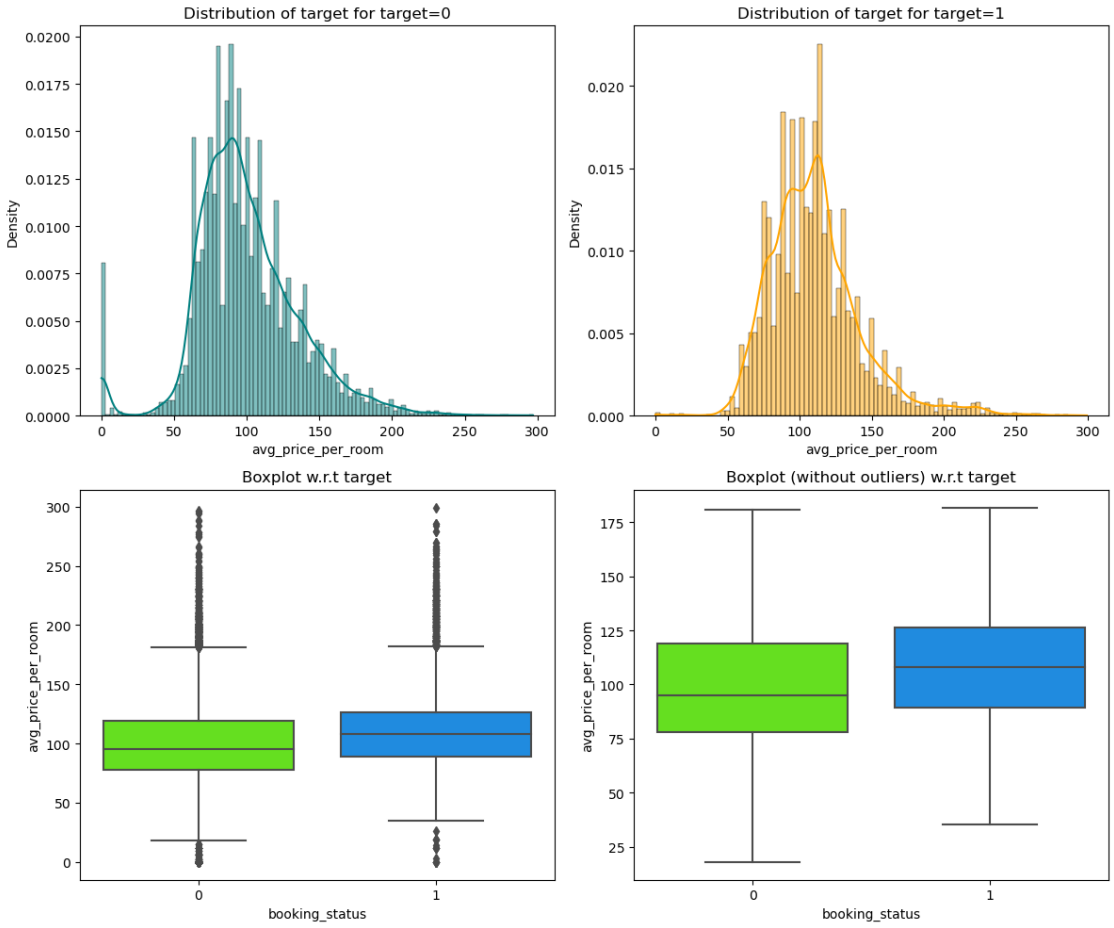


Figure 21 booking status and Average price per room

Graph of the booking status and lead time

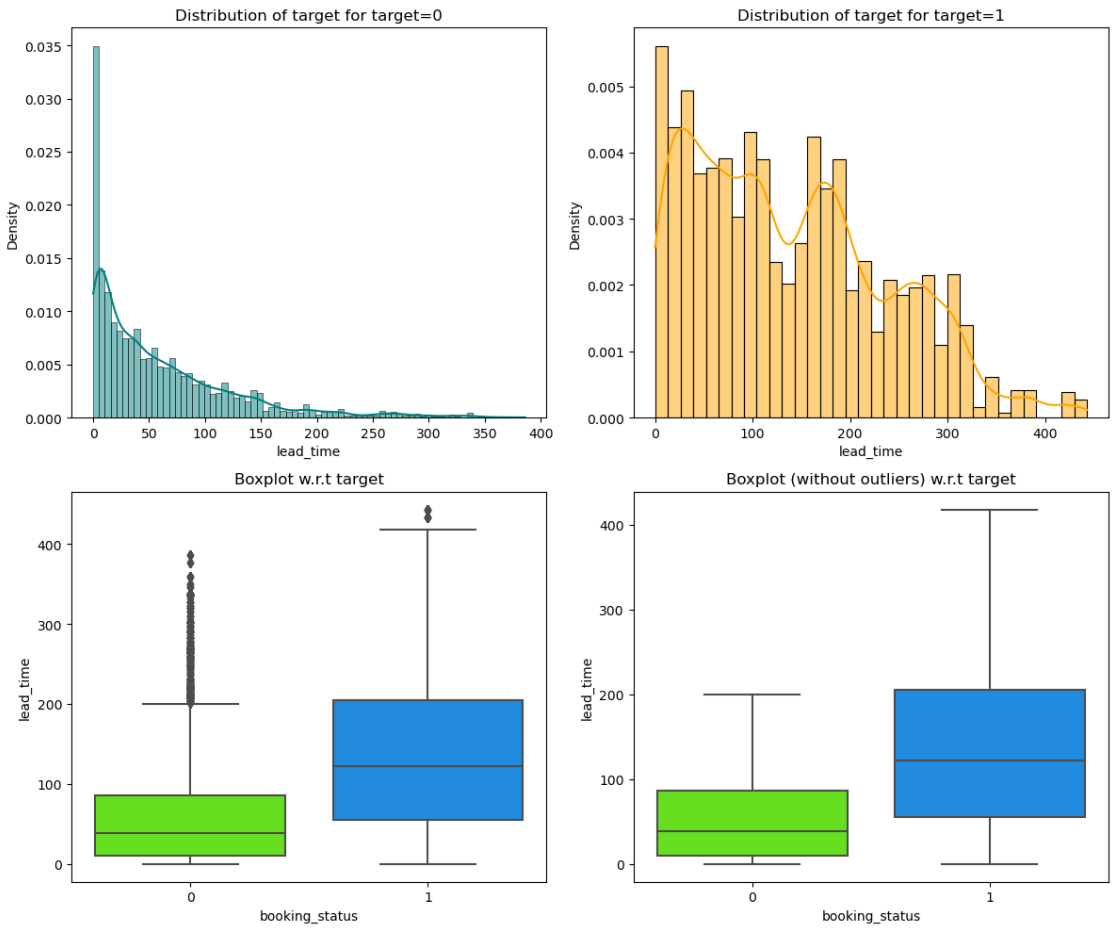


Figure 22 booking status and lead time

Creating a new dataset of the customers who travelled with the families, because people usually travel in group

```
       no_of_adults  no_of_children  no_of_weekend_nights  no_of_week_nights  \
0                 2               0                     1                  2
1                 2               0                     2                  3
3                 2               0                     0                  2
4                 2               0                     1                  1
5                 2               0                     0                  2
...             ...             ...                   ...                ...
36270             3               0                     2                  6
36271             2               0                     1                  3
36272             2               0                     2                  6
36273             2               0                     0                  3
36274             2               0                     1                  2

      type_of_meal_plan  required_car_parking_space room_type_reserved  \
0           Meal Plan 1                           0        Room_Type 1
1          Not Selected                           0        Room_Type 1
3           Meal Plan 1                           0        Room_Type 1
4          Not Selected                           0        Room_Type 1
```

```
5           Meal Plan 2                           0        Room_Type 1
...                 ...                         ...                ...
36270       Meal Plan 1                           0        Room_Type 4
36271       Meal Plan 1                           0        Room_Type 1
36272       Meal Plan 1                           0        Room_Type 1
36273      Not Selected                           0        Room_Type 1
36274       Meal Plan 1                           0        Room_Type 1

       lead_time  arrival_year  arrival_month  arrival_date  \
0            224          2017             10             2
1              5          2018             11             6
3            211          2018              5            20
4             48          2018              4            11
5            346          2018              9            13
...          ...           ...            ...           ...
36270         85          2018              8             3
36271        228          2018             10            17
36272        148          2018              7             1
36273         63          2018              4            21
36274        207          2018             12            30

      market_segment_type  repeated_guest  no_of_previous_cancellations  \
0                 Offline               0                             0
1                  Online               0                             0
3                  Online               0                             0
4                  Online               0                             0
5                  Online               0                             0
...                   ...             ...                           ...
36270              Online               0                             0
36271              Online               0                             0
36272              Online               0                             0
36273              Online               0                             0
36274             Offline               0                             0

       no_of_previous_bookings_not_canceled  avg_price_per_room  \
0                                         0            65.00000
1                                         0           106.68000
3                                         0           100.00000
4                                         0            94.50000
5                                         0           115.00000
...                                     ...                 ...
36270                                     0           167.80000
36271                                     0            90.95000
36272                                     0            98.39000
36273                                     0            94.50000
36274                                     0           161.67000

       no_of_special_requests  booking_status
0                           0               0
1                           1               0
3                           0               1
4                           0               1
5                           1               1
...                       ...             ...
36270                       1               0
```

```
36271                       2               1
36272                       2               0
36273                       0               1
36274                       0               0

[28441 rows x 18 columns]
```

```
family["no_of_family_members"] = (family["no_of_adults"] + family["no_of_children"])
family
```

```
       no_of_adults  no_of_children  no_of_weekend_nights  no_of_week_nights  \
0                 2               0                     1                  2
1                 2               0                     2                  3
3                 2               0                     0                  2
4                 2               0                     1                  1
5                 2               0                     0                  2
...             ...             ...                   ...                ...
36270             3               0                     2                  6
36271             2               0                     1                  3
36272             2               0                     2                  6
36273             2               0                     0                  3
36274             2               0                     1                  2

      type_of_meal_plan  required_car_parking_space room_type_reserved  \
0           Meal Plan 1                           0        Room_Type 1
1          Not Selected                           0        Room_Type 1
3           Meal Plan 1                           0        Room_Type 1
4          Not Selected                           0        Room_Type 1
5           Meal Plan 2                           0        Room_Type 1
...                 ...                         ...                ...
36270       Meal Plan 1                           0        Room_Type 4
36271       Meal Plan 1                           0        Room_Type 1
36272       Meal Plan 1                           0        Room_Type 1
36273      Not Selected                           0        Room_Type 1
36274       Meal Plan 1                           0        Room_Type 1

       lead_time  arrival_year  arrival_month  arrival_date  \
0            224          2017             10             2
1              5          2018             11             6
3            211          2018              5            20
4             48          2018              4            11
5            346          2018              9            13
...          ...           ...            ...           ...
36270         85          2018              8             3
36271        228          2018             10            17
36272        148          2018              7             1
36273         63          2018              4            21
36274        207          2018             12            30

      market_segment_type  repeated_guest  no_of_previous_cancellations  \
0                 Offline               0                             0
1                  Online               0                             0
3                  Online               0                             0
4                  Online               0                             0
5                  Online               0                             0
```

```
...                   ...             ...                           ...
36270              Online              0                             0
36271              Online              0                             0
36272              Online              0                             0
36273              Online              0                             0
36274             Offline              0                             0

       no_of_previous_bookings_not_canceled  avg_price_per_room  \
0                                        0            65.00000
1                                        0           106.68000
3                                        0           100.00000
4                                        0            94.50000
5                                        0           115.00000
...                                    ...                 ...
36270                                    0           167.80000
36271                                    0            90.95000
36272                                    0            98.39000
36273                                    0            94.50000
36274                                    0           161.67000

       no_of_special_requests  booking_status  no_of_family_members
0                           0               0                     2
1                           1               0                     2
3                           0               1                     2
4                           0               1                     2
5                           1               1                     2
...                       ...             ...                   ...
36270                       1               0                     3
36271                       2               1                     2
36272                       2               0                     2
36273                       0               1                     2
36274                       0               0                     2

[28441 rows x 19 columns]

booking_status            0     1    All
no_of_family_members
All                   18456  9985  28441
2                     15506  8213  23719
3                      2425  1368   3793
4                       514   398    912
5                        10     5     15
11                        0     1      1
12                        1     0      1
--------------------------------------------------------------------------------
-------------------------------------------
```

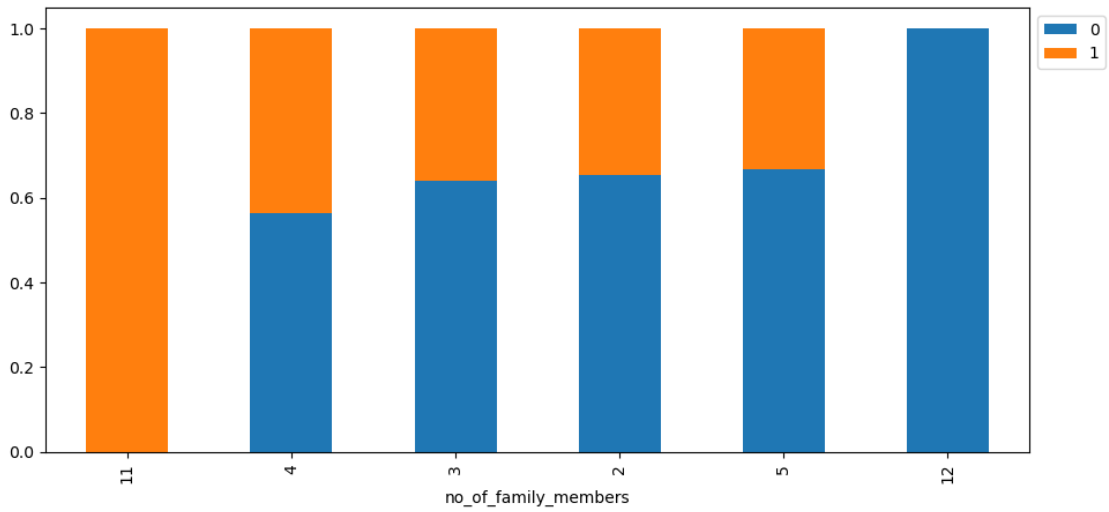


Figure 23 no. of family members

Let's look at the booking status of the customers who has reseved it for atleast a day

```
       no_of_adults  no_of_children  no_of_weekend_nights  no_of_week_nights  \
0                 2               0                     1                  2
1                 2               0                     2                  3
2                 1               0                     2                  1
4                 2               0                     1                  1
6                 2               0                     1                  3
...             ...             ...                   ...                ...
36266             2               0                     2                  2
36270             3               0                     2                  6
36271             2               0                     1                  3
36272             2               0                     2                  6
36274             2               0                     1                  2

      type_of_meal_plan  required_car_parking_space room_type_reserved  \
0           Meal Plan 1                           0        Room_Type 1
1          Not Selected                           0        Room_Type 1
2           Meal Plan 1                           0        Room_Type 1
4          Not Selected                           0        Room_Type 1
6           Meal Plan 1                           0        Room_Type 1
...                 ...                         ...                ...
36266       Meal Plan 1                           0        Room_Type 2
36270       Meal Plan 1                           0        Room_Type 4
36271       Meal Plan 1                           0        Room_Type 1
36272       Meal Plan 1                           0        Room_Type 1
36274       Meal Plan 1                           0        Room_Type 1

       lead_time  arrival_year  arrival_month  arrival_date  \
0            224          2017             10             2
1              5          2018             11             6
2              1          2018              2            28
4             48          2018              4            11
6             34          2017             10            15
```

```
...          ...           ...             ...            ...
36266          8          2018              3              4
36270         85          2018              8              3
36271        228          2018             10             17
36272        148          2018              7              1
36274        207          2018             12             30

      market_segment_type  repeated_guest  no_of_previous_cancellations  \
0                 Offline               0                             0
1                  Online               0                             0
2                  Online               0                             0
4                  Online               0                             0
6                  Online               0                             0
...                   ...             ...                           ...
36266              Online               0                             0
36270              Online               0                             0
36271              Online               0                             0
36272              Online               0                             0
36274             Offline               0                             0

       no_of_previous_bookings_not_canceled  avg_price_per_room  \
0                                         0            65.00000
1                                         0           106.68000
2                                         0            60.00000
4                                         0            94.50000
6                                         0           107.55000
...                                     ...                 ...
36266                                     0            85.96000
36270                                     0           167.80000
36271                                     0            90.95000
36272                                     0            98.39000
36274                                     0           161.67000

       no_of_special_requests  booking_status  total_days
0                           0               0           3
1                           1               0           5
2                           0               1           3
4                           0               1           2
6                           1               0           4
...                       ...             ...         ...
36266                       1               1           4
36270                       1               0           8
36271                       2               1           4
36272                       2               0           8
36274                       0               0           3

[17094 rows x 19 columns]
```

```
#Lets plot the total days with the booking status
```

booking_status	0	1	All
total_days			
All	10979	6115	17094
3	3689	2183	5872
4	2977	1387	4364
5	1593	738	2331
2	1301	639	1940
6	566	465	1031
7	590	383	973
8	100	79	179
10	51	58	109
9	58	53	111
14	5	27	32
15	5	26	31
13	3	15	18
12	9	15	24
11	24	15	39
20	3	8	11
19	1	5	6
16	1	5	6
17	1	4	5
18	0	3	3
21	1	3	4
22	0	2	2
23	1	1	2
24	0	1	1

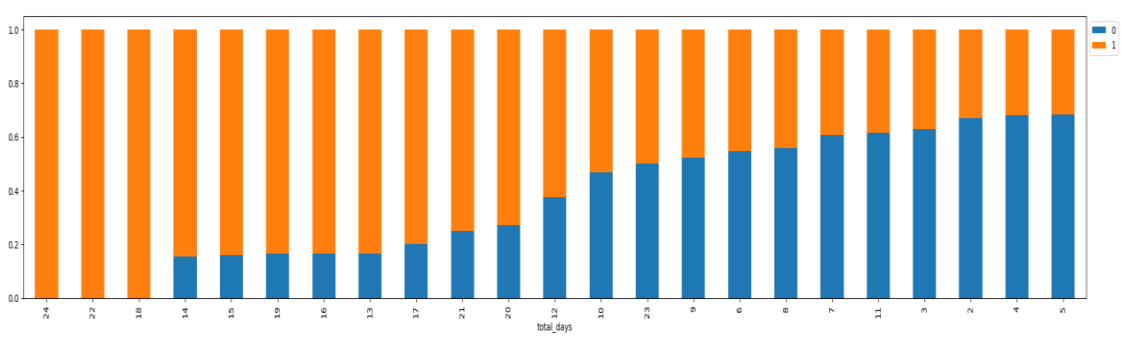


Figure 24 total days with booking status

```
#Let's plot the graph of repeated guest with the booking status
```

```
booking_status      0      1    All
repeated_guest
All             24390  11885  36275
0               23476  11869  35345
1                 914     16    930
--------------------------------------------------------------------------------
-----------------------------------------
```

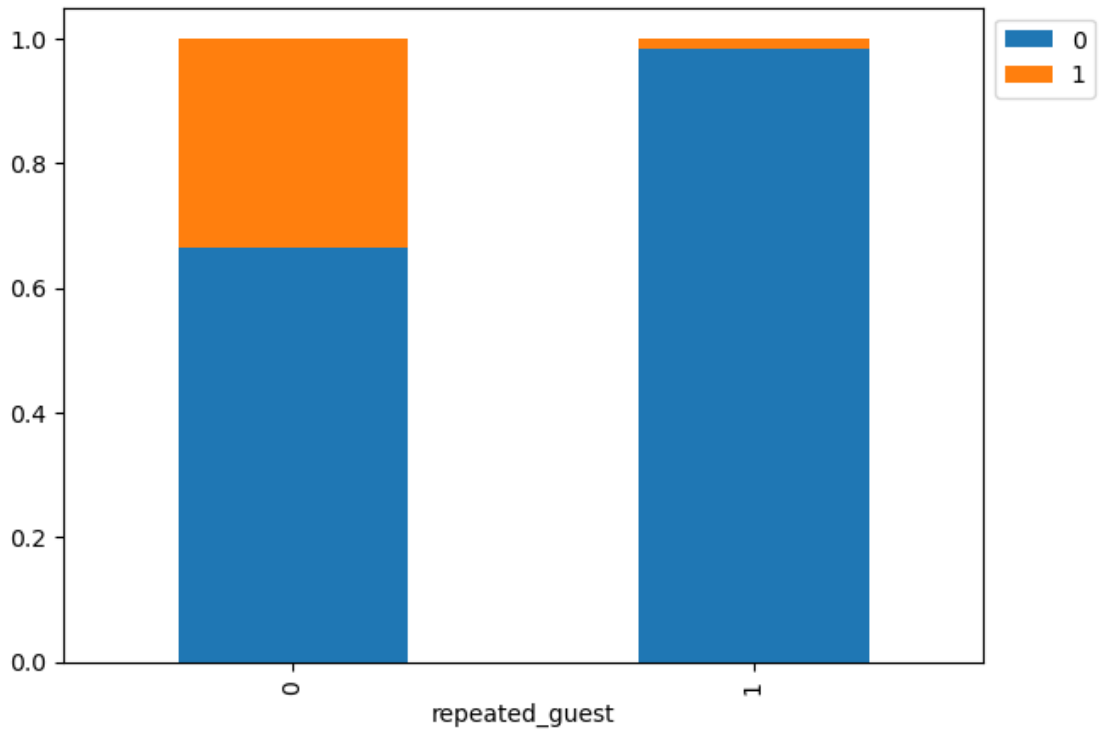


Figure 25 repeated guest with the booking status

```
#Plotting the graph to see the busiest month at the motel
```

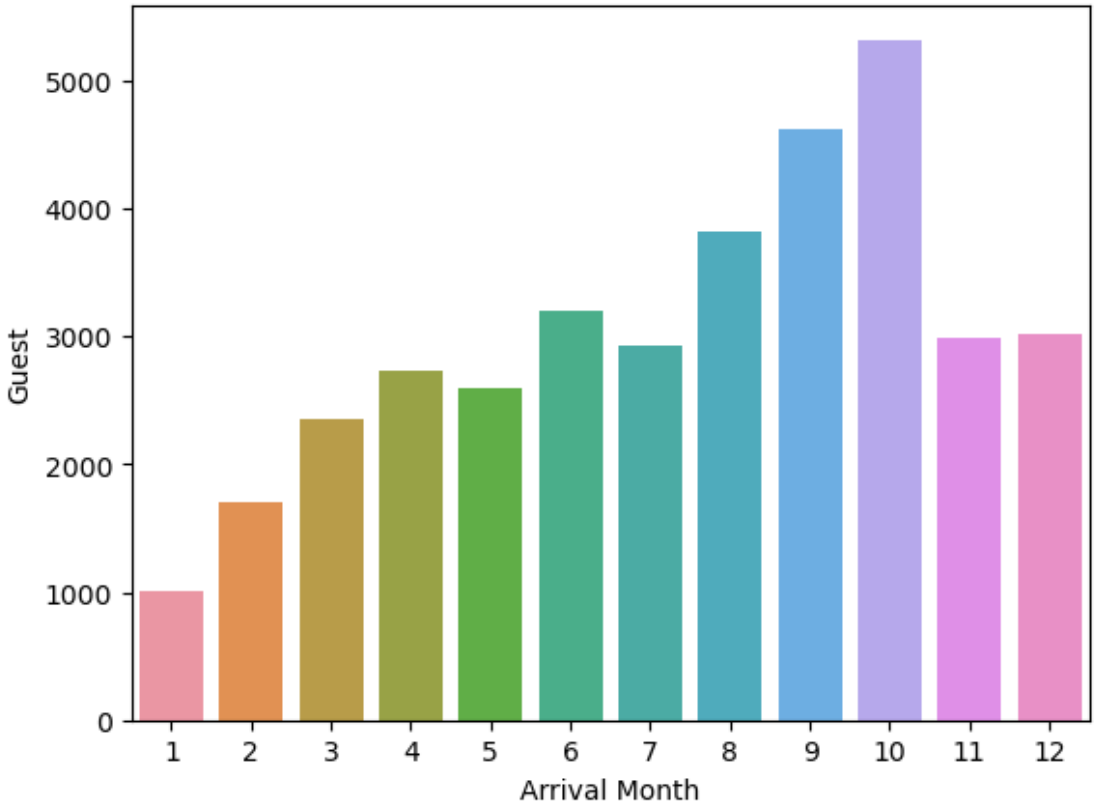


Figure 26 arrival month and guest

Looking at the booking cancelled in each month

```
booking_status      0      1    All
arrival_month
All             24390  11885  36275
10               3437   1880   5317
9                3073   1538   4611
8                2325   1488   3813
7                1606   1314   2920
6                1912   1291   3203
4                1741    995   2736
5                1650    948   2598
11               2105    875   2980
3                1658    700   2358
2                1274    430   1704
12               2619    402   3021
1                 990     24   1014
--------------------------------------------------------------------------------
-----------------------------------------
```

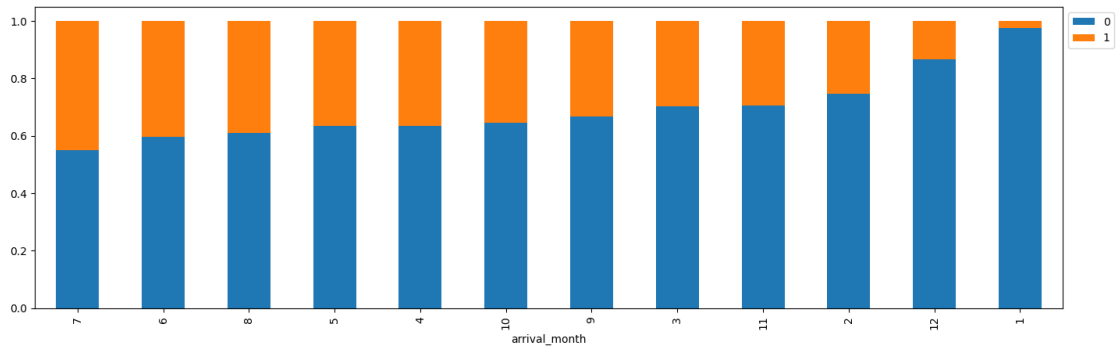


Figure 27 arrival month

```
#Average Prices per arrival month
```

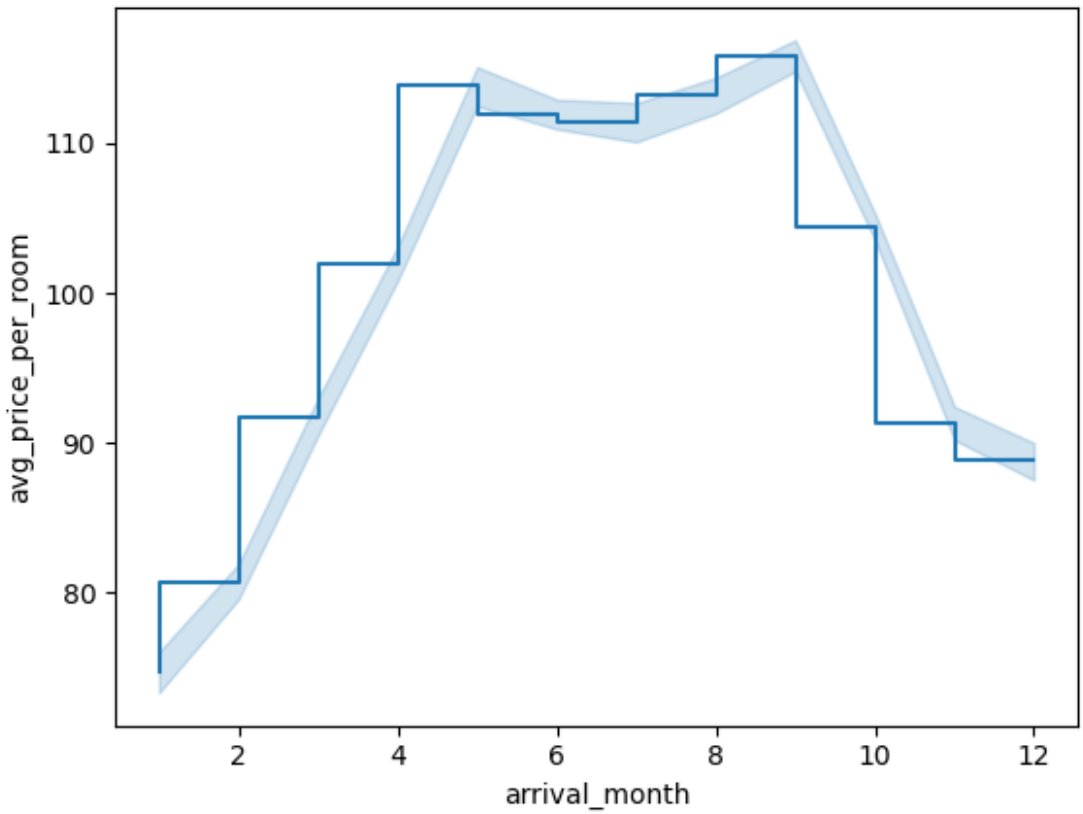


Figure 28 arrival month and avg price per room

Data Preprocessing

As we looked above there is no null values but there are outliers in the data

```
#lets plot the bar plot of the numeric data types
```

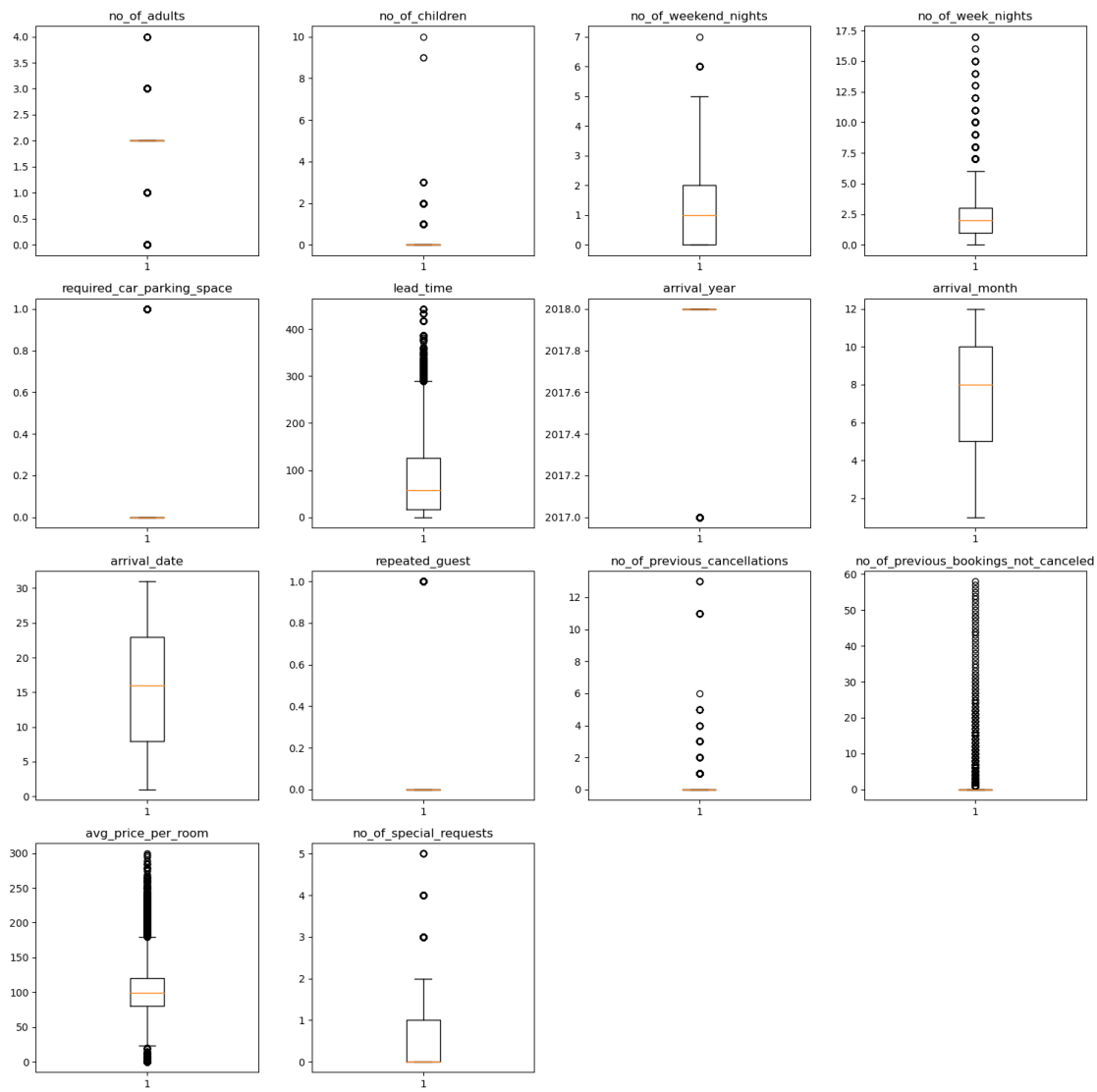


Figure 29 numeric data types plot

- Number of adults and avg price per room have lower and upper outliers
- All the other columns have upper outliers

```
#looking at the average price per room and why it is zero for some values
```

```
Complementary     354
Online            191
Name: market_segment_type, dtype: int64
```

Out of total booking of 545 with price of zero euros 354 were complementary and 191 was online which means the prices are legit

```
## substituted the outlier value with upper whisker
U_Whisker - 179.55
```

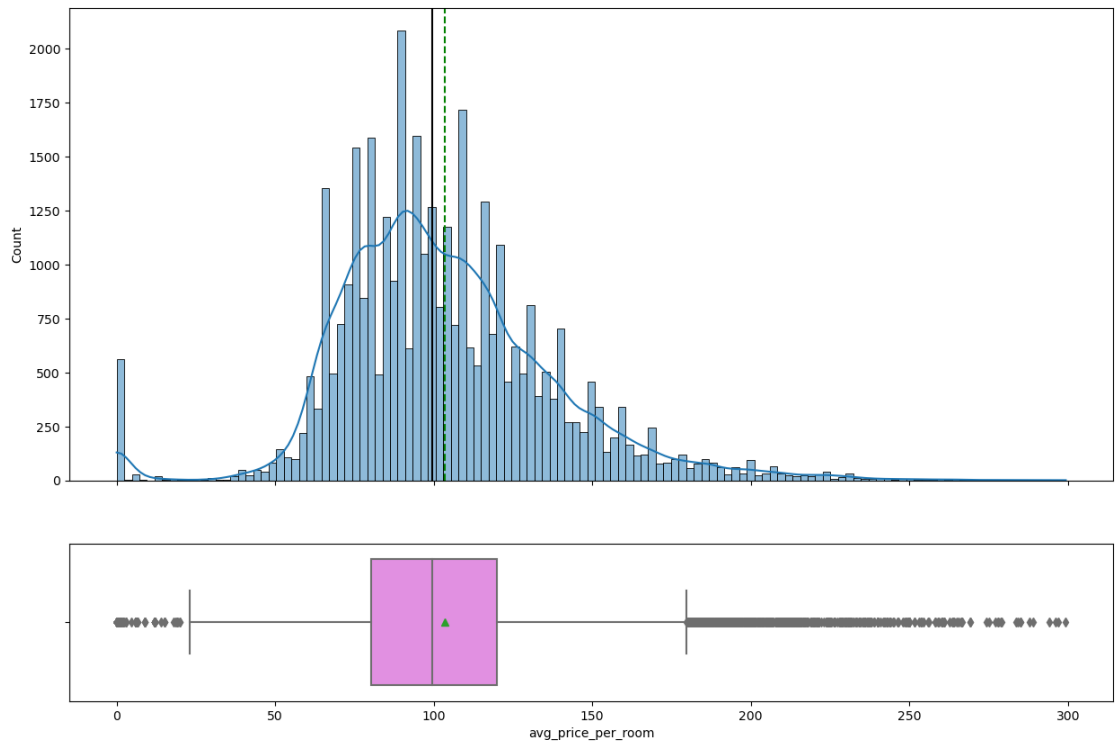


Figure 30 outlier substitution

EDA

- It is a good idea to explore the data once again after manipulating it.

```
data.describe().T
```

	count	mean	std
no_of_adults	36275.00000	1.84496	0.51871
no_of_children	36275.00000	0.10528	0.40265
no_of_weekend_nights	36275.00000	0.81072	0.87064
no_of_week_nights	36275.00000	2.20430	1.41090
required_car_parking_space	36275.00000	0.03099	0.17328
lead_time	36275.00000	85.23256	85.93082
arrival_year	36275.00000	2017.82043	0.38384
arrival_month	36275.00000	7.42365	3.06989
arrival_date	36275.00000	15.59700	8.74045
repeated_guest	36275.00000	0.02564	0.15805
no_of_previous_cancellations	36275.00000	0.02335	0.36833
no_of_previous_bookings_not_canceled	36275.00000	0.15341	1.75417
avg_price_per_room	36275.00000	103.36314	34.80147
no_of_special_requests	36275.00000	0.61966	0.78624
booking_status	36275.00000	0.32764	0.46936

```
                                          min        25%        50%  \
no_of_adults                          0.00000    2.00000    2.00000
no_of_children                        0.00000    0.00000    0.00000
no_of_weekend_nights                  0.00000    0.00000    1.00000
no_of_week_nights                     0.00000    1.00000    2.00000
required_car_parking_space            0.00000    0.00000    0.00000
lead_time                             0.00000   17.00000   57.00000
arrival_year                       2017.00000 2018.00000 2018.00000
arrival_month                         1.00000    5.00000    8.00000
arrival_date                          1.00000    8.00000   16.00000
repeated_guest                        0.00000    0.00000    0.00000
no_of_previous_cancellations          0.00000    0.00000    0.00000
no_of_previous_bookings_not_canceled  0.00000    0.00000    0.00000
avg_price_per_room                    0.00000   80.30000   99.45000
no_of_special_requests                0.00000    0.00000    0.00000
booking_status                        0.00000    0.00000    0.00000

                                          75%        max
no_of_adults                          2.00000    4.00000
no_of_children                        0.00000   10.00000
no_of_weekend_nights                  2.00000    7.00000
no_of_week_nights                     3.00000   17.00000
required_car_parking_space            0.00000    1.00000
lead_time                           126.00000  443.00000
arrival_year                       2018.00000 2018.00000
arrival_month                        10.00000   12.00000
arrival_date                         23.00000   31.00000
repeated_guest                        0.00000    1.00000
no_of_previous_cancellations          0.00000   13.00000
no_of_previous_bookings_not_canceled  0.00000   58.00000
avg_price_per_room                  120.00000  299.33000
no_of_special_requests                1.00000    5.00000
booking_status                          1.00000    1.00000
```

```
#lets look at the value counts for the numeric values
```

```
value count for no_of_adults
2    26108
1     7695
3     2317
0      139
4       16
Name: no_of_adults, dtype: int64
To verify total counts 36275
************************************************
value count for no_of_children
0     33577
1      1618
2      1058
3        19
9         2
10        1
Name: no_of_children, dtype: int64
To verify total counts 36275
************************************************
```

```
value count for no_of_weekend_nights
0    16872
1     9995
2     9071
3      153
4      129
5       34
6       20
7        1
Name: no_of_weekend_nights, dtype: int64
To verify total counts 36275
**************************************************
value count for no_of_week_nights
2     11444
1      9488
3      7839
4      2990
0      2387
5      1614
6       189
7       113
10       62
8        62
9        34
11       17
15       10
12        9
14        7
13        5
17        3
16        2
Name: no_of_week_nights, dtype: int64
To verify total counts 36275
**************************************************
value count for required_car_parking_space
0    35151
1     1124
Name: required_car_parking_space, dtype: int64
To verify total counts 36275
**************************************************
value count for lead_time
0      1297
1      1078
2       643
3       630
4       628
       ...
300       1
353       1
328       1
352       1
351       1
Name: lead_time, Length: 352, dtype: int64
To verify total counts 36275
**************************************************
```

```
value count for arrival_year
2018    29761
2017     6514
Name: arrival_year, dtype: int64
To verify total counts 36275
**************************************************
value count for arrival_month
10    5317
9     4611
8     3813
6     3203
12    3021
11    2980
7     2920
4     2736
5     2598
3     2358
2     1704
1     1014
Name: arrival_month, dtype: int64
To verify total counts 36275
**************************************************
value count for arrival_date
13    1358
17    1345
2     1331
4     1327
19    1327
16    1306
20    1281
15    1273
6     1273
18    1260
14    1242
30    1216
12    1204
8     1198
29    1190
21    1158
5     1154
26    1146
25    1146
1     1133
9     1130
28    1129
7     1110
24    1103
11    1098
3     1098
10    1089
27    1059
22    1023
23     990
31     578
Name: arrival_date, dtype: int64
```

```
To verify total counts 36275
**************************************************
value count for repeated_guest
0    35345
1      930
Name: repeated_guest, dtype: int64
To verify total counts 36275
**************************************************
value count for no_of_previous_cancellations
0     35937
1       198
2        46
3        43
11       25
5        11
4        10
13        4
6         1
Name: no_of_previous_cancellations, dtype: int64
To verify total counts 36275
**************************************************
value count for no_of_previous_bookings_not_canceled
0     35463
1       228
2       112
3        80
4        65
5        60
6        36
7        24
8        23
10       19
9        19
11       15
12       12
14        9
15        8
16        7
13        7
18        6
20        6
21        6
17        6
19        6
22        6
25        3
27        3
24        3
23        3
44        2
29        2
48        2
28        2
30        2
32        2
```

```
31        2
26        2
46        1
55        1
45        1
57        1
53        1
54        1
58        1
41        1
40        1
43        1
35        1
50        1
56        1
33        1
37        1
42        1
51        1
38        1
34        1
39        1
52        1
49        1
47        1
36        1
Name: no_of_previous_bookings_not_canceled, dtype: int64
To verify total counts 36275
**************************************************
value count for no_of_special_requests
0    19777
1    11373
2     4364
3      675
4       78
5        8
Name: no_of_special_requests, dtype: int64
To verify total counts 36275
**************************************************
value count for booking_status
0    24390
1    11885
Name: booking_status, dtype: int64
To verify total counts 36275
**************************************************
```

Based on the unique values of the columns the outliers seem reasonable as no unusual valued found

#let's Change the booking status data type to interger by one hot encoding

```
   no_of_adults  no_of_children  no_of_weekend_nights  no_of_week_nights  \
0             2               0                     1                  2
1             2               0                     2                  3
2             1               0                     2                  1
3             2               0                     0                  2
4             2               0                     1                  1

  type_of_meal_plan  required_car_parking_space room_type_reserved  lead_time  \
0       Meal Plan 1                           0        Room_Type 1        224
1      Not Selected                           0        Room_Type 1          5
2       Meal Plan 1                           0        Room_Type 1          1
3       Meal Plan 1                           0        Room_Type 1        211
4      Not Selected                           0        Room_Type 1         48

   arrival_year  arrival_month  arrival_date market_segment_type  \
0          2017             10             2             Offline
1          2018             11             6              Online
2          2018              2            28              Online
3          2018              5            20              Online
4          2018              4            11              Online

   repeated_guest  no_of_previous_cancellations  \
0               0                             0
1               0                             0
2               0                             0
3               0                             0
4               0                             0

   no_of_previous_bookings_not_canceled  avg_price_per_room  \
0                                     0            65.00000
1                                     0           106.68000
2                                     0            60.00000
3                                     0           100.00000
4                                     0            94.50000

   no_of_special_requests  booking_status
0                       0               0
1                       1               0
2                       0               1
3                       0               1
4                       0               1
```

Model Building

Logistic Regression Model

Data Preparation

- Predicting which bookings will be canceled.
- encoding categorical features.
- Splitting the data into train and test

```
        const  no_of_adults  no_of_children  no_of_weekend_nights  \
0     1.00000             2               0                     1
1     1.00000             2               0                     2
2     1.00000             1               0                     2
3     1.00000             2               0                     0
4     1.00000             2               0                     1
...       ...           ...             ...                   ...
36270 1.00000             3               0                     2
36271 1.00000             2               0                     1
36272 1.00000             2               0                     2
36273 1.00000             2               0                     0
36274 1.00000             2               0                     1

       no_of_week_nights  required_car_parking_space  lead_time  arrival_year  \
0                      2                           0        224          2017
1                      3                           0          5          2018
2                      1                           0          1          2018
3                      2                           0        211          2018
4                      1                           0         48          2018
...                  ...                         ...        ...           ...
36270                  6                           0         85          2018
36271                  3                           0        228          2018
36272                  6                           0        148          2018
36273                  3                           0         63          2018
36274                  2                           0        207          2018

       arrival_month  arrival_date  repeated_guest  \
0                 10             2               0
1                 11             6               0
2                  2            28               0
3                  5            20               0
4                  4            11               0
...              ...           ...             ...
36270              8             3               0
36271             10            17               0
36272              7             1               0
36273              4            21               0
36274             12            30               0

       no_of_previous_cancellations  no_of_previous_bookings_not_canceled  \
0                                 0                                     0
1                                 0                                     0
2                                 0                                     0
3                                 0                                     0
4                                 0                                     0
```

```
...                              ...                                   ...
36270                              0                                     0
36271                              0                                     0
36272                              0                                     0
36273                              0                                     0
36274                              0                                     0

       avg_price_per_room  no_of_special_requests  \
0               65.00000                       0
1              106.68000                       1
2               60.00000                       0
3              100.00000                       0
4               94.50000                       0
...                  ...                     ...
36270          167.80000                       1
36271           90.95000                       2
36272           98.39000                       2
36273           94.50000                       0
36274          161.67000                       0

       type_of_meal_plan_Meal Plan 2  type_of_meal_plan_Meal Plan 3  \
0                                  0                               0
1                                  0                               0
2                                  0                               0
3                                  0                               0
4                                  0                               0
...                              ...                             ...
36270                              0                               0
36271                              0                               0
36272                              0                               0
36273                              0                               0
36274                              0                               0

       type_of_meal_plan_Not Selected  room_type_reserved_Room_Type 2  \
0                                   0                                 0
1                                   1                                 0
2                                   0                                 0
3                                   0                                 0
4                                   1                                 0
...                               ...                               ...
36270                               0                                 0
36271                               0                                 0
36272                               0                                 0
36273                               1                                 0
36274                               0                                 0

       room_type_reserved_Room_Type 3  room_type_reserved_Room_Type 4  \
0                                   0                                 0
1                                   0                                 0
2                                   0                                 0
3                                   0                                 0
4                                   0                                 0
...                               ...                               ...
36270                               0                                 1
36271                               0                                 0
```

```
36272                                0                                0
36273                                0                                0
36274                                0                                0

       room_type_reserved_Room_Type 5  room_type_reserved_Room_Type 6  \
0                                    0                                0
1                                    0                                0
2                                    0                                0
3                                    0                                0
4                                    0                                0
...                                ...                              ...
36270                                0                                0
36271                                0                                0
36272                                0                                0
36273                                0                                0
36274                                0                                0

       room_type_reserved_Room_Type 7  market_segment_type_Complementary  \
0                                    0                                   0
1                                    0                                   0
2                                    0                                   0
3                                    0                                   0
4                                    0                                   0
...                                ...                                 ...
36270                                0                                   0
36271                                0                                   0
36272                                0                                   0
36273                                0                                   0
36274                                0                                   0

       market_segment_type_Corporate  market_segment_type_Offline  \
0                                  0                            1
1                                  0                            0
2                                  0                            0
3                                  0                            0
4                                  0                            0
...                              ...                          ...
36270                              0                            0
36271                              0                            0
36272                              0                            0
36273                              0                            0
36274                              0                            1

       market_segment_type_Online
0                               0
1                               1
2                               1
3                               1
4                               1
...                           ...
36270                           1
36271                           1
36272                           1
36273                           1
36274                           0
```

```
[36275 rows x 28 columns]

Shape of Training set :  (25392, 28)
Shape of test set :  (10883, 28)
Percentage of classes in training set:
0   0.67064
1   0.32936
Name: booking_status, dtype: float64
Percentage of classes in test set:
0   0.67638
1   0.32362
Name: booking_status, dtype: float64
```

- As shown above the data is split into 70% and 30% as Train and Test data respectively

```
                           Logit Regression Results
==============================================================================
=
Dep. Variable:         booking_status   No. Observations:
25392
Model:                          Logit   Df Residuals:
25364
Method:                           MLE   Df Model:
27
Date:                Thu, 24 Aug 2023   Pseudo R-squ.:
0.3297
Time:                        19:33:36   Log-Likelihood:                -
10786.
converged:                       False   LL-Null:                       -
16091.
Covariance Type:            nonrobust   LLR p-value:
0.000
==============================================================================
============================
                                            coef    std err          z
P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
---------------------------
const                                   -919.5146    120.841     -7.609
0.000   -1156.358    -682.671
no_of_adults                              0.1118      0.038      2.970
0.003       0.038       0.186
no_of_children                            0.1549      0.057      2.704
0.007       0.043       0.267
no_of_weekend_nights                      0.1072      0.020      5.418
0.000       0.068       0.146
no_of_week_nights                         0.0404      0.012      3.282
0.001       0.016       0.064
required_car_parking_space               -1.5900      0.138    -11.554
```

```
0.000      -1.860      -1.320
lead_time                                0.0157      0.000     58.895
0.000       0.015       0.016
arrival_year                             0.4544      0.060      7.589
0.000       0.337       0.572
arrival_month                           -0.0420      0.006     -6.481
0.000      -0.055      -0.029
arrival_date                             0.0005      0.002      0.281
0.779      -0.003       0.004
repeated_guest                          -2.3434      0.617     -3.799
0.000      -3.552      -1.134
no_of_previous_cancellations             0.2666      0.086      3.110
0.002       0.099       0.435
no_of_previous_bookings_not_canceled    -0.1723      0.153     -1.130
0.259      -0.471       0.127
avg_price_per_room                       0.0191      0.001     25.779
0.000       0.018       0.021
no_of_special_requests                  -1.4709      0.030    -48.824
0.000      -1.530      -1.412
type_of_meal_plan_Meal Plan 2            0.1760      0.067      2.644
0.008       0.046       0.307
type_of_meal_plan_Meal Plan 3           21.6364   3.44e+04      0.001
0.999   -6.74e+04    6.75e+04
type_of_meal_plan_Not Selected           0.2823      0.053      5.319
0.000       0.178       0.386
room_type_reserved_Room_Type 2          -0.3579      0.131     -2.735
0.006      -0.614      -0.101
room_type_reserved_Room_Type 3          -0.0025      1.310     -0.002
0.998      -2.570       2.565
room_type_reserved_Room_Type 4          -0.2887      0.053     -5.427
0.000      -0.393      -0.184
room_type_reserved_Room_Type 5          -0.7292      0.209     -3.487
0.000      -1.139      -0.319
room_type_reserved_Room_Type 6          -0.9518      0.147     -6.480
0.000      -1.240      -0.664
room_type_reserved_Room_Type 7          -1.4233      0.294     -4.845
0.000      -1.999      -0.847
market_segment_type_Complementary      -36.1001   3.52e+04     -0.001
0.999    -6.9e+04    6.89e+04
market_segment_type_Corporate           -1.1944      0.266     -4.489
0.000      -1.716      -0.673
market_segment_type_Offline             -2.1966      0.255     -8.626
0.000      -2.696      -1.698
market_segment_type_Online              -0.4031      0.251     -1.604
0.109      -0.896       0.089
==============================================================================
==========================
```

- Coefficinets in the model indicate the change in dependent variable

- `arrival_date`,`type_of_meal_plan_Meal Plan 3`,`no_of_previous_bookings_not_canceled`,`market_segment_type_Complementary` and `market_segment_type_Online` have p value > 0.05
- Unit increase in no_of_repeated_guest while other variables constant decreases the probability of cancellation by 2.34

Model evaluation criteria

Model can make wrong predictions in two ways:

1. Predicting a guest booking will not cancel, but in reality, booking cancels (False Positive)
2. Predicting a guest will cancel, but in reality, booking does not cancel (False Negative)

Determine Importance:

Both cases are important:

- If we predict that a booking will not cancel and booking gets canceled, hotel loses revenue and has to bear additional costs of distribution channels, lowering room price

- If we predict that a booking will cancel and booking isn't canceled, hotel might be unable to provide satisfactory services to guests by not being adequately staffed or prepared to host. This might damage brand equity/guest satisfaction and lead to opportunity loss

How to reduce the losses?

- Since both scenarios are to be addressed equally, hotel would want F1 Score to be maximized
- Greater the F1 score, higher the chances of minimizing both False Negatives and False Positives
- F1_score = (2 * P * R)/(P + R)

Let's create functions to calculate the performance metrics and confsion matrix

- model_performance_classification_statsmodels function (To check performance of models)
- confusion_matrix_statsmodels function (To plot confusion matrix)

```
#Function to plot the confusion matrix of the classification model
```

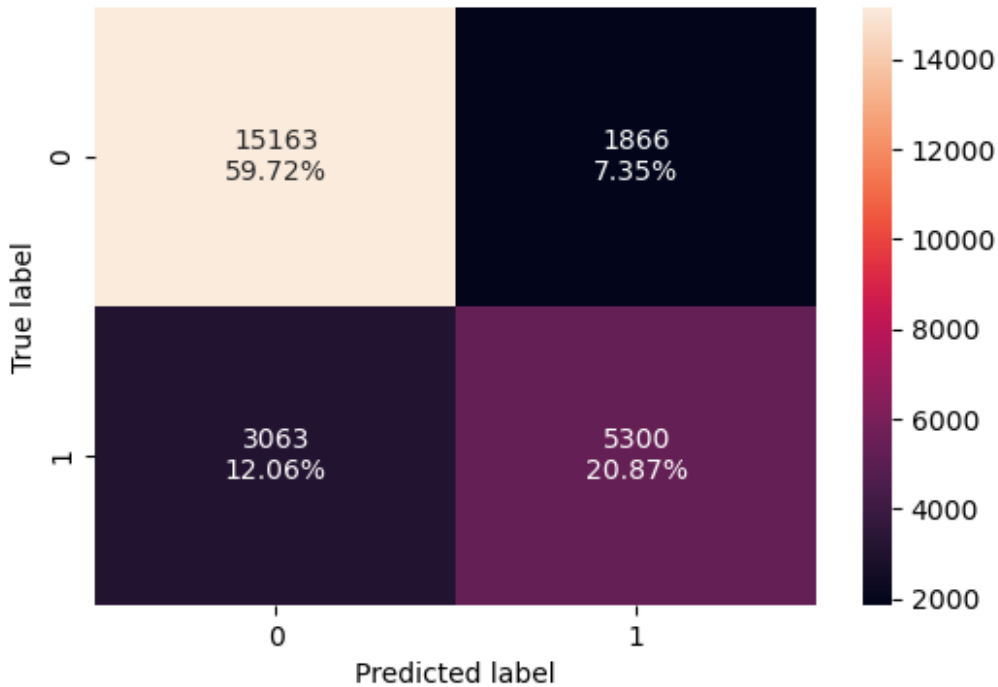


Figure 31 initial confusion matrix

```
Performance of the classification model

   Accuracy  Recall  Precision      F1
0   0.80588 0.63374    0.73960 0.68259
```

- As shown in the performance of the classification model F1 score is 0.68 we need to maximize it
- The model needs to be checked for variable multicollinearity and needs to be removed to obtain reliable p values

Multicollinearity Check

- To check for Multicollinearity we need to use VIF
- IF VIF is around 1, there is little or No Correlation between independent variables
- IF VIF > 5 There is moderate collinearity
- IF VIF => 10 there is high_multi Collinearity
- Let's create a function to calculate the VIF

```
#calculating VIF
```

```
                                feature            VIF
0                                 const 39484236.78442
1                          no_of_adults        1.34886
2                        no_of_children        1.97845
3                  no_of_weekend_nights        1.06960
4                     no_of_week_nights        1.09580
5             required_car_parking_space        1.03994
6                              lead_time        1.39497
7                          arrival_year        1.43142
8                         arrival_month        1.27615
9                          arrival_date        1.00673
10                       repeated_guest        1.78372
11          no_of_previous_cancellations        1.39569
12  no_of_previous_bookings_not_canceled        1.65200
13                    avg_price_per_room        2.07087
14                no_of_special_requests        1.24739
15         type_of_meal_plan_Meal Plan 2        1.27154
16         type_of_meal_plan_Meal Plan 3        1.02521
17        type_of_meal_plan_Not Selected        1.27300
18        room_type_reserved_Room_Type 2        1.10158
19        room_type_reserved_Room_Type 3        1.00330
20        room_type_reserved_Room_Type 4        1.36352
21        room_type_reserved_Room_Type 5        1.02820
22        room_type_reserved_Room_Type 6        1.97507
23        room_type_reserved_Room_Type 7        1.11515
24     market_segment_type_Complementary        4.50521
25         market_segment_type_Corporate       16.92864
26           market_segment_type_Offline       64.11476
27            market_segment_type_Online       71.17687
```

- Only some dummy variables show multicollinearity which can be ignored

Dropping the variables with high p-values

- We have to drop the variables with high p-values one by one and repeat the steps until we get the variables with less than 0.05 p-values
- Creating a function to drop the variable with high p_values one by one

```
                           Logit Regression Results
==========================================================================
=
Dep. Variable:         booking_status   No. Observations:
25392
Model:                          Logit   Df Residuals:
25370
Method:                           MLE   Df Model:
21
Date:                Thu, 24 Aug 2023   Pseudo R-squ.:
```

```
0.3287
Time:                        19:33:38   Log-Likelihood:                -
10802.
converged:                       True   LL-Null:                       -
16091.
Covariance Type:            nonrobust   LLR p-value:
0.000
==============================================================================
=====================
                                   coef    std err          z      P>|z|
[0.025      0.975]
------------------------------------------------------------------------------
---------------------
const                           -912.5938    120.484     -7.574      0.000
-1148.738    -676.449
no_of_adults                      0.1069      0.037      2.865      0.004
0.034       0.180
no_of_children                    0.1508      0.057      2.634      0.008
0.039       0.263
no_of_weekend_nights              0.1090      0.020      5.519      0.000
0.070       0.148
no_of_week_nights                 0.0423      0.012      3.445      0.001
0.018       0.066
required_car_parking_space       -1.5902      0.138    -11.553      0.000
-1.860      -1.320
lead_time                         0.0157      0.000     59.240      0.000
0.015       0.016
arrival_year                      0.4508      0.060      7.550      0.000
0.334       0.568
arrival_month                    -0.0428      0.006     -6.629      0.000
-0.055      -0.030
repeated_guest                   -2.7309      0.557     -4.901      0.000
-3.823      -1.639
no_of_previous_cancellations      0.2289      0.077      2.983      0.003
0.079       0.379
avg_price_per_room                0.0195      0.001     26.705      0.000
0.018       0.021
no_of_special_requests           -1.4719      0.030    -48.927      0.000
-1.531      -1.413
type_of_meal_plan_Meal Plan 2     0.1650      0.066      2.482      0.013
0.035       0.295
type_of_meal_plan_Not Selected    0.2897      0.053      5.475      0.000
0.186       0.393
room_type_reserved_Room_Type 2   -0.3530      0.131     -2.699      0.007
-0.609      -0.097
room_type_reserved_Room_Type 4   -0.2890      0.053     -5.448      0.000
-0.393      -0.185
room_type_reserved_Room_Type 5   -0.7463      0.208     -3.581      0.000
-1.155      -0.338
room_type_reserved_Room_Type 6   -0.9704      0.147     -6.613      0.000
```

```
-1.258      -0.683
room_type_reserved_Room_Type 7    -1.4568      0.293     -4.965      0.000
-2.032      -0.882
market_segment_type_Corporate     -0.7899      0.103     -7.678      0.000
-0.992      -0.588
market_segment_type_Offline       -1.7841      0.052    -34.328      0.000
-1.886      -1.682
==============================================================================
=====================
```

- We have treated for Multicollinearity, the model lg1 has dropped 6 variables with p-value>0.05
- All the categorical variables in the model lg1 had less the 0.05 p-values so lg1 is final model

Let's look at the model performance

```
#lg1 Model Performance
```

```
   Accuracy  Recall  Precision      F1
0   0.80529 0.63255    0.73872 0.68153
```

```
#Let's try the model on the test data
```

```
   Accuracy  Recall  Precision      F1
0   0.80410 0.63061    0.72772 0.67569
```

As we can see the model lg1 gives comparable values for train and test data, with F1-score of 0.68 we will try to improve it

We will convert the coefficient to odd (probability max = 1)

- The Coefficients of logistic regression model are in terms of log(odds), to find the odds we have to take exponential of the coefficients.Therefore, odds = exp(b)
- The Percentage change in odds is given as odds = (exp(b) - 1) * 100

```
                                  Odds  Change_odd%
const                          0.00000   -100.00000
no_of_adults                   1.11280     11.28010
no_of_children                 1.16277     16.27660
no_of_weekend_nights           1.11519     11.51891
no_of_week_nights              1.04320      4.32010
required_car_parking_space     0.20389    -79.61080
lead_time                      1.01586      1.58561
arrival_year                   1.56956     56.95574
arrival_month                  0.95812     -4.18775
repeated_guest                 0.06516    -93.48406
no_of_previous_cancellations   1.25728     25.72753
avg_price_per_room             1.01968      1.96805
no_of_special_requests         0.22949    -77.05093
```

```
type_of_meal_plan_Meal Plan 2  1.17936     17.93642
type_of_meal_plan_Not Selected 1.33602     33.60222
room_type_reserved_Room_Type 2 0.70260    -29.73968
room_type_reserved_Room_Type 4 0.74898    -25.10182
room_type_reserved_Room_Type 5 0.47411    -52.58887
room_type_reserved_Room_Type 6 0.37891    -62.10862
room_type_reserved_Room_Type 7 0.23298    -76.70209
market_segment_type_Corporate  0.45389    -54.61097
market_segment_type_Offline    0.16795    -83.20494
```

Checking model performance on training set

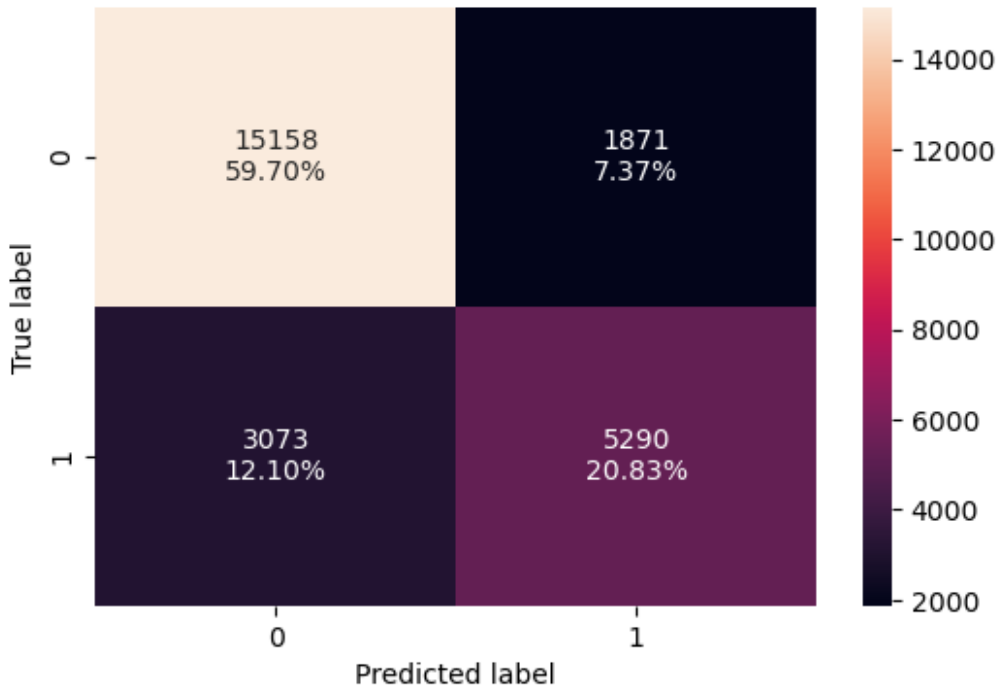


Figure 32 initial training set confusion matrix

```
#Check performance to X_train1 and y_train
```

```
Training performance:

   Accuracy  Recall  Precision      F1
0   0.80529 0.63255    0.73872 0.68153
```

Let's look at ROC-AUC on Training set

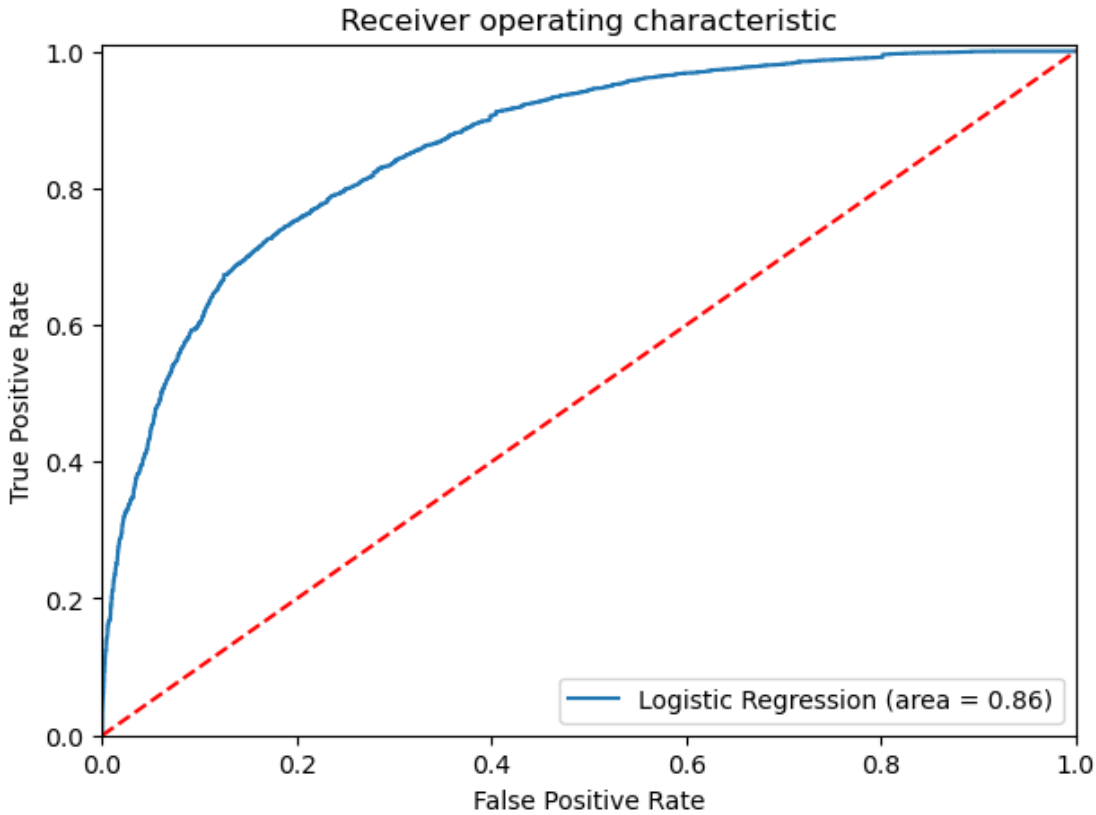


Figure 33 ROC-AUC curve

Model Performance improvement

- Let's see if the recall score can be improved further, by changing the model threshold using AUC-ROC curve.

Optimal threshold using AUC-ROC curve

```
# Optimal threshold as per AUC-ROC curve
```

```
0.36059930727333694
```

```
# The optimal cut off would be where tpr is high and fpr is low
```

```
# Creating confusion matrix
```

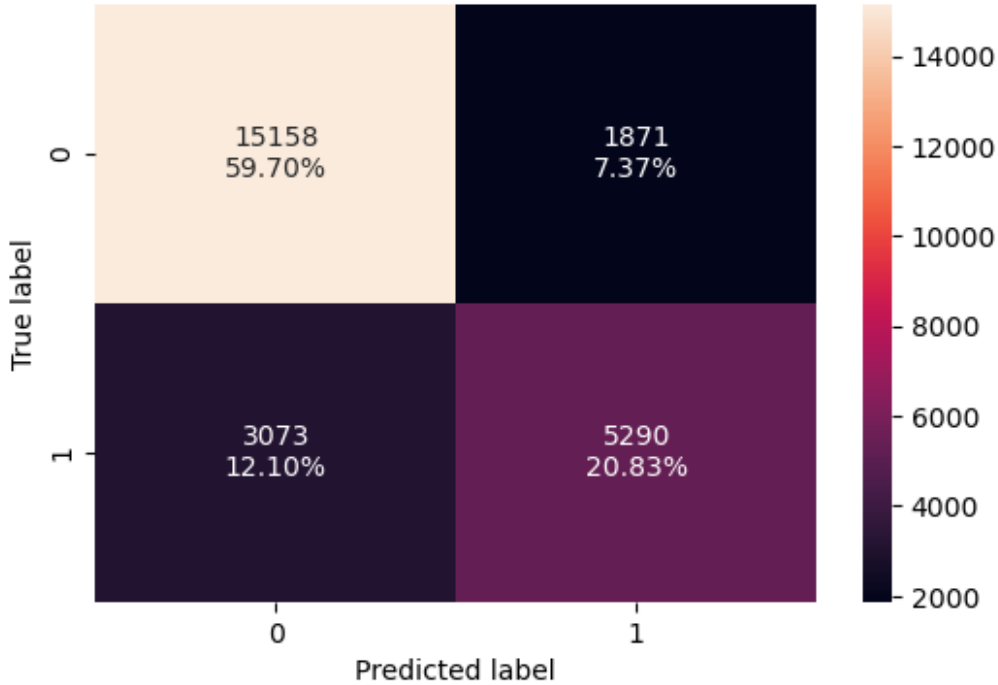


Figure 34 model change 1 confusion matrix

```
# Checking the model performance for this model
```

```
Training Model  performance:

   Accuracy  Recall  Precision      F1
0   0.78989 0.74327    0.66100 0.69972
```

Let's see Precision-Recall curve to see if we can find a better threshold

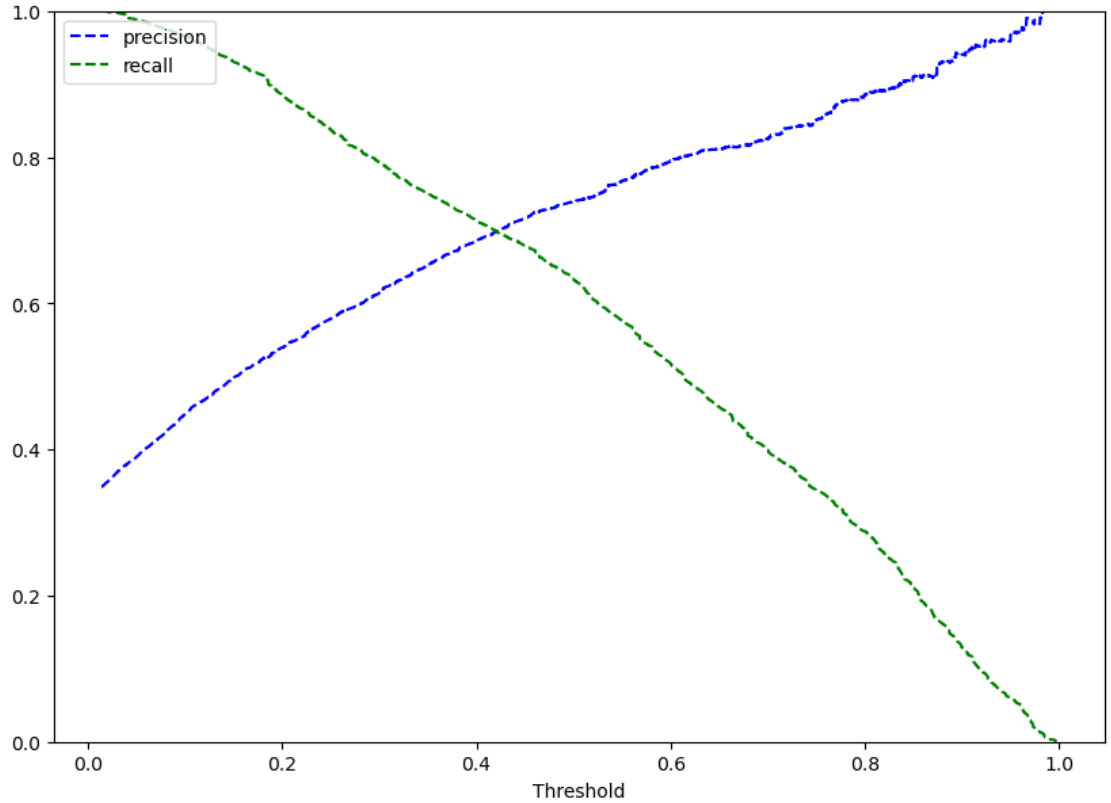


Figure 35 precision-recall curve

```
#setting up the threshold
optimal_threshold_curve = 0.42
```

At the threshold of 0.42 we can get balanced recall and precision

Checking model performance on training set

Now,let's set the threshold to 0.42 and find out the model performance

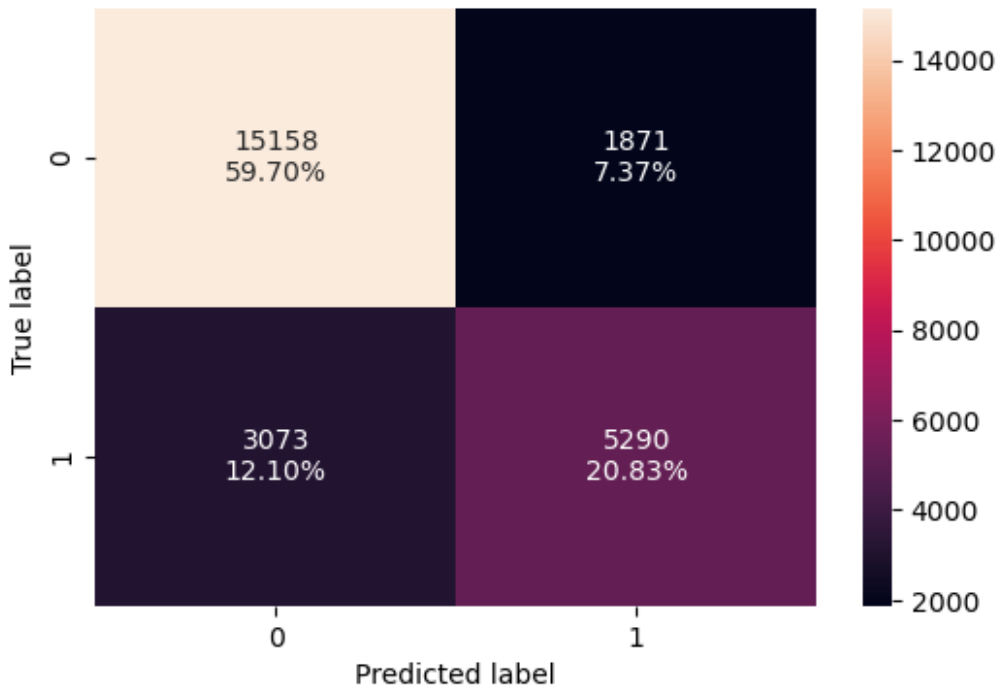


Figure 36 0.42 threshold confusion matrix

```
Training performance:

   Accuracy  Recall  Precision      F1
0   0.80108 0.69903    0.69761 0.69832
```

Recall has reduced to 0.70 and Precision increased to 0.69 but the F1 metrics is still the same

Model is still good

Let's check the model on test data

Using model with default threshold

```
# creating confusion matrix
```

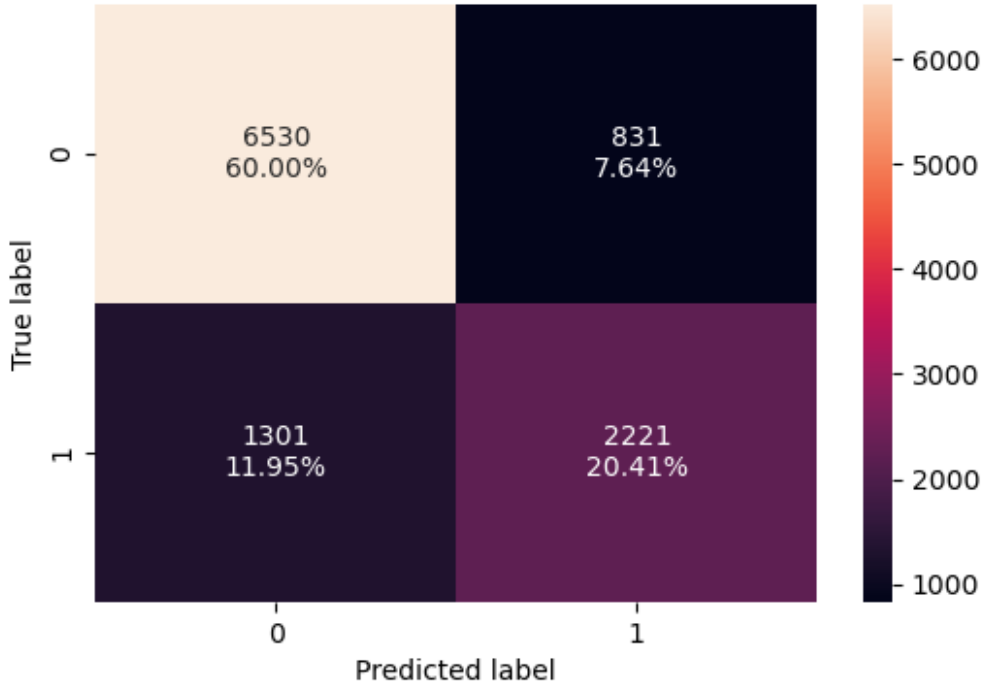


Figure 37 default threshold confusion matrix

```
#performance on X_test1 and y_test
```

```
Model Test performance:

   Accuracy  Recall  Precision      F1
0   0.80410 0.63061    0.72772 0.67569
```

- F1-score-0.67
- Accuracy - 0.80
- Recall - 0.63
- Precision - 0.72
- Model is working well on the test data

ROC curve on the test data

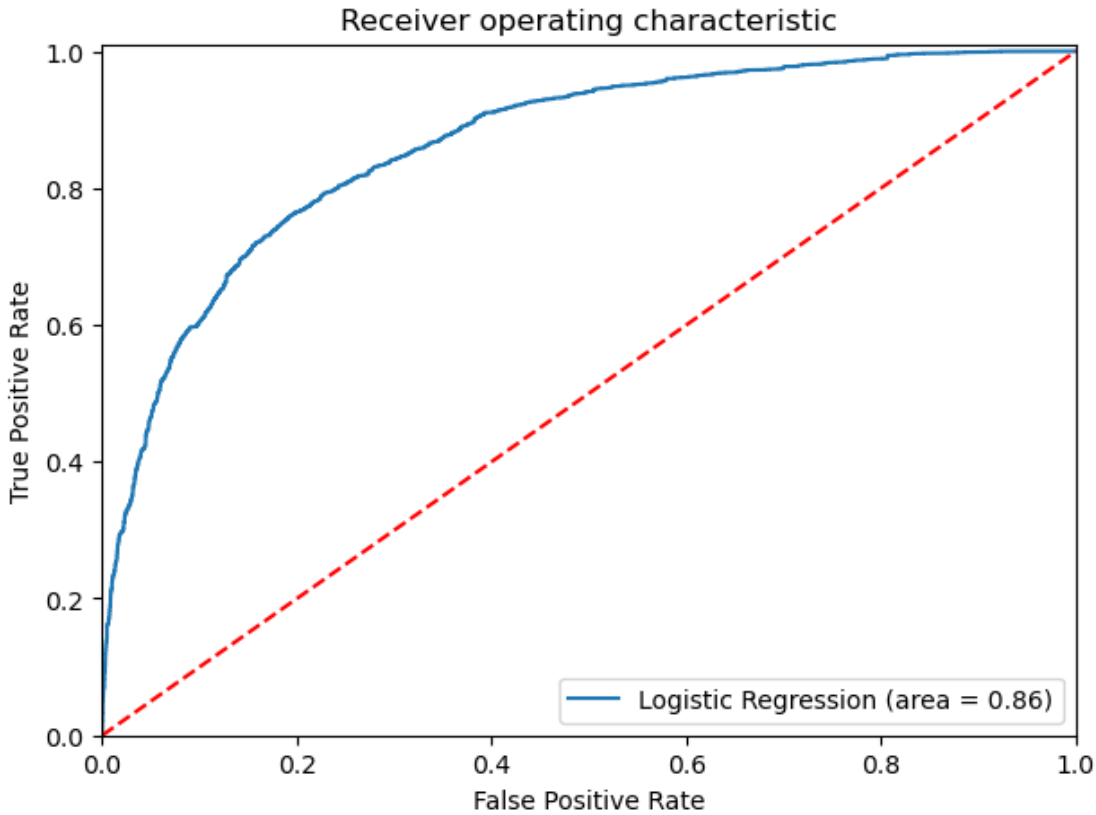


Figure 38 ROC curve test data

Using the Optimal threshold = 0.37 on the model with test data

```
#plotting confusion matrix for optimal threshold 0.37
```

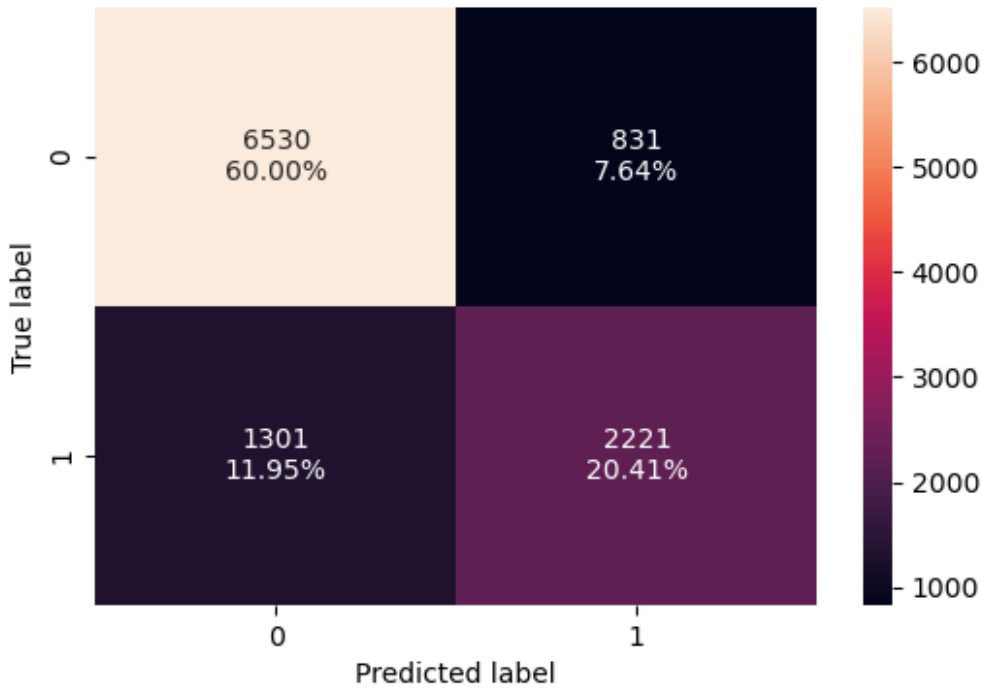


Figure 39 0.37 threshold confusion matrix

```
#checking the model performance
```

```
Test performance:

   Accuracy  Recall  Precision      F1
0   0.79371 0.74730    0.66015 0.70103
```

Using the threshold = 0.42 on the model with test data

```
#creating the confusion matrix
```

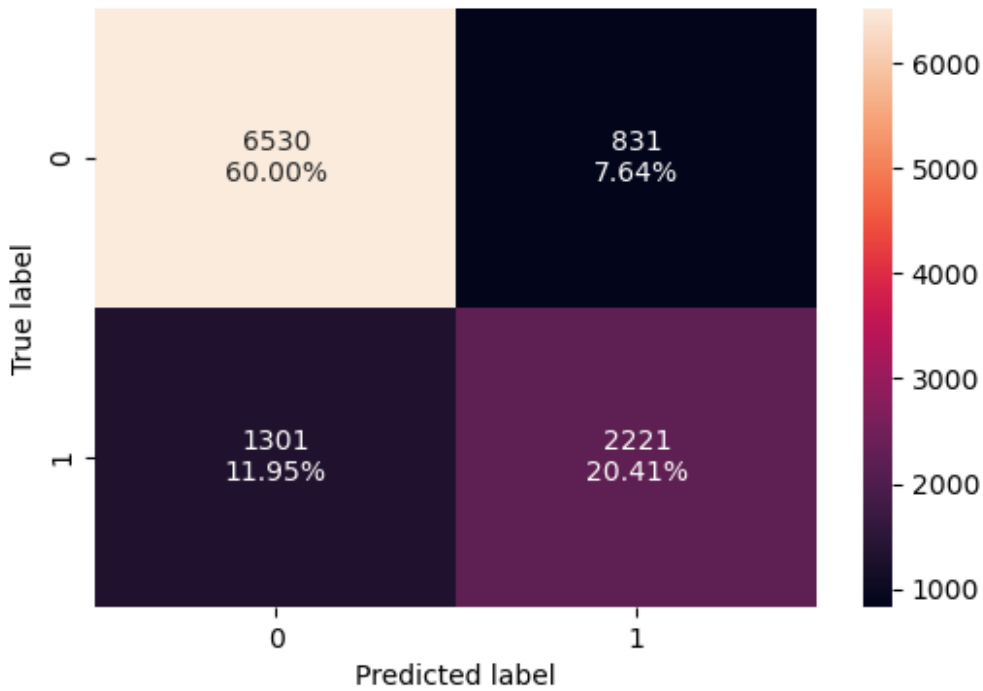


Figure 40 0.42 threshold confusion matrix

```
#checking the model performance
```

```
Test performance:

   Accuracy  Recall  Precision      F1
0   0.80355 0.70329    0.69384 0.69853
```

Building a Logistic Regression model

Model performance evaluation

```
#training model performance comparison
```

```
Training performance comparison:

           Logistic Regression-default Threshold  \
Accuracy                                 0.80529
Recall                                   0.63255
Precision                                0.73872
F1                                       0.68153

           Logistic Regression-0.37 Threshold  \
```

```
Accuracy                                0.78989
Recall                                  0.74327
Precision                               0.66100
F1                                      0.69972

           Logistic Regression-0.42 Threshold
Accuracy                                0.80108
Recall                                  0.69903
Precision                               0.69761
F1                                      0.69832
```

```
# test performance comparison


Test set performance comparison:

           Logistic Regression-default Threshold (0.5)  \
Accuracy                                         0.80410
Recall                                           0.63061
Precision                                        0.72772
F1                                               0.67569

           Logistic Regression-0.37 Threshold  \
Accuracy                                0.79371
Recall                                  0.74730
Precision                               0.66015
F1                                      0.70103

           Logistic Regression-0.42 Threshold
Accuracy                                0.80355
Recall                                  0.70329
Precision                               0.69384
F1                                      0.69853
```

Final Model SUmmary

- All three models are performing well on both the training and test data without overfitting
- Model with threshold value of 0.37 is giving best F1-score

Building a Decision Tree model

Data Preparation for modelling(Decision Tree)

- We want to predict which bookings will be canceled
- We need to encode categorical features
- We will split the data into train and test to be able to evaluate the model

Function to calculcate performance metrics and confusion matrix

Building Decision Tree

Check the model on training set

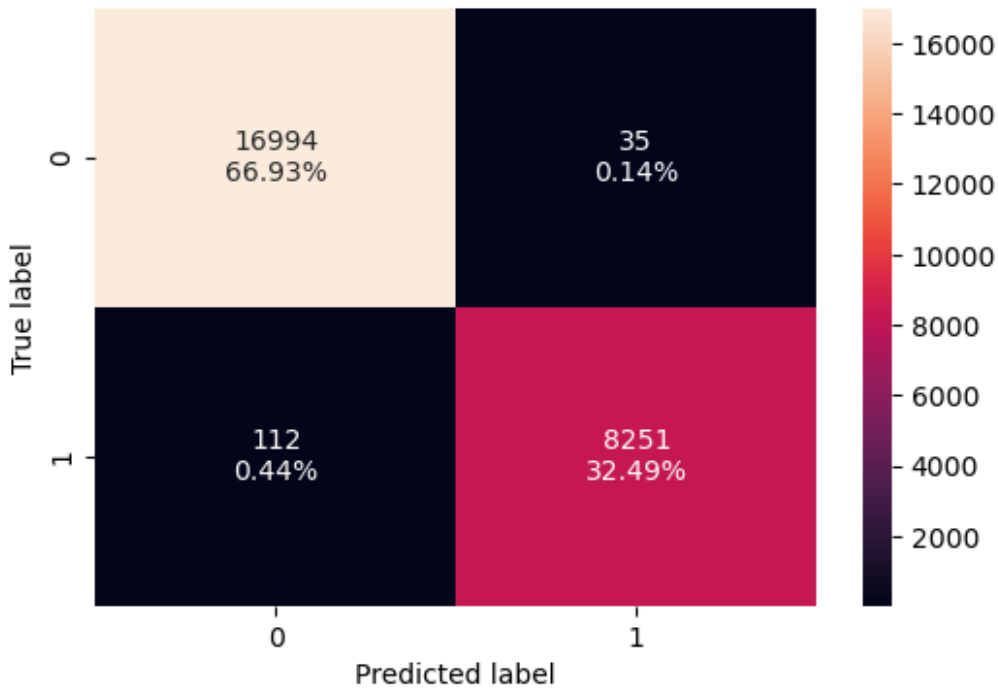


Figure 41 training model test confusion matrix

	Accuracy	Recall	Precision	F1
0	0.99421	0.98661	0.99578	0.99117

Check the model on testing set

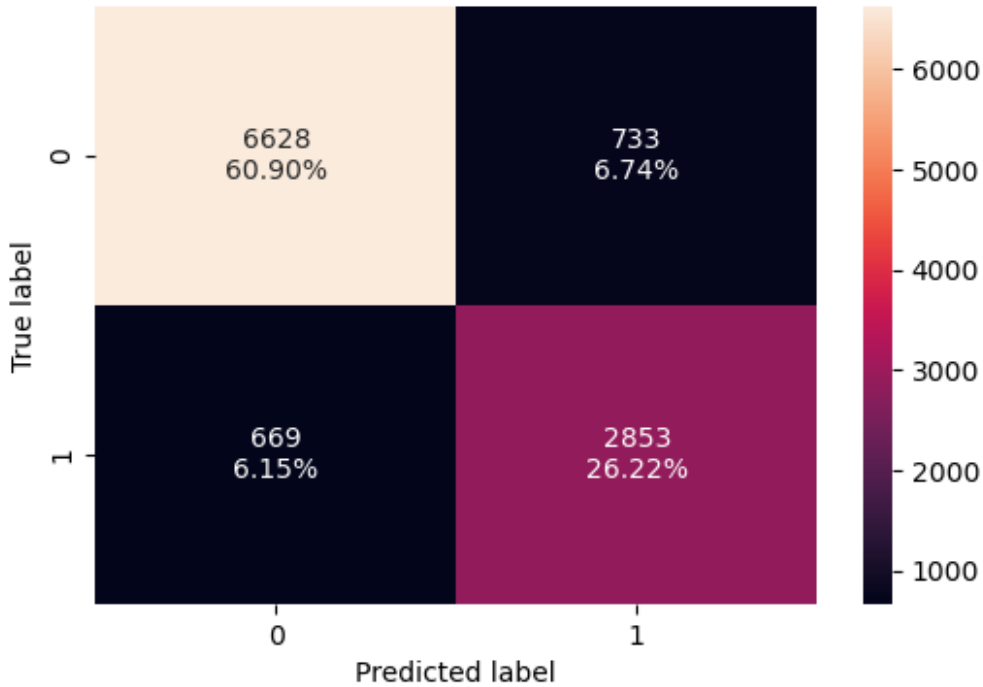


Figure 42 test set model test confusion matrix

```
#decision_tree_perf_test
```

```
   Accuracy  Recall  Precision      F1
0   0.87118 0.81005    0.79559 0.80276
```

```
#decision_tree_perf_train
```

```
   Accuracy  Recall  Precision      F1
0   0.99421 0.98661    0.99578 0.99117
```

- As shown in the comparison there is high discrepancy between the test and training data in terms of Accuracy, Recall,Precision and F1-Score
- This means model is overfitting

Let's prune the tree

- Before pruning the tree lets check important features

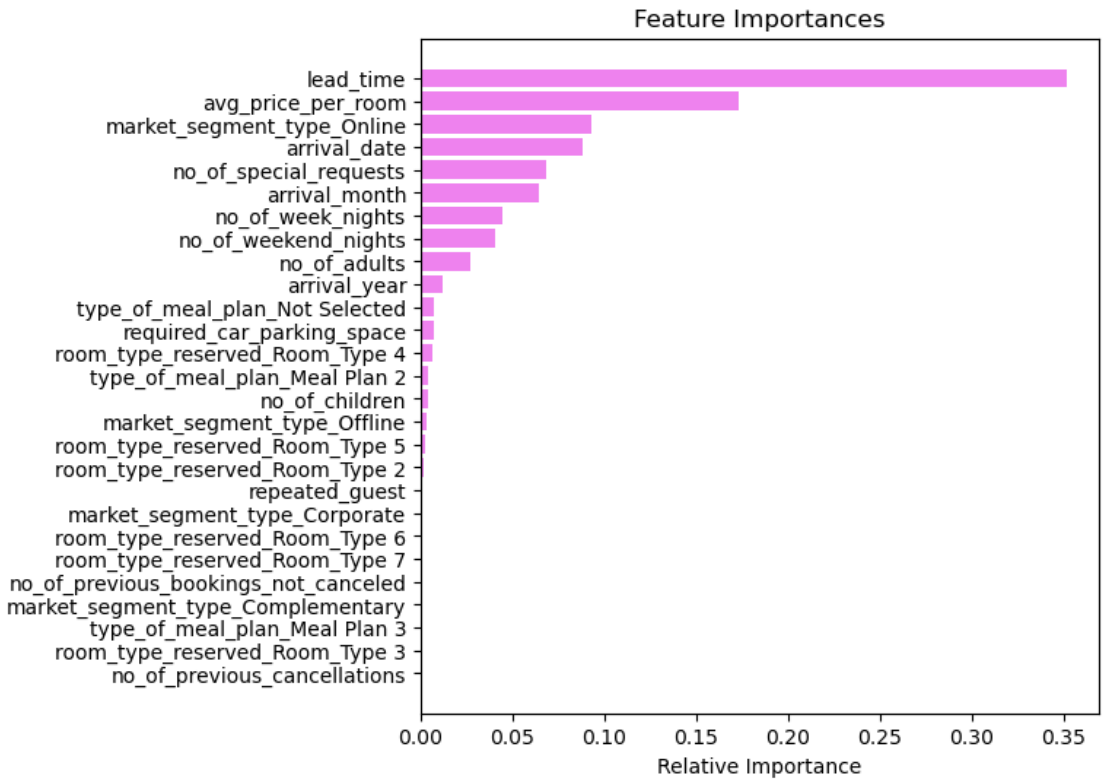


Figure 43 features importance check

Pre-Pruning

The objective was to fine-tune a decision tree model for booking cancellation prediction by exploring various hyperparameter settings using grid search with cross-validation. The classifier was configured with a fixed random state (set to 1) and a "balanced" class weight to address any imbalances in cancellation records.

The hyperparameters explored include:

- **Max Depth:** Values of 2, 4, and 6 were considered. This range provides a balance, ensuring the tree is neither too shallow (which might underfit the data) nor too deep (which can lead to overfitting).
- **Max Leaf Nodes:** Options of 50, 75, 150, and 250 were tested. Controlling the number of terminal nodes helps regulate the complexity of the model, allowing us to find a sweet spot that captures the important patterns without capturing noise.
- **Minimum Samples Split:** Values of 10, 30, 50, and 70 were used. By varying this parameter, the model is encouraged to only perform splits when there is a sufficient amount of data, thereby reducing the risk of splitting on anomalies and overfitting.

These parameter choices are well suited because they directly control the trade-off between bias and variance. In combination with the F1 score being used as the evaluation metric—which balances precision and recall—the grid search effectively identifies a model configuration that is both robust and generalizes well to unseen data.

After Pre-pruning let's see the performance of the training set

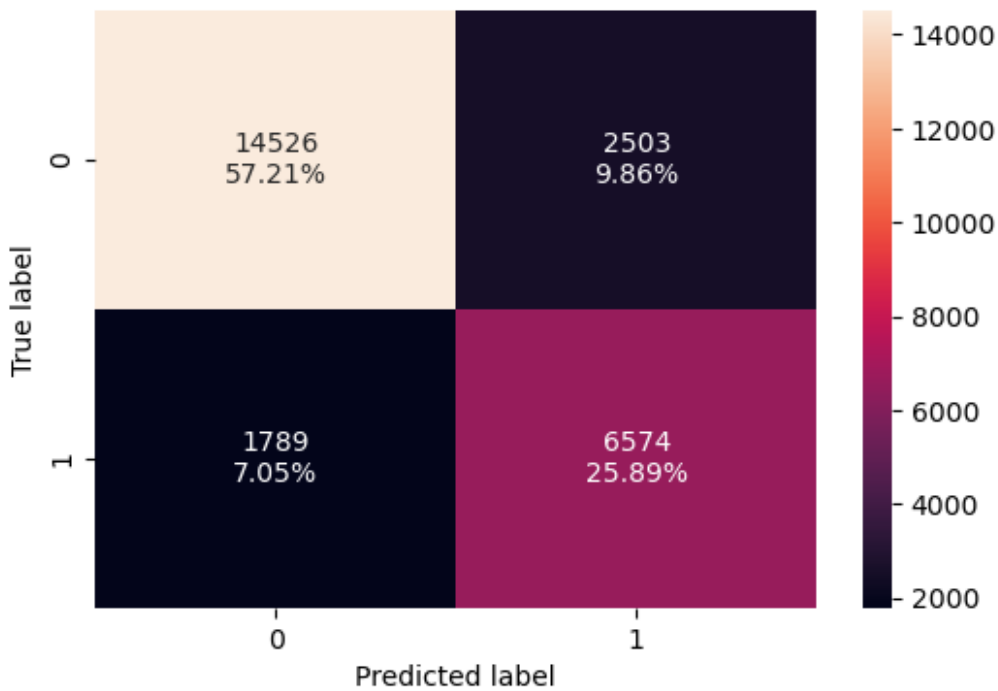


Figure 44 pre pruned training confusion matrix

	Accuracy	Recall	Precision	F1
0	0.83097	0.78608	0.72425	0.75390

After Pre-pruning let's see the performance of the testing set

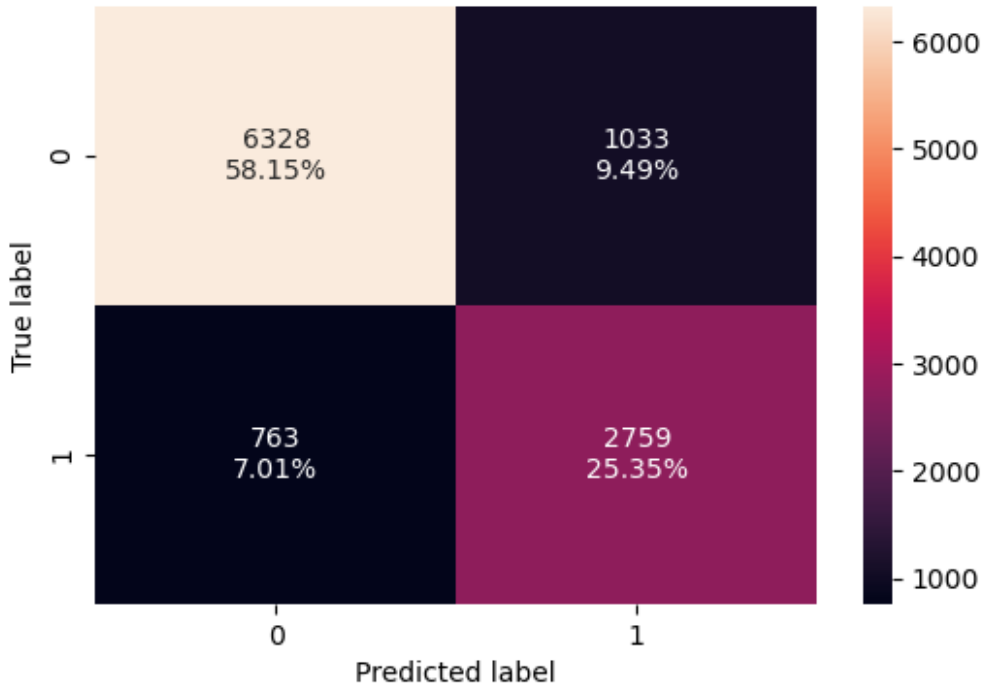


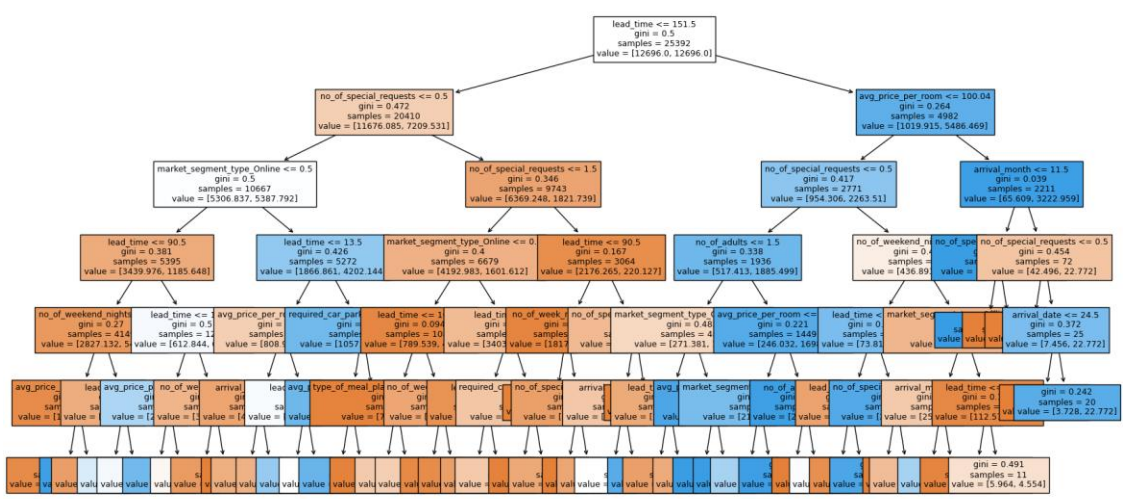
Figure 45 pre pruned testing confusion matrix

	Accuracy	Recall	Precision	F1
0	0.83497	0.78336	0.72758	0.75444

As we can see after pre-pruning

- The parameters between train and test data have comparable values

Visualizing the Decision Tree



```
#Printing the text report of decision tree
```

```
|--- lead_time <= 151.50
|   |--- no_of_special_requests <= 0.50
|   |   |--- market_segment_type_Online <= 0.50
|   |   |   |--- lead_time <= 90.50
|   |   |   |   |--- no_of_weekend_nights <= 0.50
|   |   |   |   |   |--- avg_price_per_room <= 196.50
|   |   |   |   |   |   |--- weights: [1736.39, 133.59] class: 0
|   |   |   |   |   |--- avg_price_per_room >  196.50
|   |   |   |   |   |   |--- weights: [0.75, 24.29] class: 1
|   |   |   |   |--- no_of_weekend_nights >  0.50
|   |   |   |   |   |--- lead_time <= 68.50
|   |   |   |   |   |   |--- weights: [960.27, 223.16] class: 0
|   |   |   |   |   |--- lead_time >  68.50
|   |   |   |   |   |   |--- weights: [129.73, 160.92] class: 1
|   |   |   |--- lead_time >  90.50
|   |   |   |   |--- lead_time <= 117.50
|   |   |   |   |   |--- avg_price_per_room <= 93.58
|   |   |   |   |   |   |--- weights: [214.72, 227.72] class: 1
|   |   |   |   |   |--- avg_price_per_room >  93.58
|   |   |   |   |   |   |--- weights: [82.76, 285.41] class: 1
|   |   |   |   |--- lead_time >  117.50
|   |   |   |   |   |--- no_of_week_nights <= 1.50
|   |   |   |   |   |   |--- weights: [87.23, 81.98] class: 0
|   |   |   |   |   |--- no_of_week_nights >  1.50
|   |   |   |   |   |   |--- weights: [228.14, 48.58] class: 0
|   |   |--- market_segment_type_Online >  0.50
|   |   |   |--- lead_time <= 13.50
|   |   |   |   |--- avg_price_per_room <= 99.44
|   |   |   |   |   |--- arrival_month <= 1.50
|   |   |   |   |   |   |--- weights: [92.45, 0.00] class: 0
|   |   |   |   |   |--- arrival_month >  1.50
|   |   |   |   |   |   |--- weights: [363.83, 132.08] class: 0
|   |   |   |   |--- avg_price_per_room >  99.44
|   |   |   |   |   |--- lead_time <= 3.50
|   |   |   |   |   |   |--- weights: [219.94, 85.01] class: 0
|   |   |   |   |   |--- lead_time >  3.50
|   |   |   |   |   |   |--- weights: [132.71, 280.85] class: 1
|   |   |   |--- lead_time >  13.50
|   |   |   |   |--- required_car_parking_space <= 0.50
|   |   |   |   |   |--- avg_price_per_room <= 71.92
|   |   |   |   |   |   |--- weights: [158.80, 159.40] class: 1
|   |   |   |   |   |--- avg_price_per_room >  71.92
|   |   |   |   |   |   |--- weights: [850.67, 3543.28] class: 1
|   |   |   |   |--- required_car_parking_space >  0.50
|   |   |   |   |   |--- weights: [48.46, 1.52] class: 0
|   |--- no_of_special_requests >  0.50
|   |   |--- no_of_special_requests <= 1.50
```

```
|   |   |   |--- market_segment_type_Online <= 0.50
|   |   |   |   |--- lead_time <= 102.50
|   |   |   |   |   |--- type_of_meal_plan_Not Selected <= 0.50
|   |   |   |   |   |   |--- weights: [697.09, 9.11] class: 0
|   |   |   |   |   |--- type_of_meal_plan_Not Selected >  0.50
|   |   |   |   |   |   |--- weights: [15.66, 9.11] class: 0
|   |   |   |   |--- lead_time >  102.50
|   |   |   |   |   |--- no_of_week_nights <= 2.50
|   |   |   |   |   |   |--- weights: [32.06, 19.74] class: 0
|   |   |   |   |   |--- no_of_week_nights >  2.50
|   |   |   |   |   |   |--- weights: [44.73, 3.04] class: 0
|   |   |   |--- market_segment_type_Online >  0.50
|   |   |   |   |--- lead_time <= 8.50
|   |   |   |   |   |--- lead_time <= 4.50
|   |   |   |   |   |   |--- weights: [498.03, 44.03] class: 0
|   |   |   |   |   |--- lead_time >  4.50
|   |   |   |   |   |   |--- weights: [258.71, 63.76] class: 0
|   |   |   |   |--- lead_time >  8.50
|   |   |   |   |   |--- required_car_parking_space <= 0.50
|   |   |   |   |   |   |--- weights: [2512.51, 1451.32] class: 0
|   |   |   |   |   |--- required_car_parking_space >  0.50
|   |   |   |   |   |   |--- weights: [134.20, 1.52] class: 0
|   |   |--- no_of_special_requests >  1.50
|   |   |   |--- lead_time <= 90.50
|   |   |   |   |--- no_of_week_nights <= 3.50
|   |   |   |   |   |--- weights: [1585.04, 0.00] class: 0
|   |   |   |   |--- no_of_week_nights >  3.50
|   |   |   |   |   |--- no_of_special_requests <= 2.50
|   |   |   |   |   |   |--- weights: [180.42, 57.69] class: 0
|   |   |   |   |   |--- no_of_special_requests >  2.50
|   |   |   |   |   |   |--- weights: [52.19, 0.00] class: 0
|   |   |   |--- lead_time >  90.50
|   |   |   |   |--- no_of_special_requests <= 2.50
|   |   |   |   |   |--- arrival_month <= 8.50
|   |   |   |   |   |   |--- weights: [184.90, 56.17] class: 0
|   |   |   |   |   |--- arrival_month >  8.50
|   |   |   |   |   |   |--- weights: [106.61, 106.27] class: 0
|   |   |   |   |--- no_of_special_requests >  2.50
|   |   |   |   |   |--- weights: [67.10, 0.00] class: 0
|--- lead_time >  151.50
|   |--- avg_price_per_room <= 100.04
|   |   |--- no_of_special_requests <= 0.50
|   |   |   |--- no_of_adults <= 1.50
|   |   |   |   |--- market_segment_type_Online <= 0.50
|   |   |   |   |   |--- lead_time <= 163.50
|   |   |   |   |   |   |--- weights: [3.73, 24.29] class: 1
|   |   |   |   |   |--- lead_time >  163.50
|   |   |   |   |   |   |--- weights: [257.96, 62.24] class: 0
|   |   |   |   |--- market_segment_type_Online >  0.50
|   |   |   |   |   |--- avg_price_per_room <= 2.50
```

```
|   |   |   |   |   |   |--- weights: [8.95, 3.04] class: 0
|   |   |   |   |   |--- avg_price_per_room >  2.50
|   |   |   |   |   |   |--- weights: [0.75, 97.16] class: 1
|   |   |   |--- no_of_adults >  1.50
|   |   |   |   |--- avg_price_per_room <= 82.47
|   |   |   |   |   |--- market_segment_type_Offline <= 0.50
|   |   |   |   |   |   |--- weights: [2.98, 282.37] class: 1
|   |   |   |   |   |--- market_segment_type_Offline >  0.50
|   |   |   |   |   |   |--- weights: [213.97, 385.60] class: 1
|   |   |   |   |--- avg_price_per_room >  82.47
|   |   |   |   |   |--- no_of_adults <= 2.50
|   |   |   |   |   |   |--- weights: [23.86, 1030.80] class: 1
|   |   |   |   |   |--- no_of_adults >  2.50
|   |   |   |   |   |   |--- weights: [5.22, 0.00] class: 0
|   |   |--- no_of_special_requests >  0.50
|   |   |   |--- no_of_weekend_nights <= 0.50
|   |   |   |   |--- lead_time <= 180.50
|   |   |   |   |   |--- lead_time <= 159.50
|   |   |   |   |   |   |--- weights: [7.46, 7.59] class: 1
|   |   |   |   |   |--- lead_time >  159.50
|   |   |   |   |   |   |--- weights: [37.28, 4.55] class: 0
|   |   |   |   |--- lead_time >  180.50
|   |   |   |   |   |--- no_of_special_requests <= 2.50
|   |   |   |   |   |   |--- weights: [20.13, 212.54] class: 1
|   |   |   |   |   |--- no_of_special_requests >  2.50
|   |   |   |   |   |   |--- weights: [8.95, 0.00] class: 0
|   |   |   |--- no_of_weekend_nights >  0.50
|   |   |   |   |--- market_segment_type_Offline <= 0.50
|   |   |   |   |   |--- arrival_month <= 11.50
|   |   |   |   |   |   |--- weights: [231.12, 110.82] class: 0
|   |   |   |   |   |--- arrival_month >  11.50
|   |   |   |   |   |   |--- weights: [19.38, 34.92] class: 1
|   |   |   |   |--- market_segment_type_Offline >  0.50
|   |   |   |   |   |--- lead_time <= 348.50
|   |   |   |   |   |   |--- weights: [106.61, 3.04] class: 0
|   |   |   |   |   |--- lead_time >  348.50
|   |   |   |   |   |   |--- weights: [5.96, 4.55] class: 0
|   |--- avg_price_per_room >  100.04
|   |   |--- arrival_month <= 11.50
|   |   |   |--- no_of_special_requests <= 2.50
|   |   |   |   |--- weights: [0.00, 3200.19] class: 1
|   |   |   |--- no_of_special_requests >  2.50
|   |   |   |   |--- weights: [23.11, 0.00] class: 0
|   |   |--- arrival_month >  11.50
|   |   |   |--- no_of_special_requests <= 0.50
|   |   |   |   |--- weights: [35.04, 0.00] class: 0
|   |   |   |--- no_of_special_requests >  0.50
|   |   |   |   |--- arrival_date <= 24.50
|   |   |   |   |   |--- weights: [3.73, 0.00] class: 0
|   |   |   |   |--- arrival_date >  24.50
```

```
|   |   |   |   |   |--- weights: [3.73, 22.77] class: 1
```

Decision Tree output understanding

- If the `leadtime is`<=151.5,
- `no_of_special_requests`<= 0.5,
- `no_of_weekights`<=0.5,
- `avg_room_price` >= 196.50,
- `market_segment_online`<=0.5,

Then the booking is likely to get cancelled

```
#lets see the important features after Pre-pruning
```

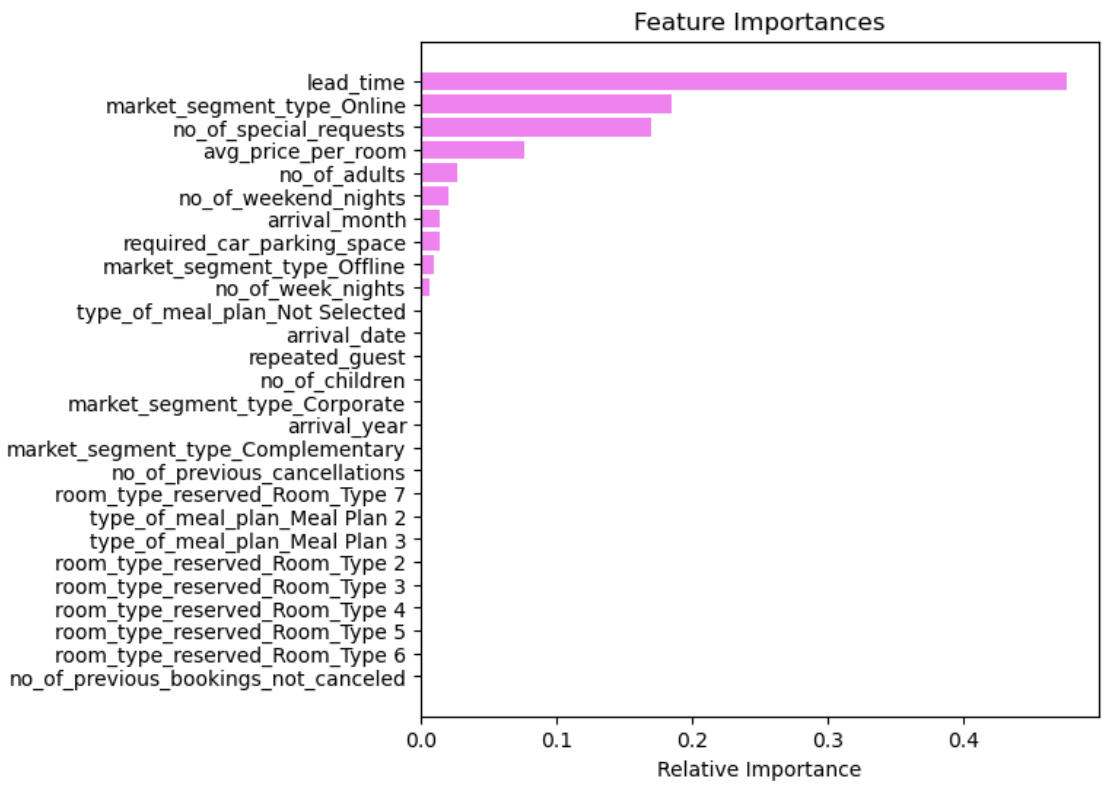


Figure 46 pre pruned feature importance check

- After Pre-Pruning, `lead_time lead time`,`market_segment_online`,`special requests`,`average_room_price` are the top important features

Cost Complexity Pruning

- Cost Complexity Pruning is a post pruning method to control tree size
- we will use `DecisionTreeClassifier`

```
      ccp_alphas  impurities
0        0.00000     0.00838
1        0.00000     0.00838
2        0.00000     0.00838
3        0.00000     0.00838
4        0.00000     0.00838
...          ...         ...
1837     0.00890     0.32806
1838     0.00980     0.33786
1839     0.01272     0.35058
1840     0.03412     0.41882
1841     0.08118     0.50000

[1842 rows x 2 columns]
```

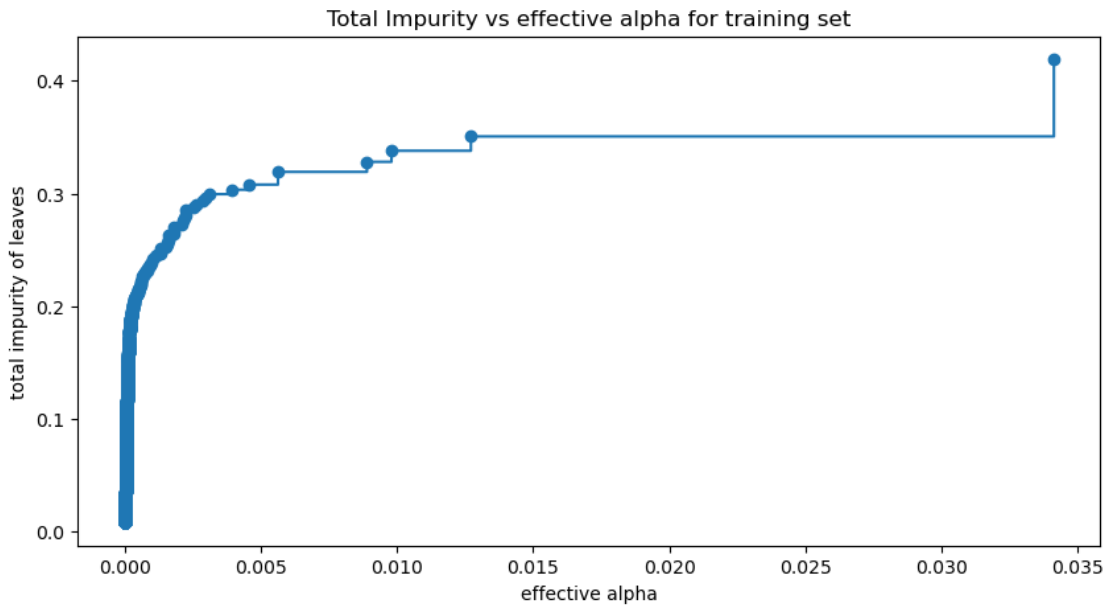


Figure 47 total impurity vs effective alpha training set

Next, we train a decision tree using effective alphas. The last value in `ccp_alphas` is the alpha value that prunes the whole tree, leaving the tree, clfs[-1], with one node.

```
Number of nodes in the last tree is: 1 with ccp_alpha: 0.08117914389136932
```

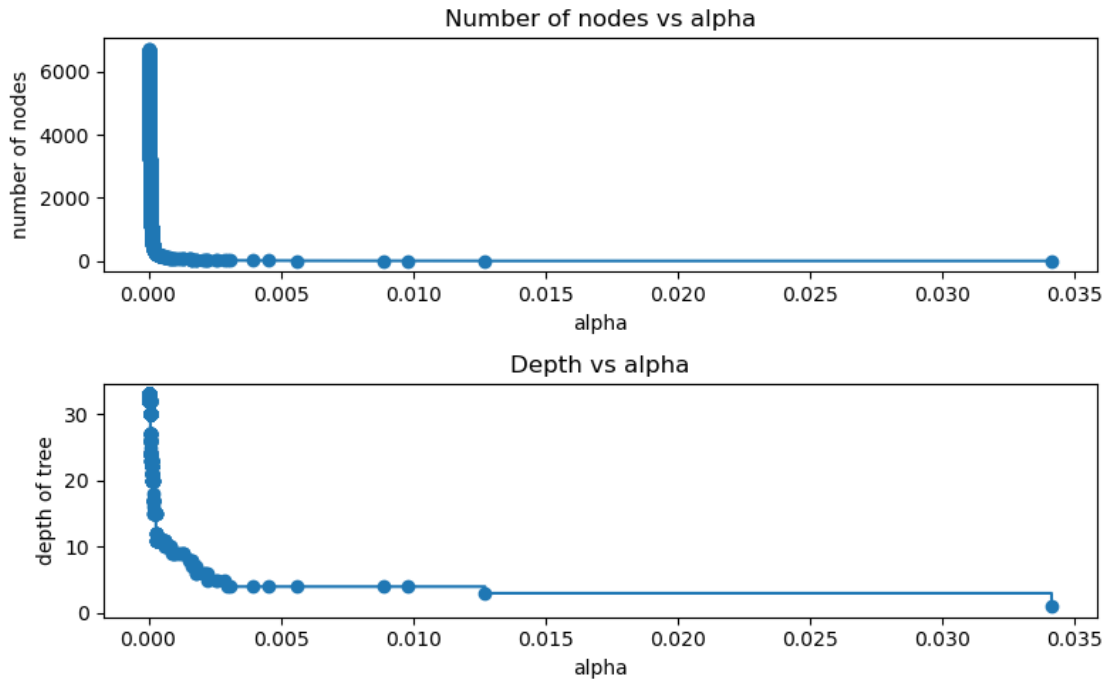


Figure 48 alpha vs nodes and alpha vs depth

F1 Score vs Alpha for training sets

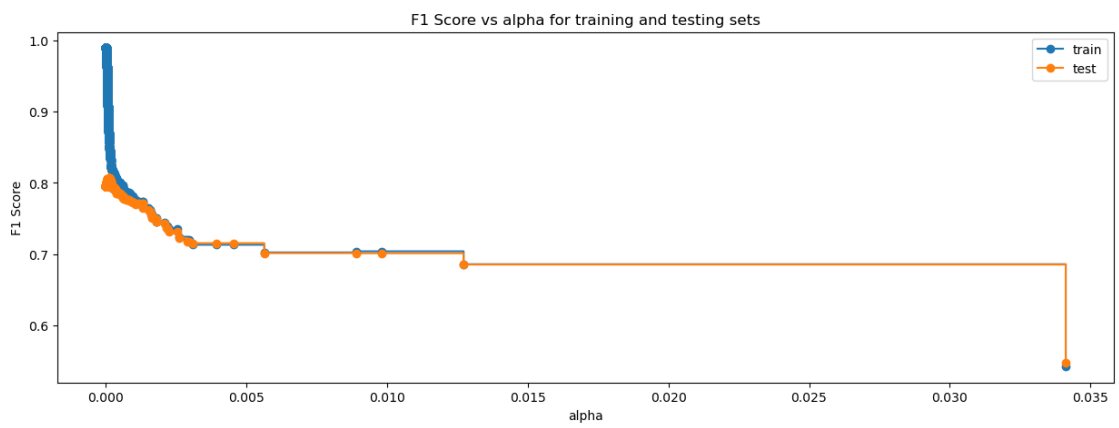


Figure 49 F1 score vs alpha training set

Checking model on training set

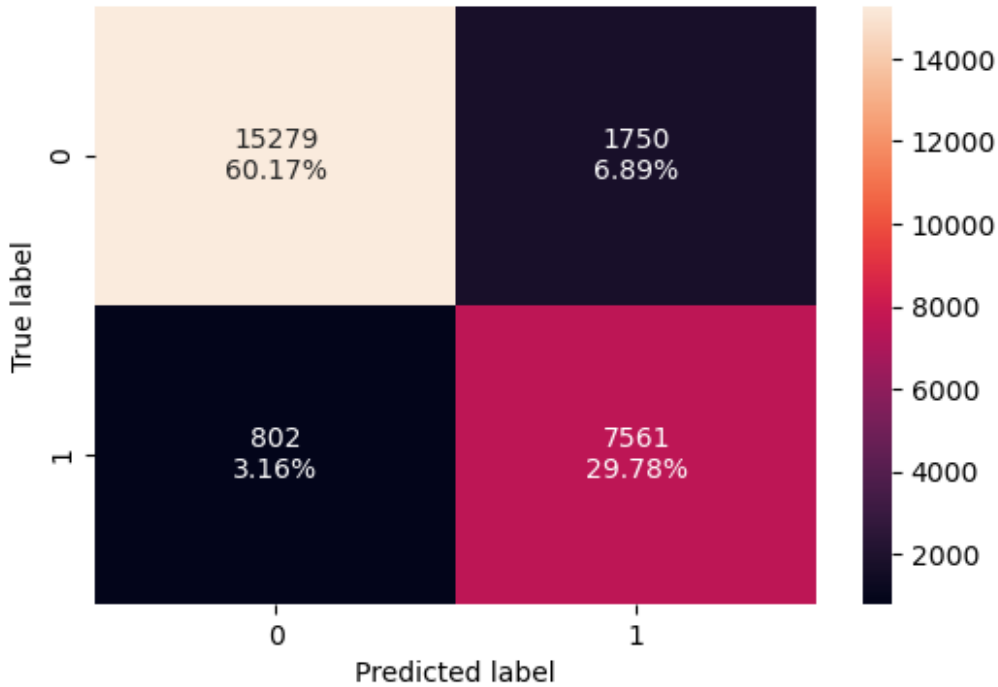


Figure 50 pre pruned confusion matrix training set

	Accuracy	Recall	Precision	F1
0	0.89950	0.90410	0.81205	0.85561

Checking model on test data set

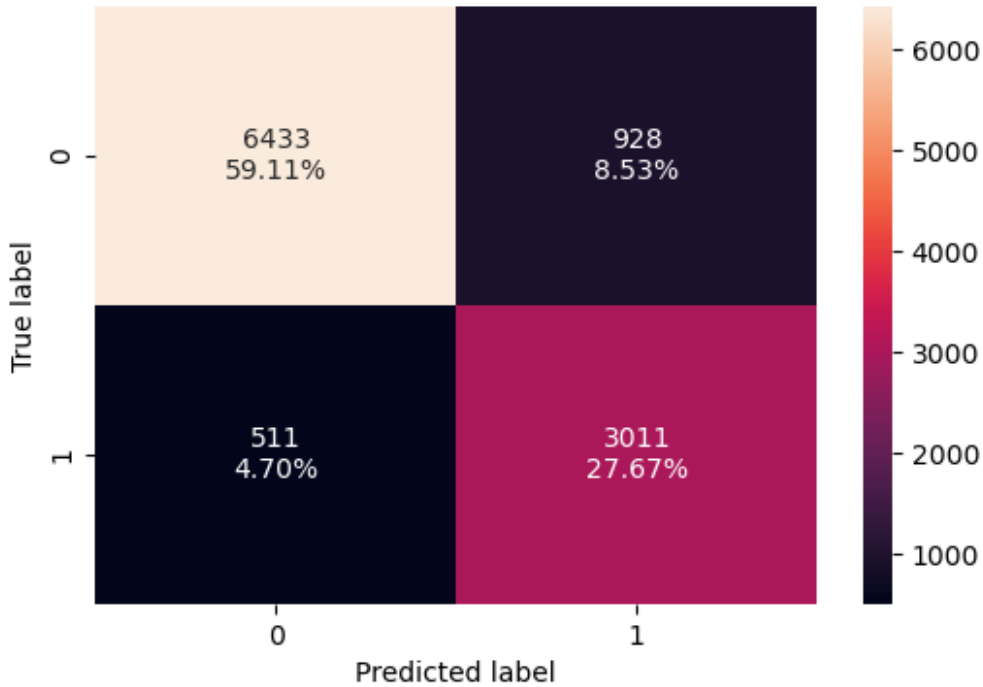


Figure 51 pre pruned confusion matrix test set

	Accuracy	Recall	Precision	F1
0	0.86778	0.85491	0.76441	0.80713

```
#Let's plot the tree plot
```

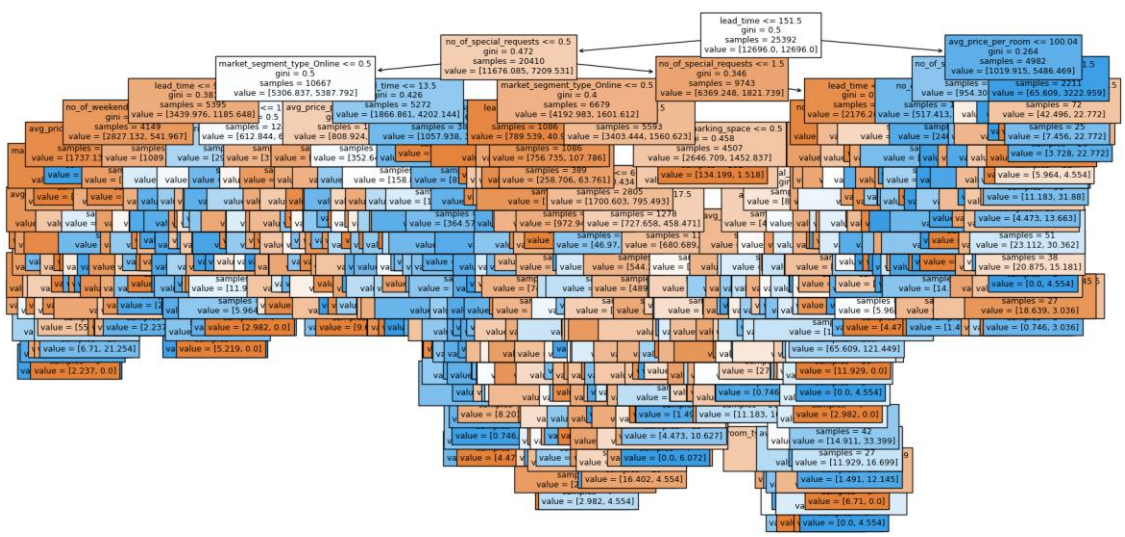


Figure 52 decision tree

```
#Let's see the decision tree report
```

```
|--- lead_time <= 151.50
|   |--- no_of_special_requests <= 0.50
|   |   |--- market_segment_type_Online <= 0.50
|   |   |   |--- lead_time <= 90.50
|   |   |   |   |--- no_of_weekend_nights <= 0.50
|   |   |   |   |   |--- avg_price_per_room <= 196.50
|   |   |   |   |   |   |--- market_segment_type_Offline <= 0.50
|   |   |   |   |   |   |   |--- lead_time <= 16.50
|   |   |   |   |   |   |   |   |--- avg_price_per_room <= 68.50
|   |   |   |   |   |   |   |   |   |--- weights: [207.26, 10.63] class: 0
|   |   |   |   |   |   |   |   |--- avg_price_per_room >  68.50
|   |   |   |   |   |   |   |   |   |--- arrival_date <= 29.50
|   |   |   |   |   |   |   |   |   |   |--- no_of_adults <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- no_of_adults >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |--- arrival_date >  29.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [2.24, 7.59] class: 1
|   |   |   |   |   |   |   |--- lead_time >  16.50
|   |   |   |   |   |   |   |   |--- avg_price_per_room <= 135.00
|   |   |   |   |   |   |   |   |   |--- arrival_month <= 11.50
|   |   |   |   |   |   |   |   |   |   |--- repeated_guest <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |--- repeated_guest >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [11.18, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |--- arrival_month >  11.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [21.62, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- avg_price_per_room >  135.00
|   |   |   |   |   |   |   |   |   |--- weights: [0.00, 12.14] class: 1
|   |   |   |   |   |   |--- market_segment_type_Offline >  0.50
|   |   |   |   |   |   |   |--- weights: [1199.59, 1.52] class: 0
|   |   |   |   |   |--- avg_price_per_room >  196.50
|   |   |   |   |   |   |--- weights: [0.75, 24.29] class: 1
|   |   |   |   |--- no_of_weekend_nights >  0.50
|   |   |   |   |   |--- lead_time <= 68.50
|   |   |   |   |   |   |--- arrival_month <= 9.50
|   |   |   |   |   |   |   |--- avg_price_per_room <= 63.29
|   |   |   |   |   |   |   |   |--- arrival_date <= 20.50
|   |   |   |   |   |   |   |   |   |--- type_of_meal_plan_Not Selected <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [41.75, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |--- type_of_meal_plan_Not Selected >  0.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [0.75, 3.04] class: 1
|   |   |   |   |   |   |   |   |--- arrival_date >  20.50
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 59.75
|   |   |   |   |   |   |   |   |   |   |--- arrival_date <= 23.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [1.49, 12.14] class: 1
|   |   |   |   |   |   |   |   |   |   |--- arrival_date >  23.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [14.91, 1.52] class: 0
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  59.75
|   |   |   |   |   |   |   |   |   |   |--- lead_time <= 44.00
```

```
|   |   |   |   |   |   |   |   |   |   |   |   |--- weights: [0.75, 59.21] class: 1
|   |   |   |   |   |   |   |   |   |   |   |--- lead_time >  44.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- weights: [3.73, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- avg_price_per_room >  63.29
|   |   |   |   |   |   |   |   |   |--- no_of_weekend_nights <= 3.50
|   |   |   |   |   |   |   |   |   |   |--- lead_time <= 59.50
|   |   |   |   |   |   |   |   |   |   |   |--- arrival_month <= 7.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- arrival_month >  7.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- lead_time >  59.50
|   |   |   |   |   |   |   |   |   |   |   |--- arrival_month <= 5.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- arrival_month >  5.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- weights: [20.13, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |--- no_of_weekend_nights >  3.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [0.75, 15.18] class: 1
|   |   |   |   |   |   |   |--- arrival_month >  9.50
|   |   |   |   |   |   |   |   |--- weights: [413.04, 27.33] class: 0
|   |   |   |   |   |   |--- lead_time >  68.50
|   |   |   |   |   |   |   |--- avg_price_per_room <= 99.98
|   |   |   |   |   |   |   |   |--- arrival_month <= 3.50
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 62.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [15.66, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  62.50
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 80.38
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [8.20, 25.81] class: 1
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  80.38
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [3.73, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- arrival_month >  3.50
|   |   |   |   |   |   |   |   |   |--- no_of_week_nights <= 2.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [55.17, 3.04] class: 0
|   |   |   |   |   |   |   |   |   |--- no_of_week_nights >  2.50
|   |   |   |   |   |   |   |   |   |   |--- lead_time <= 73.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [0.00, 4.55] class: 1
|   |   |   |   |   |   |   |   |   |   |--- lead_time >  73.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [21.62, 4.55] class: 0
|   |   |   |   |   |   |   |--- avg_price_per_room >  99.98
|   |   |   |   |   |   |   |   |--- arrival_year <= 2017.50
|   |   |   |   |   |   |   |   |   |--- weights: [8.95, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- arrival_year >  2017.50
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 132.43
|   |   |   |   |   |   |   |   |   |   |--- weights: [9.69, 122.97] class: 1
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  132.43
|   |   |   |   |   |   |   |   |   |   |--- weights: [6.71, 0.00] class: 0
|   |   |   |   |--- lead_time >  90.50
|   |   |   |   |   |--- lead_time <= 117.50
|   |   |   |   |   |   |--- avg_price_per_room <= 93.58
|   |   |   |   |   |   |   |--- avg_price_per_room <= 75.07
|   |   |   |   |   |   |   |   |--- no_of_week_nights <= 2.50
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 58.75
|   |   |   |   |   |   |   |   |   |   |--- weights: [5.96, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  58.75
|   |   |   |   |   |   |   |   |   |   |--- repeated_guest <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- arrival_month <= 4.50
```

```
|   |   |   |   |   |   |   |   |   |   |   |   |--- weights: [2.24, 118.41] class: 1
|   |   |   |   |   |   |   |   |   |   |   |--- arrival_month >  4.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |--- repeated_guest >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [4.47, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- no_of_week_nights >  2.50
|   |   |   |   |   |   |   |   |   |--- arrival_date <= 11.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [31.31, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |--- arrival_date >  11.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [29.08, 15.18] class: 0
|   |   |   |   |   |   |   |--- avg_price_per_room >  75.07
|   |   |   |   |   |   |   |   |--- arrival_month <= 3.50
|   |   |   |   |   |   |   |   |   |--- weights: [59.64, 3.04] class: 0
|   |   |   |   |   |   |   |   |--- arrival_month >  3.50
|   |   |   |   |   |   |   |   |   |--- arrival_month <= 4.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [1.49, 16.70] class: 1
|   |   |   |   |   |   |   |   |   |--- arrival_month >  4.50
|   |   |   |   |   |   |   |   |   |   |--- no_of_adults <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 86.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- weights: [2.24, 16.70] class: 1
|   |   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  86.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- weights: [8.95, 3.04] class: 0
|   |   |   |   |   |   |   |   |   |   |--- no_of_adults >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- arrival_date <= 22.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- weights: [44.73, 4.55] class: 0
|   |   |   |   |   |   |   |   |   |   |   |--- arrival_date >  22.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |--- avg_price_per_room >  93.58
|   |   |   |   |   |   |   |--- arrival_date <= 11.50
|   |   |   |   |   |   |   |   |--- no_of_week_nights <= 1.50
|   |   |   |   |   |   |   |   |   |--- weights: [16.40, 39.47] class: 1
|   |   |   |   |   |   |   |   |--- no_of_week_nights >  1.50
|   |   |   |   |   |   |   |   |   |--- weights: [20.13, 6.07] class: 0
|   |   |   |   |   |   |   |--- arrival_date >  11.50
|   |   |   |   |   |   |   |   |--- avg_price_per_room <= 102.09
|   |   |   |   |   |   |   |   |   |--- weights: [5.22, 144.22] class: 1
|   |   |   |   |   |   |   |   |--- avg_price_per_room >  102.09
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 109.50
|   |   |   |   |   |   |   |   |   |   |--- no_of_week_nights <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [0.75, 16.70] class: 1
|   |   |   |   |   |   |   |   |   |   |--- no_of_week_nights >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [33.55, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  109.50
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 124.25
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [2.98, 75.91] class: 1
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  124.25
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [3.73, 3.04] class: 0
|   |   |   |   |   |--- lead_time >  117.50
|   |   |   |   |   |   |--- no_of_week_nights <= 1.50
|   |   |   |   |   |   |   |--- arrival_date <= 7.50
|   |   |   |   |   |   |   |   |--- weights: [38.02, 0.00] class: 0
|   |   |   |   |   |   |   |--- arrival_date >  7.50
|   |   |   |   |   |   |   |   |--- avg_price_per_room <= 93.58
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 65.38
|   |   |   |   |   |   |   |   |   |   |--- weights: [0.00, 4.55] class: 1
```

```
|   |   |   |   |   |   |   |   |--- avg_price_per_room >  65.38
|   |   |   |   |   |   |   |   |   |--- weights: [24.60, 3.04] class: 0
|   |   |   |   |   |   |   |--- avg_price_per_room >  93.58
|   |   |   |   |   |   |   |   |--- arrival_date <= 28.00
|   |   |   |   |   |   |   |   |   |--- weights: [14.91, 72.87] class: 1
|   |   |   |   |   |   |   |   |--- arrival_date >  28.00
|   |   |   |   |   |   |   |   |   |--- weights: [9.69, 1.52] class: 0
|   |   |   |   |   |--- no_of_week_nights >  1.50
|   |   |   |   |   |   |--- no_of_adults <= 1.50
|   |   |   |   |   |   |   |--- weights: [84.25, 0.00] class: 0
|   |   |   |   |   |   |--- no_of_adults >  1.50
|   |   |   |   |   |   |   |--- lead_time <= 125.50
|   |   |   |   |   |   |   |   |--- avg_price_per_room <= 90.85
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 87.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [13.42, 13.66] class: 1
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  87.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [0.00, 15.18] class: 1
|   |   |   |   |   |   |   |   |--- avg_price_per_room >  90.85
|   |   |   |   |   |   |   |   |   |--- weights: [10.44, 0.00] class: 0
|   |   |   |   |   |   |   |--- lead_time >  125.50
|   |   |   |   |   |   |   |   |--- arrival_date <= 19.50
|   |   |   |   |   |   |   |   |   |--- weights: [58.15, 18.22] class: 0
|   |   |   |   |   |   |   |   |--- arrival_date >  19.50
|   |   |   |   |   |   |   |   |   |--- weights: [61.88, 1.52] class: 0
|   |   |--- market_segment_type_Online >  0.50
|   |   |   |--- lead_time <= 13.50
|   |   |   |   |--- avg_price_per_room <= 99.44
|   |   |   |   |   |--- arrival_month <= 1.50
|   |   |   |   |   |   |--- weights: [92.45, 0.00] class: 0
|   |   |   |   |   |--- arrival_month >  1.50
|   |   |   |   |   |   |--- arrival_month <= 8.50
|   |   |   |   |   |   |   |--- no_of_weekend_nights <= 1.50
|   |   |   |   |   |   |   |   |--- avg_price_per_room <= 70.05
|   |   |   |   |   |   |   |   |   |--- weights: [31.31, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- avg_price_per_room >  70.05
|   |   |   |   |   |   |   |   |   |--- lead_time <= 5.50
|   |   |   |   |   |   |   |   |   |   |--- no_of_adults <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [38.77, 1.52] class: 0
|   |   |   |   |   |   |   |   |   |   |--- no_of_adults >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |--- lead_time >  5.50
|   |   |   |   |   |   |   |   |   |   |--- arrival_date <= 3.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [6.71, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |   |--- arrival_date >  3.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [34.30, 40.99] class: 1
|   |   |   |   |   |   |   |   |--- no_of_weekend_nights >  1.50
|   |   |   |   |   |   |   |   |   |--- no_of_adults <= 1.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [0.00, 19.74] class: 1
|   |   |   |   |   |   |   |   |   |--- no_of_adults >  1.50
|   |   |   |   |   |   |   |   |   |   |--- lead_time <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 74.21
|   |   |   |   |   |   |   |   |   |   |   |   |--- weights: [0.75, 3.04] class: 1
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  74.21
|   |   |   |   |   |   |   |   |   |   |   |   |--- weights: [9.69, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |--- lead_time >  2.50
```

```
|   |   |   |   |   |   |   |   |   |   |--- weights: [4.47, 10.63] class: 1
|   |   |   |   |   |   |   |--- arrival_month >  8.50
|   |   |   |   |   |   |   |   |--- no_of_week_nights <= 3.50
|   |   |   |   |   |   |   |   |   |--- weights: [155.07, 6.07] class: 0
|   |   |   |   |   |   |   |   |--- no_of_week_nights >  3.50
|   |   |   |   |   |   |   |   |   |--- arrival_month <= 11.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [3.73, 10.63] class: 1
|   |   |   |   |   |   |   |   |   |--- arrival_month >  11.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [7.46, 0.00] class: 0
|   |   |   |   |   |--- avg_price_per_room >  99.44
|   |   |   |   |   |   |--- lead_time <= 3.50
|   |   |   |   |   |   |   |--- avg_price_per_room <= 202.67
|   |   |   |   |   |   |   |   |--- no_of_week_nights <= 4.50
|   |   |   |   |   |   |   |   |   |--- arrival_month <= 5.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [63.37, 30.36] class: 0
|   |   |   |   |   |   |   |   |   |--- arrival_month >  5.50
|   |   |   |   |   |   |   |   |   |   |--- arrival_date <= 20.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [115.56, 12.14] class: 0
|   |   |   |   |   |   |   |   |   |   |--- arrival_date >  20.50
|   |   |   |   |   |   |   |   |   |   |   |--- arrival_date <= 24.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- arrival_date >  24.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- weights: [28.33, 3.04] class: 0
|   |   |   |   |   |   |   |   |--- no_of_week_nights >  4.50
|   |   |   |   |   |   |   |   |   |--- weights: [0.00, 6.07] class: 1
|   |   |   |   |   |   |   |--- avg_price_per_room >  202.67
|   |   |   |   |   |   |   |   |--- weights: [0.75, 22.77] class: 1
|   |   |   |   |   |   |--- lead_time >  3.50
|   |   |   |   |   |   |   |--- arrival_month <= 8.50
|   |   |   |   |   |   |   |   |--- avg_price_per_room <= 119.25
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 118.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [18.64, 59.21] class: 1
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  118.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [8.20, 1.52] class: 0
|   |   |   |   |   |   |   |   |--- avg_price_per_room >  119.25
|   |   |   |   |   |   |   |   |   |--- weights: [34.30, 171.55] class: 1
|   |   |   |   |   |   |   |--- arrival_month >  8.50
|   |   |   |   |   |   |   |   |--- arrival_year <= 2017.50
|   |   |   |   |   |   |   |   |   |--- weights: [26.09, 1.52] class: 0
|   |   |   |   |   |   |   |   |--- arrival_year >  2017.50
|   |   |   |   |   |   |   |   |   |--- arrival_month <= 11.50
|   |   |   |   |   |   |   |   |   |   |--- arrival_date <= 14.00
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [9.69, 36.43] class: 1
|   |   |   |   |   |   |   |   |   |   |--- arrival_date >  14.00
|   |   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 208.67
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  208.67
|   |   |   |   |   |   |   |   |   |   |   |   |--- weights: [0.00, 4.55] class: 1
|   |   |   |   |   |   |   |   |   |--- arrival_month >  11.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [15.66, 0.00] class: 0
|   |   |   |   |--- lead_time >  13.50
|   |   |   |   |   |--- required_car_parking_space <= 0.50
|   |   |   |   |   |   |--- avg_price_per_room <= 71.92
|   |   |   |   |   |   |   |--- avg_price_per_room <= 59.43
|   |   |   |   |   |   |   |   |--- lead_time <= 84.50
```

```
|   |   |   |   |   |   |   |   |--- weights: [50.70, 7.59] class: 0
|   |   |   |   |   |   |   |--- lead_time >  84.50
|   |   |   |   |   |   |   |   |--- arrival_year <= 2017.50
|   |   |   |   |   |   |   |   |   |--- arrival_date <= 27.00
|   |   |   |   |   |   |   |   |   |   |--- lead_time <= 131.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [0.75, 15.18] class: 1
|   |   |   |   |   |   |   |   |   |   |--- lead_time >  131.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [2.24, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |--- arrival_date >  27.00
|   |   |   |   |   |   |   |   |   |   |--- weights: [3.73, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- arrival_year >  2017.50
|   |   |   |   |   |   |   |   |   |--- weights: [10.44, 0.00] class: 0
|   |   |   |   |   |   |--- avg_price_per_room >  59.43
|   |   |   |   |   |   |   |--- lead_time <= 25.50
|   |   |   |   |   |   |   |   |--- weights: [20.88, 6.07] class: 0
|   |   |   |   |   |   |   |--- lead_time >  25.50
|   |   |   |   |   |   |   |   |--- avg_price_per_room <= 71.34
|   |   |   |   |   |   |   |   |   |--- arrival_month <= 3.50
|   |   |   |   |   |   |   |   |   |   |--- lead_time <= 68.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [15.66, 78.94] class: 1
|   |   |   |   |   |   |   |   |   |   |--- lead_time >  68.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |--- arrival_month >  3.50
|   |   |   |   |   |   |   |   |   |   |--- lead_time <= 102.00
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- lead_time >  102.00
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [12.67, 3.04] class: 0
|   |   |   |   |   |   |   |   |--- avg_price_per_room >  71.34
|   |   |   |   |   |   |   |   |   |--- weights: [11.18, 0.00] class: 0
|   |   |   |   |   |--- avg_price_per_room >  71.92
|   |   |   |   |   |   |--- arrival_year <= 2017.50
|   |   |   |   |   |   |   |--- lead_time <= 65.50
|   |   |   |   |   |   |   |   |--- avg_price_per_room <= 120.45
|   |   |   |   |   |   |   |   |   |--- weights: [79.77, 9.11] class: 0
|   |   |   |   |   |   |   |   |--- avg_price_per_room >  120.45
|   |   |   |   |   |   |   |   |   |--- no_of_week_nights <= 1.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [3.73, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |--- no_of_week_nights >  1.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [3.73, 12.14] class: 1
|   |   |   |   |   |   |   |--- lead_time >  65.50
|   |   |   |   |   |   |   |   |--- type_of_meal_plan_Meal Plan 2 <= 0.50
|   |   |   |   |   |   |   |   |   |--- arrival_date <= 27.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [16.40, 47.06] class: 1
|   |   |   |   |   |   |   |   |   |--- arrival_date >  27.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [3.73, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- type_of_meal_plan_Meal Plan 2 >  0.50
|   |   |   |   |   |   |   |   |   |--- weights: [0.00, 63.76] class: 1
|   |   |   |   |   |   |--- arrival_year >  2017.50
|   |   |   |   |   |   |   |--- avg_price_per_room <= 104.31
|   |   |   |   |   |   |   |   |--- lead_time <= 25.50
|   |   |   |   |   |   |   |   |   |--- arrival_month <= 11.50
|   |   |   |   |   |   |   |   |   |   |--- arrival_month <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [16.40, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |   |--- arrival_month >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [38.77, 118.41] class: 1
```

```
|   |   |   |   |   |   |   |   |   |--- arrival_month >  11.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [23.11, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- lead_time >  25.50
|   |   |   |   |   |   |   |   |   |--- type_of_meal_plan_Not Selected <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- no_of_week_nights <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [39.51, 185.21] class: 1
|   |   |   |   |   |   |   |   |   |   |--- no_of_week_nights >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |--- type_of_meal_plan_Not Selected >  0.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [73.81, 411.41] class: 1
|   |   |   |   |   |   |   |--- avg_price_per_room >  104.31
|   |   |   |   |   |   |   |   |--- arrival_month <= 10.50
|   |   |   |   |   |   |   |   |   |--- room_type_reserved_Room_Type 5 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 195.30
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 9
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  195.30
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [0.75, 138.15] class: 1
|   |   |   |   |   |   |   |   |   |--- room_type_reserved_Room_Type 5 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- arrival_date <= 22.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [11.18, 6.07] class: 0
|   |   |   |   |   |   |   |   |   |   |--- arrival_date >  22.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [0.75, 9.11] class: 1
|   |   |   |   |   |   |   |   |--- arrival_month >  10.50
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 168.06
|   |   |   |   |   |   |   |   |   |   |--- lead_time <= 22.00
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- lead_time >  22.00
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [17.15, 83.50] class: 1
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  168.06
|   |   |   |   |   |   |   |   |   |   |--- weights: [12.67, 6.07] class: 0
|   |   |   |   |--- required_car_parking_space >  0.50
|   |   |   |   |   |--- weights: [48.46, 1.52] class: 0
|   |--- no_of_special_requests >  0.50
|   |   |--- no_of_special_requests <= 1.50
|   |   |   |--- market_segment_type_Online <= 0.50
|   |   |   |   |--- lead_time <= 102.50
|   |   |   |   |   |--- type_of_meal_plan_Not Selected <= 0.50
|   |   |   |   |   |   |--- weights: [697.09, 9.11] class: 0
|   |   |   |   |   |--- type_of_meal_plan_Not Selected >  0.50
|   |   |   |   |   |   |--- lead_time <= 63.00
|   |   |   |   |   |   |   |--- weights: [15.66, 1.52] class: 0
|   |   |   |   |   |   |--- lead_time >  63.00
|   |   |   |   |   |   |   |--- weights: [0.00, 7.59] class: 1
|   |   |   |   |--- lead_time >  102.50
|   |   |   |   |   |--- no_of_week_nights <= 2.50
|   |   |   |   |   |   |--- arrival_month <= 8.50
|   |   |   |   |   |   |   |--- weights: [31.31, 13.66] class: 0
|   |   |   |   |   |   |--- arrival_month >  8.50
|   |   |   |   |   |   |   |--- weights: [0.75, 6.07] class: 1
|   |   |   |   |   |--- no_of_week_nights >  2.50
|   |   |   |   |   |   |--- weights: [44.73, 3.04] class: 0
|   |   |   |--- market_segment_type_Online >  0.50
|   |   |   |   |--- lead_time <= 8.50
|   |   |   |   |   |--- lead_time <= 4.50
|   |   |   |   |   |   |--- no_of_week_nights <= 10.00
```

```
|   |   |   |   |   |   |   |--- weights: [498.03, 40.99] class: 0
|   |   |   |   |   |   |--- no_of_week_nights >  10.00
|   |   |   |   |   |   |   |--- weights: [0.00, 3.04] class: 1
|   |   |   |   |   |--- lead_time >  4.50
|   |   |   |   |   |   |--- arrival_date <= 13.50
|   |   |   |   |   |   |   |--- arrival_month <= 9.50
|   |   |   |   |   |   |   |   |--- weights: [58.90, 36.43] class: 0
|   |   |   |   |   |   |   |--- arrival_month >  9.50
|   |   |   |   |   |   |   |   |--- weights: [33.55, 1.52] class: 0
|   |   |   |   |   |   |--- arrival_date >  13.50
|   |   |   |   |   |   |   |--- type_of_meal_plan_Not Selected <= 0.50
|   |   |   |   |   |   |   |   |--- weights: [123.76, 9.11] class: 0
|   |   |   |   |   |   |   |--- type_of_meal_plan_Not Selected >  0.50
|   |   |   |   |   |   |   |   |--- avg_price_per_room <= 126.33
|   |   |   |   |   |   |   |   |   |--- weights: [32.80, 3.04] class: 0
|   |   |   |   |   |   |   |   |--- avg_price_per_room >  126.33
|   |   |   |   |   |   |   |   |   |--- weights: [9.69, 13.66] class: 1
|   |   |   |   |--- lead_time >  8.50
|   |   |   |   |   |--- required_car_parking_space <= 0.50
|   |   |   |   |   |   |--- avg_price_per_room <= 118.55
|   |   |   |   |   |   |   |--- lead_time <= 61.50
|   |   |   |   |   |   |   |   |--- arrival_month <= 11.50
|   |   |   |   |   |   |   |   |   |--- arrival_month <= 1.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [70.08, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |--- arrival_month >  1.50
|   |   |   |   |   |   |   |   |   |   |--- no_of_week_nights <= 4.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 11
|   |   |   |   |   |   |   |   |   |   |--- no_of_week_nights >  4.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |--- arrival_month >  11.50
|   |   |   |   |   |   |   |   |   |--- weights: [126.74, 1.52] class: 0
|   |   |   |   |   |   |   |--- lead_time >  61.50
|   |   |   |   |   |   |   |   |--- arrival_year <= 2017.50
|   |   |   |   |   |   |   |   |   |--- arrival_month <= 7.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [4.47, 57.69] class: 1
|   |   |   |   |   |   |   |   |   |--- arrival_month >  7.50
|   |   |   |   |   |   |   |   |   |   |--- lead_time <= 66.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [5.22, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |   |--- lead_time >  66.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |--- arrival_year >  2017.50
|   |   |   |   |   |   |   |   |   |--- arrival_month <= 9.50
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 71.93
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [54.43, 3.04] class: 0
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  71.93
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 10
|   |   |   |   |   |   |   |   |   |--- arrival_month >  9.50
|   |   |   |   |   |   |   |   |   |   |--- no_of_week_nights <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |--- no_of_week_nights >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |--- avg_price_per_room >  118.55
|   |   |   |   |   |   |   |--- arrival_month <= 8.50
|   |   |   |   |   |   |   |   |--- arrival_date <= 19.50
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 182.20
```

```
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 118.98
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  118.98
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  182.20
|   |   |   |   |   |   |   |   |   |   |--- arrival_date <= 7.00
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [5.96, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |   |--- arrival_date >  7.00
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |--- arrival_date >  19.50
|   |   |   |   |   |   |   |   |   |--- arrival_date <= 27.50
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 121.20
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [18.64, 6.07] class: 0
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  121.20
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |--- arrival_date >  27.50
|   |   |   |   |   |   |   |   |   |   |--- lead_time <= 55.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- lead_time >  55.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |--- arrival_month >  8.50
|   |   |   |   |   |   |   |   |--- arrival_year <= 2017.50
|   |   |   |   |   |   |   |   |   |--- arrival_month <= 9.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [11.93, 10.63] class: 0
|   |   |   |   |   |   |   |   |   |--- arrival_month >  9.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [37.28, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- arrival_year >  2017.50
|   |   |   |   |   |   |   |   |   |--- arrival_month <= 11.50
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 119.20
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [9.69, 28.84] class: 1
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  119.20
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 12
|   |   |   |   |   |   |   |   |   |--- arrival_month >  11.50
|   |   |   |   |   |   |   |   |   |   |--- lead_time <= 100.00
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [49.95, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |   |--- lead_time >  100.00
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [0.75, 18.22] class: 1
|   |   |   |   |   |--- required_car_parking_space >  0.50
|   |   |   |   |   |   |--- weights: [134.20, 1.52] class: 0
|   |   |--- no_of_special_requests >  1.50
|   |   |   |--- lead_time <= 90.50
|   |   |   |   |--- no_of_week_nights <= 3.50
|   |   |   |   |   |--- weights: [1585.04, 0.00] class: 0
|   |   |   |   |--- no_of_week_nights >  3.50
|   |   |   |   |   |--- no_of_special_requests <= 2.50
|   |   |   |   |   |   |--- no_of_week_nights <= 9.50
|   |   |   |   |   |   |   |--- lead_time <= 6.50
|   |   |   |   |   |   |   |   |--- weights: [32.06, 0.00] class: 0
|   |   |   |   |   |   |   |--- lead_time >  6.50
|   |   |   |   |   |   |   |   |--- arrival_month <= 11.50
|   |   |   |   |   |   |   |   |   |--- arrival_date <= 5.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [23.11, 1.52] class: 0
|   |   |   |   |   |   |   |   |   |--- arrival_date >  5.50
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 93.09
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
```

```
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  93.09
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [77.54, 27.33] class: 0
|   |   |   |   |   |   |   |   |   |--- arrival_month >  11.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [19.38, 0.00] class: 0
|   |   |   |   |   |   |   |--- no_of_week_nights >  9.50
|   |   |   |   |   |   |   |   |--- weights: [0.00, 3.04] class: 1
|   |   |   |   |   |   |--- no_of_special_requests >  2.50
|   |   |   |   |   |   |   |--- weights: [52.19, 0.00] class: 0
|   |   |   |--- lead_time >  90.50
|   |   |   |   |--- no_of_special_requests <= 2.50
|   |   |   |   |   |--- arrival_month <= 8.50
|   |   |   |   |   |   |--- avg_price_per_room <= 202.95
|   |   |   |   |   |   |   |--- arrival_year <= 2017.50
|   |   |   |   |   |   |   |   |--- arrival_month <= 7.50
|   |   |   |   |   |   |   |   |   |--- weights: [1.49, 9.11] class: 1
|   |   |   |   |   |   |   |   |--- arrival_month >  7.50
|   |   |   |   |   |   |   |   |   |--- weights: [8.20, 3.04] class: 0
|   |   |   |   |   |   |   |--- arrival_year >  2017.50
|   |   |   |   |   |   |   |   |--- lead_time <= 150.50
|   |   |   |   |   |   |   |   |   |--- weights: [175.20, 28.84] class: 0
|   |   |   |   |   |   |   |   |--- lead_time >  150.50
|   |   |   |   |   |   |   |   |   |--- weights: [0.00, 4.55] class: 1
|   |   |   |   |   |   |--- avg_price_per_room >  202.95
|   |   |   |   |   |   |   |--- weights: [0.00, 10.63] class: 1
|   |   |   |   |   |--- arrival_month >  8.50
|   |   |   |   |   |   |--- avg_price_per_room <= 153.15
|   |   |   |   |   |   |   |--- room_type_reserved_Room_Type 2 <= 0.50
|   |   |   |   |   |   |   |   |--- avg_price_per_room <= 71.12
|   |   |   |   |   |   |   |   |   |--- weights: [3.73, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- avg_price_per_room >  71.12
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 90.42
|   |   |   |   |   |   |   |   |   |   |--- arrival_month <= 11.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- arrival_month >  11.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [12.67, 7.59] class: 0
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  90.42
|   |   |   |   |   |   |   |   |   |   |--- weights: [64.12, 60.72] class: 0
|   |   |   |   |   |   |   |--- room_type_reserved_Room_Type 2 >  0.50
|   |   |   |   |   |   |   |   |--- weights: [5.96, 0.00] class: 0
|   |   |   |   |   |   |--- avg_price_per_room >  153.15
|   |   |   |   |   |   |   |--- weights: [12.67, 3.04] class: 0
|   |   |   |   |--- no_of_special_requests >  2.50
|   |   |   |   |   |--- weights: [67.10, 0.00] class: 0
|--- lead_time >  151.50
|   |--- avg_price_per_room <= 100.04
|   |   |--- no_of_special_requests <= 0.50
|   |   |   |--- no_of_adults <= 1.50
|   |   |   |   |--- market_segment_type_Online <= 0.50
|   |   |   |   |   |--- lead_time <= 163.50
|   |   |   |   |   |   |--- no_of_week_nights <= 1.50
|   |   |   |   |   |   |   |--- weights: [2.98, 0.00] class: 0
|   |   |   |   |   |   |--- no_of_week_nights >  1.50
|   |   |   |   |   |   |   |--- weights: [0.75, 24.29] class: 1
|   |   |   |   |   |--- lead_time >  163.50
|   |   |   |   |   |   |--- lead_time <= 341.00
```

```
|   |   |   |   |   |   |   |--- lead_time <= 173.00
|   |   |   |   |   |   |   |   |--- arrival_date <= 3.50
|   |   |   |   |   |   |   |   |   |--- weights: [46.97, 9.11] class: 0
|   |   |   |   |   |   |   |   |--- arrival_date >  3.50
|   |   |   |   |   |   |   |   |   |--- no_of_weekend_nights <= 1.00
|   |   |   |   |   |   |   |   |   |   |--- weights: [0.00, 13.66] class: 1
|   |   |   |   |   |   |   |   |   |--- no_of_weekend_nights >  1.00
|   |   |   |   |   |   |   |   |   |   |--- weights: [2.24, 0.00] class: 0
|   |   |   |   |   |   |   |--- lead_time >  173.00
|   |   |   |   |   |   |   |   |--- arrival_month <= 5.50
|   |   |   |   |   |   |   |   |   |--- arrival_date <= 7.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [0.00, 4.55] class: 1
|   |   |   |   |   |   |   |   |   |--- arrival_date >  7.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [6.71, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- arrival_month >  5.50
|   |   |   |   |   |   |   |   |   |--- weights: [188.62, 7.59] class: 0
|   |   |   |   |   |   |--- lead_time >  341.00
|   |   |   |   |   |   |   |--- weights: [13.42, 27.33] class: 1
|   |   |   |   |--- market_segment_type_Online >  0.50
|   |   |   |   |   |--- avg_price_per_room <= 2.50
|   |   |   |   |   |   |--- lead_time <= 285.50
|   |   |   |   |   |   |   |--- weights: [8.20, 0.00] class: 0
|   |   |   |   |   |   |--- lead_time >  285.50
|   |   |   |   |   |   |   |--- weights: [0.75, 3.04] class: 1
|   |   |   |   |   |--- avg_price_per_room >  2.50
|   |   |   |   |   |   |--- weights: [0.75, 97.16] class: 1
|   |   |   |--- no_of_adults >  1.50
|   |   |   |   |--- avg_price_per_room <= 82.47
|   |   |   |   |   |--- market_segment_type_Offline <= 0.50
|   |   |   |   |   |   |--- weights: [2.98, 282.37] class: 1
|   |   |   |   |   |--- market_segment_type_Offline >  0.50
|   |   |   |   |   |   |--- arrival_month <= 11.50
|   |   |   |   |   |   |   |--- lead_time <= 244.00
|   |   |   |   |   |   |   |   |--- no_of_week_nights <= 1.50
|   |   |   |   |   |   |   |   |   |--- no_of_weekend_nights <= 1.50
|   |   |   |   |   |   |   |   |   |   |--- lead_time <= 166.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [2.24, 0.00] class: 0
|   |   |   |   |   |   |   |   |   |   |--- lead_time >  166.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [2.24, 57.69] class: 1
|   |   |   |   |   |   |   |   |   |--- no_of_weekend_nights >  1.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [17.89, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- no_of_week_nights >  1.50
|   |   |   |   |   |   |   |   |   |--- no_of_weekend_nights <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- arrival_month <= 9.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [11.18, 3.04] class: 0
|   |   |   |   |   |   |   |   |   |   |--- arrival_month >  9.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [0.00, 12.14] class: 1
|   |   |   |   |   |   |   |   |   |--- no_of_weekend_nights >  0.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [75.30, 12.14] class: 0
|   |   |   |   |   |   |   |--- lead_time >  244.00
|   |   |   |   |   |   |   |   |--- arrival_year <= 2017.50
|   |   |   |   |   |   |   |   |   |--- weights: [25.35, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- arrival_year >  2017.50
|   |   |   |   |   |   |   |   |   |--- avg_price_per_room <= 80.38
|   |   |   |   |   |   |   |   |   |   |--- no_of_week_nights <= 3.50
```

```
|   |   |   |   |   |   |   |   |   |   |   |   |--- weights: [11.18, 264.15] class: 1
|   |   |   |   |   |   |   |   |   |   |   |--- no_of_week_nights >  3.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- avg_price_per_room >  80.38
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [7.46, 0.00] class: 0
|   |   |   |   |   |   |   |--- arrival_month >  11.50
|   |   |   |   |   |   |   |   |--- weights: [46.22, 0.00] class: 0
|   |   |   |   |   |--- avg_price_per_room >  82.47
|   |   |   |   |   |   |--- no_of_adults <= 2.50
|   |   |   |   |   |   |   |--- lead_time <= 324.50
|   |   |   |   |   |   |   |   |--- arrival_month <= 11.50
|   |   |   |   |   |   |   |   |   |--- room_type_reserved_Room_Type 4 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [7.46, 986.78] class: 1
|   |   |   |   |   |   |   |   |   |--- room_type_reserved_Room_Type 4 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- market_segment_type_Offline <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [0.00, 10.63] class: 1
|   |   |   |   |   |   |   |   |   |   |--- market_segment_type_Offline >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [4.47, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- arrival_month >  11.50
|   |   |   |   |   |   |   |   |   |--- market_segment_type_Offline <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [0.00, 19.74] class: 1
|   |   |   |   |   |   |   |   |   |--- market_segment_type_Offline >  0.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [5.22, 0.00] class: 0
|   |   |   |   |   |   |   |--- lead_time >  324.50
|   |   |   |   |   |   |   |   |--- avg_price_per_room <= 89.00
|   |   |   |   |   |   |   |   |   |--- weights: [5.96, 0.00] class: 0
|   |   |   |   |   |   |   |   |--- avg_price_per_room >  89.00
|   |   |   |   |   |   |   |   |   |--- weights: [0.75, 13.66] class: 1
|   |   |   |   |   |   |--- no_of_adults >  2.50
|   |   |   |   |   |   |   |--- weights: [5.22, 0.00] class: 0
|   |   |   |--- no_of_special_requests >  0.50
|   |   |   |   |--- no_of_weekend_nights <= 0.50
|   |   |   |   |   |--- lead_time <= 180.50
|   |   |   |   |   |   |--- lead_time <= 159.50
|   |   |   |   |   |   |   |--- arrival_month <= 8.50
|   |   |   |   |   |   |   |   |--- weights: [5.96, 0.00] class: 0
|   |   |   |   |   |   |   |--- arrival_month >  8.50
|   |   |   |   |   |   |   |   |--- weights: [1.49, 7.59] class: 1
|   |   |   |   |   |   |--- lead_time >  159.50
|   |   |   |   |   |   |   |--- arrival_date <= 1.50
|   |   |   |   |   |   |   |   |--- weights: [1.49, 3.04] class: 1
|   |   |   |   |   |   |   |--- arrival_date >  1.50
|   |   |   |   |   |   |   |   |--- weights: [35.79, 1.52] class: 0
|   |   |   |   |   |--- lead_time >  180.50
|   |   |   |   |   |   |--- no_of_special_requests <= 2.50
|   |   |   |   |   |   |   |--- market_segment_type_Online <= 0.50
|   |   |   |   |   |   |   |   |--- avg_price_per_room <= 96.37
|   |   |   |   |   |   |   |   |   |--- weights: [12.67, 3.04] class: 0
|   |   |   |   |   |   |   |   |--- avg_price_per_room >  96.37
|   |   |   |   |   |   |   |   |   |--- weights: [0.00, 3.04] class: 1
|   |   |   |   |   |   |   |--- market_segment_type_Online >  0.50
|   |   |   |   |   |   |   |   |--- weights: [7.46, 206.46] class: 1
|   |   |   |   |   |   |--- no_of_special_requests >  2.50
|   |   |   |   |   |   |   |--- weights: [8.95, 0.00] class: 0
|   |   |   |   |--- no_of_weekend_nights >  0.50
```

```
|   |   |   |   |--- market_segment_type_Offline <= 0.50
|   |   |   |   |   |--- arrival_month <= 11.50
|   |   |   |   |   |   |--- avg_price_per_room <= 76.48
|   |   |   |   |   |   |   |--- weights: [46.97, 4.55] class: 0
|   |   |   |   |   |   |--- avg_price_per_room >  76.48
|   |   |   |   |   |   |   |--- no_of_week_nights <= 6.50
|   |   |   |   |   |   |   |   |--- arrival_date <= 27.50
|   |   |   |   |   |   |   |   |   |--- lead_time <= 233.00
|   |   |   |   |   |   |   |   |   |   |--- lead_time <= 152.50
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [1.49, 4.55] class: 1
|   |   |   |   |   |   |   |   |   |   |--- lead_time >  152.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |--- lead_time >  233.00
|   |   |   |   |   |   |   |   |   |   |--- weights: [23.11, 19.74] class: 0
|   |   |   |   |   |   |   |   |--- arrival_date >  27.50
|   |   |   |   |   |   |   |   |   |--- no_of_week_nights <= 1.50
|   |   |   |   |   |   |   |   |   |   |--- weights: [2.24, 15.18] class: 1
|   |   |   |   |   |   |   |   |   |--- no_of_week_nights >  1.50
|   |   |   |   |   |   |   |   |   |   |--- lead_time <= 269.00
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- lead_time >  269.00
|   |   |   |   |   |   |   |   |   |   |   |--- weights: [0.00, 4.55] class: 1
|   |   |   |   |   |   |   |--- no_of_week_nights >  6.50
|   |   |   |   |   |   |   |   |--- weights: [4.47, 13.66] class: 1
|   |   |   |   |   |   |--- arrival_month >  11.50
|   |   |   |   |   |   |   |--- arrival_date <= 14.50
|   |   |   |   |   |   |   |   |--- weights: [8.20, 3.04] class: 0
|   |   |   |   |   |   |   |--- arrival_date >  14.50
|   |   |   |   |   |   |   |   |--- weights: [11.18, 31.88] class: 1
|   |   |   |   |--- market_segment_type_Offline >  0.50
|   |   |   |   |   |--- lead_time <= 348.50
|   |   |   |   |   |   |--- weights: [106.61, 3.04] class: 0
|   |   |   |   |   |--- lead_time >  348.50
|   |   |   |   |   |   |--- weights: [5.96, 4.55] class: 0
|   |--- avg_price_per_room >  100.04
|   |   |--- arrival_month <= 11.50
|   |   |   |--- no_of_special_requests <= 2.50
|   |   |   |   |--- weights: [0.00, 3200.19] class: 1
|   |   |   |--- no_of_special_requests >  2.50
|   |   |   |   |--- weights: [23.11, 0.00] class: 0
|   |   |--- arrival_month >  11.50
|   |   |   |--- no_of_special_requests <= 0.50
|   |   |   |   |--- weights: [35.04, 0.00] class: 0
|   |   |   |--- no_of_special_requests >  0.50
|   |   |   |   |--- arrival_date <= 24.50
|   |   |   |   |   |--- weights: [3.73, 0.00] class: 0
|   |   |   |   |--- arrival_date >  24.50
|   |   |   |   |   |--- weights: [3.73, 22.77] class: 1
```

```
#Let's plot important features
```

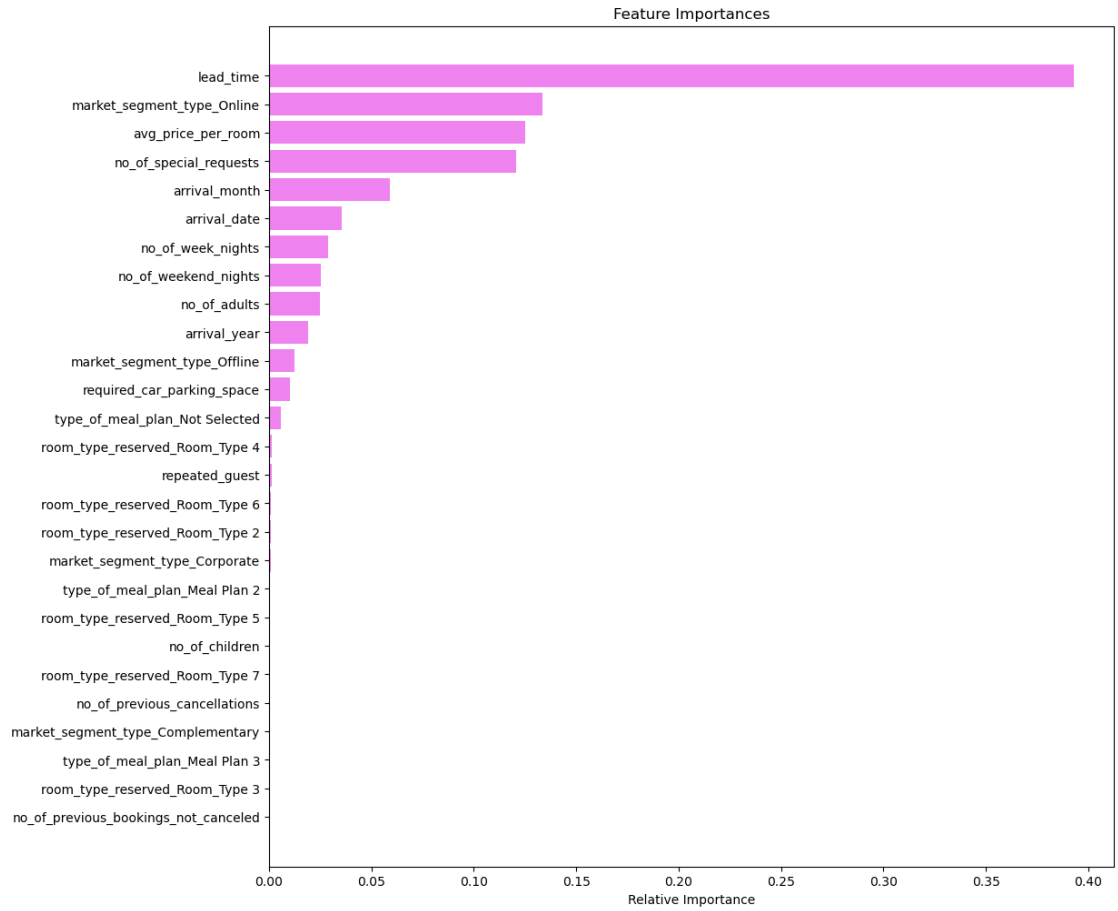


Figure 53 pre pruned feature importance check

- `lead_time` and `market_segment_type_online` are still top features in the pruned tree

Model Performance Comparison and Conclusions

Model performance comparison on the training data

```
Training Model performance comparison:

           Decision Tree sklearn  Decision Tree (Pre-Pruning)  \
Accuracy                 0.99421                      0.83097
Recall                   0.98661                      0.78608
Precision                0.99578                      0.72425
F1                       0.99117                      0.75390

           Decision Tree (Post-Pruning)
```

```
Accuracy                        0.89950
Recall                          0.90410
Precision                       0.81205
F1                              0.85561
```

Model performance comparison on the testing data

```
Testing Model performance comparison:

           Decision Tree sklearn  Decision Tree (Pre-Pruning)  \
Accuracy                 0.87118                      0.83497
Recall                   0.81005                      0.78336
Precision                0.79559                      0.72758
F1                       0.80276                      0.75444

           Decision Tree (Post-Pruning)
Accuracy                        0.86778
Recall                          0.85491
Precision                       0.76441
F1                              0.80713
```

- The parameters Accuracy,Recall,Precision and F1-score have better values in the post pruned tree than Pre-pruned tree

Conclusion

- Usually atleast 2 person are inculded in the booking
- People try to book the rooms for an average of 2 nights during weekdays and 1 night during weekends
- Usually booking are done around 3 months (85 days) in advance
- From the graph it is visible that room type 1 is more preferred
- Average room price is 104 Euros
- Customers making the reservations mostly first timers as number of previous cancellations is very low
- Majority guests book the room online

Actionable Insights and Recommendations

- **Cancellation Rate:** Roughly 33% of bookings are canceled, with higher cancellation percentages observed in the online market segment (around 38%) and very low rates among repeat guests (approximately 2%).
- **Lead Time:** Reservations with shorter lead times tend to have higher cancellation rates.
- **Special Requests:** Bookings with multiple special requests show markedly lower cancellation risk.
- **Market Segmentation:** Corporate and aviation channels have much lower cancellation rates compared to the online or offline channels.

—we can recommend a set of profitable policies and broader strategic recommendations to better manage cancellations and refunds.

Profitable Cancellation and Refund Policies

1. **Tiered Cancellation Windows & Fees:**
    - **Early Cancellation Window:** Allow free or low-penalty cancellations (i.e., full refund) if the guest cancels well in advance (for example, 7 days or more before arrival). This provides the hotel enough time to re-sell the room.
    - **Last-Minute Cancellations:** For cancellations made within a shorter window (e.g., within 48–72 hours before arrival), impose a steeper penalty—such as charging one night's rate or a fixed percentage of the total booking—reflecting the difficulty in rebooking at the last minute.
    - **Dynamic Fee Structure:** Adjust cancellation fees based on booking attributes. For high-risk segments (e.g., online bookings or bookings with short lead times), require either a non-refundable deposit or a higher cancellation fee. Conversely, for low-risk groups (such as loyal, repeated guests who often make special requests), consider offering more flexible terms to help build loyalty.
2. **Non-Refundable or Partially Refundable Rates:**
    - **Offer Discounted Non-Refundable Rooms:** Provide room rates at a discounted price in exchange for the non-refundable option. This approach benefits the hotel by guaranteeing revenue and is attractive to guests who are sure of their plans.
    - **Partially Refundable Options:** Introduce a policy where guests receive a partial refund if they cancel within a specified period. For instance, canceling between a set number of days (e.g., 72 hours) and check-in might result in a 50% refund, while cancellations beyond this window lead to no refund. This gradual penalty system incentivizes early cancellation and ensures some recovery of revenue.
3. **Deposit-Based Bookings:**
    - **Advance Deposit Requirement:** Require a deposit at the time of booking, especially for reservation categories that historically show higher cancellation rates. The deposit would be forfeited if the booking is canceled last minute, thereby mitigating revenue loss.

Additional Recommendations to Enhance Profitability and Operational Efficiency

1. **Implement Predictive Analytics:**
    - **Cancellations Forecasting:** Use machine learning models (as developed from the dataset) to predict the likelihood of cancellation on a per-booking basis. By identifying high-risk bookings upfront, the hotel can apply stricter policies (e.g., a required deposit or non-refundable conditions) on those reservations.
    - **Overbooking Strategy:** Based on predicted cancellation probabilities, adopt dynamic overbooking strategies to ensure maximal occupancy without jeopardizing guest satisfaction.
2. **Segmented Cancellation Policies:**
    - **Market-Specific Policies:** Tailor cancellation and refund policies to different market segments. For example, offer more lenient policies for corporate bookings (which tend to have very low cancellation rates) while enforcing stricter terms for online bookings.
    - **Loyalty Incentives:** Reward repeat guests by providing flexible cancellation provisions exclusively for them. This can foster loyalty and reduce the overall cancellation rate, as repeat guests have already demonstrated lower cancellation behavior.
3. **Enhanced Communication and Transparency:**
    - **Clear Communication:** Ensure that cancellation policies are clearly presented at the time of booking so that guests understand the financial implications of a cancellation. This clarity can reduce cancellations made "just in case."
    - **Reminder Systems:** Implement automated reminders leading up to the check-in date to confirm the guest's intent, which may help reduce cancellations and no-shows.
4. **Dynamic Pricing and Ancillary Revenue:**
    - **Dynamic Pricing Models:** Adopt revenue management tools to adjust room prices dynamically based on demand, booking patterns, and recent cancellation trends.
    - **Ancillary Services:** Diversify revenue streams by offering services like dining packages, spa treatments, or bundled experiences, which can provide additional revenue that partly offsets the impact of cancellations.
5. **Operational Adjustments:**
    - **Resource Planning:** Use historical data on cancellations to better forecast staffing needs and operational costs, reducing waste when cancellations occur.
    - **Flexible Rebooking:** Develop systems that allow for rapid redistribution of canceled bookings—such as offering last-minute deals as soon as a cancellation is detected—to recapture lost revenue.

Conclusion

By combining a scientifically informed cancellation and refund policy with predictive analytics and segmented approaches, the hotel can strategically reduce revenue losses from cancellations while keeping guest satisfaction high. These measures, along with enhancements to dynamic pricing, operational planning, and loyalty programs, ensure that the hotel's revenue management is both robust and adaptive to changing market conditions.